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Fluid models

Fluid models are used to model the dynamic behavior of the hydrocarbons and other fluids inside the reservoir during the life cycle of the field.

To describe the behavior of hydrocarbons, it is important to represent the pressure volume temperature (PVT) characteristics of the fluids within the simulation. This means defining how the volumes (and hence saturations) of the fluids vary with changing pressure and temperature. Three different types of fluid models can be used to define the PVT behavior:

- Black oil
- Compositional
- Thermal

Note: In both the *Black oil* and *Compositional* fluid models, any variation in the temperature of the grid cell is assumed to have little impact on the PVT behavior of the fluid and remains fixed during the simulation.

The results of fluid measurements are drawn on a phase diagram which shows where the fluids are liquids (above the bubble point line), where they are gases (below the dew point line) and where they are mixed liquid/gas (within the phase envelope enclosed by the bubble and dew point lines). By plotting the reservoir and surface conditions on a phase diagram, you can characterize your reservoir fluids and determine whether to use a black oil or compositional model.

Black oil

In a *Black oil* representation there are two hydrocarbon components, a liquid 'oil' phase and a vapor 'gas' phase. The model uses PVT tables to define the variation of physical properties such as density, viscosity and compressibility with temperature and pressure. They also contain data that relate the surface and reservoir volumes of the oil and gas components, including the oil and gas formation volume factors, the amount of solution (dissolved) gas, the solution gas/oil ratio (the ratio of dissolved gas to oil) and the gas/oil ratio (the ratio of all gas to oil). Similar PVT properties are defined for a single water component. The tables are obtained either from laboratory experiments, correlations, or a matched equation of state (EoS). The model assumes that there are no compositional changes with pressure and temperature. However, by including some variation in gas/oil and oil/gas ratios in the model, you can mimic compositional changes. For more information on the equations used in black oil modeling, see [Formulation of the equations](#).

Eclipse provides some extensions to the basic black oil model:

The [API tracking](#) facility enables Eclipse to model the mixing of different types of oil, which have different PVT properties.

The [Gi pseudo-compositional model](#) extends the black-oil model for use in gas condensate or volatile oil reservoirs to take account of fluid property changes during gas injection. The conventional model ignores the effect of light liquid components being stripped into the vapor phase during gas injection. The Gi parameter in the extended model accounts for this behavior.

Compositional

In a *Compositional* representation, the fluid model is defined by an equation of state and its critical parameters. These parameters are entered for each component, allowing the liquid and vapor phases to be broken down into their constituent components. This makes it possible, for example, to determine the percentage of methane that constitutes the gas phase. A *compositional* fluid model is more computationally expensive to solve but allows the hydrocarbons of different compositions to mix during the simulation.

A compositional model typically specifies the properties of the lighter hydrocarbon components and then creates a single pseudo-component, or lumped component for the C_7+ or $C_{10}+$ components (those containing seven or more, or ten or more carbon atoms). This pseudo-component is defined by its apparent molecular weight and specific gravity. The model can also specify the properties of non-hydrocarbons such as nitrogen and sulfur dioxide. You can also use a multi-component water model where the water in different parts of the reservoir contains different dissolved compounds.

For more information on the Eclipse compositional modeling, see the [Equations of state](#). You may want to model more than one fluid, for example where you have two separate reservoirs coupled by the production system. In this case you can assign different equations of state to different sets of grid cells in your model as described in [Equations of state regions](#).

Thermal

The *Thermal* fluid representation allows the temperature as well as the pressures to vary during the simulation, for example where steam or hot water is injected into the reservoir to increase production. This model accounts for the effects of water evaporating into steam and for the changes in physical properties such as oil density and viscosity at different temperatures. The thermal model is also useful for modeling the recovery of heavy oil using thermal techniques such as steam assisted gravity drainage, which uses heat to reduce oil viscosity and gravity to collect the lighter oil components as they fall through the reservoir.

For information on the Eclipse thermal modeling, see the [Temperature option](#) and the [Thermal model](#).

API tracking

x	Eclipse 100
	Eclipse 300

The API tracking facility enables Eclipse to model the mixing of different types of oil, having different surface densities and PVT properties.

Without the API tracking facility, the presence of different types of oil in the reservoir could be handled with the aid of PVT region numbers. For instance, oil in PVT region 1 would have its properties determined from PVT table number 1, and so on. However, this method cannot model the mixing of oil types. So, for example, oil flowing from region 1 into region 2 would appear to take on the properties associated with region 2.

The API tracking facility essentially replaces the concept of PVT regions for oil. The PVT tables used for determining the oil properties are selected at each timestep according to the average API of the oil in each grid block (or to be more precise, its average surface density). A mass conservation equation is solved at the end of each timestep to update the oil surface density in each grid block, in order to model the mixing of the different oil types.

The API tracking facility is activated by specifying the [API](#) keyword in the RUNSPEC section.

In dead oil systems the keyword [RSCONSTT](#) cannot be used with API tracking, but [RSCONST](#) (defining a uniform R_s value over the whole field) can be used.

The API tracking facility

PVT properties

Two or more sets of PVT tables should be supplied, each set being associated with a particular value of the API. The oil property tables are entered in the usual way (with keywords [PVTO](#) or [PVCO](#) for live oil problems, or keyword [PVDO](#) for dead oil problems), but with the requirement that all the [PVTO](#) or [PVCO](#) tables must have the same upper and lower R_s values. From these tables, Eclipse constructs a set of internal tables that have a common set of R_s nodes, which includes the R_s nodes of all the input tables. This allows Eclipse to use an efficient method of interpolating between tables, for intermediate values of the API. The restriction that the upper and lower R_s values must be the same for each table prevents extrapolation of the input data during this process, which may cause unphysical behavior.

The API values associated with each PVT table are set using the keyword [GRAVITY](#), which inputs the oil, water and gas gravities for each PVT table number. The API values are converted internally to oil surface densities, using the formula given in "[Conversion factors](#)". The oil API values must **decrease** monotonically with the table number. Alternatively, the surface densities associated with each table number can be input directly using the keyword [DENSITY](#). In this case the oil surface densities must **increase** monotonically with the table number.

The PVT properties of water and gas are not affected by the API Tracking option. The same number of tables must be entered in all the PVT data keywords and the [ROCK](#) keyword, but normally only the first table is actually used in all, except the oil PVT data. The unused PVT and rock compressibility tables can be defaulted by typing a single slash (/) on a new line for each unused table in the keyword.

Grouping tables into PVT regions

The API Tracking facility overrides the concept of PVT table regions for oil. The tables used to determine the oil PVT properties are selected according to the surface density of the oil in the grid block, instead of

the block's PVT region number. Thus, the `PVTNUM` keyword in the `REGIONS` section is not normally required with the API Tracking facility.

If, however, the concept of PVT regions for different types of oil is still required in an API Tracking run, the oil PVT tables can be partitioned into groups for use in different regions of the reservoir. To accomplish this, the keyword `APIGROUP` must be entered to specify the maximum number of groups of oil PVT tables. The grouping of the oil PVT tables is then determined by the order in which they are entered. Within each group, the oil API gravities should decrease (or the oil surface densities should increase) monotonically with the table number. A break in the required monotonicity indicates that the subsequent tables belong to the next API group. An error is flagged if the total number of API groups exceeds the maximum specified in the `APIGROUP` keyword.

The separate groups of oil PVT tables can then be used in different areas of the reservoir. The `REGIONS` section keyword `PVTNUM` is used to specify which API group is used for each grid block. For example, a cell in `PVTNUM` region 2 uses the second API group to obtain the oil properties.

For water and gas PVT properties, the `PVTNUM` regions correspond to the actual table numbers, which is similar to runs that do not use API Tracking. For instance, a cell in `PVTNUM` region 2 still uses table number 2 for its water and gas properties.

Setting the initial conditions

The initial API distribution throughout the reservoir is specified in the `SOLUTION` section.

If the initial conditions are determined by equilibration, the API can vary with depth independently in each equilibration region. Keyword `APIVD` supplies a table of API values versus depth for each equilibration region.

When setting the initial conditions by enumeration (keywords `PRESSURE` and `SWAT` for example), the keyword `OILAPI` is used to supply the initial API values for each grid block.

The API tracking calculation

The initial API values in the grid blocks are immediately converted into oil surface density. The hydrostatic pressure gradient calculation takes account of the varying surface density of the oil. The variation of other PVT properties with surface density (bubble point pressure, formation volume factor and viscosity) is handled by interpolating between PVT tables. The two adjacent PVT tables whose oil surface densities straddle the oil surface density value in the grid block are located, and their properties ($1/B_o$, $1/(B_o\mu_o)$) are interpolated linearly in oil surface density.

To calculate the bubble point, the two PVT tables corresponding to oil surface gravities either side of the current grid block API value are located. The bubble point is then obtained by linearly interpolating between the bubble points in these PVT tables at the given grid block dissolved gas-oil ratio.

The oil surface density in each grid block and well is held constant over the timestep. When a converged solution for the timestep has been found, and the inter-block flows determined, a mass conservation equation is solved to update the oil surface densities. The new densities are used in the next timestep.

The oil surface densities are converted back into API values for the output reports. The mnemonic `OILAPI` in the `RPTSCHEG` keyword provides an output of the grid block API values, and the well reports include the API values in the wells.

The `APILIM` keyword can be used to monitor the numbers of grid blocks with API values that are outside specified limits and additionally to constrain the API values to within these limits.

API tracking in wells

The oil surface density in each well is calculated at the end of the timestep to reflect a flow-weighted average surface density of the inflowing oil. A crossflowing well reinjects oil of this average surface density back into the formation. However, the hydrostatic head calculation in the wellbore is based on a (more accurate) flow-weighted average of **upstream** connection inflows if **WELSP ECS** item 12 is ' SEG ', allowing the oil density to vary with position in the wellbore. (If required, the pre-98A treatment of wellbore hydrostatic head, which uses the well's average oil surface density, can be restored by setting [item 35](#) in the **OPTIONS** keyword.)

If the well uses a VFP table to calculate the tubing pressure losses, this should be calculated using a suitable value of the oil surface density. However, it is possible to take account of the variation of the well's producing API over time by defining the fifth lookup variable, the ALQ, to represent the surface density of the produced oil (see keyword [WALQCALC](#)). The VFP table should be prepared using the same definition of the ALQ, with values spanning the expected range of oil surface densities. The VFP *i* program contains an option to prepare tables with variable oil surface density.

Behavior in wet gas systems

By default, in wet gas systems, the surface density property (API) of the oil is transported in both the oil and gas phases. In other words, the API is interpreted as a property of the oil component rather than the liquid oil phase. However, it is the API in the liquid oil phase that is used in subsequent calculations involving the surface density of oil.

By using [item 58](#) of the **OPTIONS** keyword this behavior can be modified. When the item is set greater than zero, the API property is only transported in the liquid oil phase. In this case, the API property can be thought of as a heavy component that does not vaporize. However, some care needs to be taken when using this option as it is possible to vaporize all the oil in a cell, leading to high surface densities and negative API values.

Brine tracking

x	Eclipse 100
x	Eclipse 300

Eclipse 100

The brine tracking facility enables Eclipse to model the mixing of waters with different salinities, as well as the effect of low versus high salinity on the flow performance with the low salinity option.

The salinities can be made of different salt species. In that case different effective salinities are calculated depending on the features used in conjunction with the brine option. For the polymer and surfactant options, ions exchanges are solved to determine the effective salinities used in the adsorption isotherms.

The concentration of salt in the water phase can modify both the water density and the viscosity, and, with the low salinity option, the saturation and relative permeability end points for oil and water and the water-oil capillary pressure.

For example, if low salinity water is injected into a reservoir with a significantly higher natural salinity, the lighter injected water would be expected to override the initial aquifer water.

The Brine Tracking facility is activated by the keyword **BRINE** in the RUNSPEC section, and the low salinity option is activated by the **LOWSALT** in the RUNSPEC as well.

The multi-component brine option is activated by the keyword **BRINE** in conjunction with the keyword **ECLMC** in the RUNSPEC section.

The Brine Model

Brine conservation equation

The distribution of brine is modeled by solving a mass conservation equation for the salt concentration in each grid block. Brine is assumed to exist solely in the water phase and is modeled internally as a water phase tracer using the equation

$$\frac{d}{dt} \left(\frac{V S_w C_s}{B_w} \right) = \sum \left[\frac{T k_{rw}}{B_w \mu_{s \text{ eff}}} (\delta P_w - \rho_w g D_z) \right] C_s + Q_w C_s \quad \text{Eq. 4.1}$$

where

ρ_w denotes the water density

$\sum$ denotes the sum over neighboring cells

C_s denotes the salt concentration in the aqueous phase

$\mu_{s \text{ eff}}$ denotes the effective viscosity of the salt

D_z is the cell center depth.

B_w is the water formation volume

T is the transmissibility

k_{rw} is the water relative permeability

S_w is the water saturation

V is the block pore volume

Q_w is the water production rate

P_w is the water pressure

g is the gravity acceleration

By default, the salt concentrations are updated at the end of a timestep after the inter-block phase flows have been determined. If the salinity effect of the flow properties is very strong, the salt concentrations can be solved in each Newton iteration so that the tracer concentrations are updated simultaneously with the phase equations by setting the [item 126](#) of the `OPTIONS` keyword to 1. See ["Solving the conservation equations"](#) for more details. The new salt concentrations are then used to update the water properties for the subsequent iterations.

Note: Small errors in the material balance may lead to large fluctuations in the reported concentrations where the fraction of the host phase is very small. These fluctuations can be limited by reducing the maximum material balance and linear convergence errors using the [TUNING](#) keyword.

Multi-component brine model

In the standard brine tracking option all salt species are lumped into a single salt. The multi-component brine model enables the modeling of processes where different salt species affect fluid properties in different ways. For example, when used in conjunction with the surfactant and polymer options, effective salinities are calculated based on the transport and exchange of different ions. These effective salinities can then be used when modeling surfactant phase behavior and polymer/surfactant adsorption.

Other salinity effects (for example, the low salinity option discussed below) do not use the multi-component brine model, and they are not active when the multi-component brine model is active. Certain combinations are explicitly banned; in particular, the [SALTNODE](#) keyword and other keywords depending on it will not work with [ECLMC](#).

In the current multi-component brine model three different ion species are considered: anions, monovalent cations, and divalent cations. All salts are assumed to be fully dissociated into their ion constituents, and salt precipitation/dissolution is therefore not accounted for. In this release, the ion species are transported as water phase tracers according to:

$$\frac{d}{dt} \left(\frac{VS_w(C_{++} + C_{++}^S)}{B_w} + VC_{++}^r \right) = \Sigma \left[\frac{Tk_{rw}}{B_w\mu_w} (\delta P_w - \rho_w g D_z) \right] C_{++} + Q_w C_{++} \quad \text{Eq. 4.2}$$

$$\frac{d}{dt} \left(\frac{VS_w C_-}{B_w} \right) = \Sigma \left[\frac{Tk_{rw}}{B_w\mu_w} (\delta P_w - \rho_w g D_z) \right] C_- + Q_w C_- \quad \text{Eq. 4.3}$$

$$C_+ = C_- - C_{++} \quad \text{Eq. 4.4}$$

where

C_{++} denotes the equivalent molar concentration of divalent cations

C_+ denotes the equivalent molar concentration of monovalent cations

C_- denotes the equivalent molar concentration of anions

C_{++}^s denotes the concentration of divalent cations associated with surfactant

C_{++}^r denotes the concentration of divalent cations associated with rock

The condition of electroneutrality in the water phase, $C_- = C_{++} + C_+$, is eliminated explicitly, and therefore only two of the three ion concentration transport equations need solving. Hence, when activating the Multi-component brine model an additional two conservation equations (4.2 and 4.3) are solved for the anion and divalent cation concentrations, respectively. As for the standard brine tracking option the ion concentrations are updated at the end of a timestep after solving the primary equations for the phase flows.

The multi-component brine model is activated using keyword **BRINE** in the RUNSPEC section in conjunction with the **ECLMC** keyword. The salts are specified in the **BRINE** keyword by their chemical formula and must be chosen from the list given in table 4.1. Internally, all salt concentrations are converted and lumped into equivalent molar concentrations of anions, monovalent cations, and divalent cations. The equivalent molar concentration of an ion is the molar concentration multiplied by the ion valence. Thus, a calcium ion (Ca^{2+}) concentration of 0.1 kg-M/m^3 corresponds to an equivalent concentration of 0.2 kg-M/m^3 . All reporting of ion concentrations is done in equivalent concentrations.

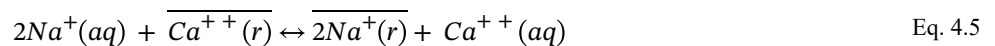
Salt name	Chemical formula	Molecular weight [kg/kg-M]
Sodium Chloride	NaCl	58.44
Potassium Chloride	KCl	74.55
Calcium Chloride	CaCl_2	110.98
Magnesium Chloride	MgCl_2	95.21
Sodium Carbonate	Na_2CO_3	105.99
Potassium Carbonate	K_2CO_3	138.20
Calcium Carbonate	CaCO_3	100.09
Magnesium Carbonate	MgCO_3	84.31
Sodium Sulfate	Na_2SO_4	142.04
Potassium Sulfate	K_2SO_4	174.26
Calcium Sulfate	CaSO_4	136.14
Magnesium Sulfate	MgSO_4	120.37

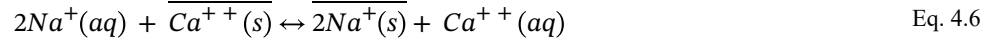
Table 4.1: List of available salts in the Multi-Component Brine mode.

Cation exchange

In surfactant processes the activity of the surfactant changes with changing salt concentrations. Divalent cations have a much greater effect on surfactant activity, and thereby on phase behavior, than monovalent cations. The cation exchange with rock surfaces will change the flowing concentrations of monovalent and divalent cations and is therefore important to the performance of surfactant floods. The exchange capacity of the rock generally depends on the clay content in the reservoir. In the same way that rock surfaces act as cation exchangers, anionic surfactant micelles can also act as cation exchangers. The cation exchange model accounts for exchange of cations on both rock surfaces and surfactant micelles.

The exchange equilibrium reactions, exemplified here for sodium and calcium, are





where the bar indicates that the ion is associated with either the rock surface (r) or surfactant micelles (s) and (aq) stands for aqueous. The exchange equilibria are modeled as

$$\frac{(C_{++})(C_+^r)^2}{(C_{++}^r)(C_+)^2} = \beta^r q^r, \quad \text{Eq. 4.7}$$

$$\frac{(C_{++})(C_+^s)^2}{(C_{++}^s)(C_+)^2} = \beta^s C^s, \quad \text{Eq. 4.8}$$

where

q^r denotes the rock cation exchange capacity

C^s denotes the equivalent molar concentration of surfactant

β^r denotes the cation exchange constant for rock

β^s denotes the cation exchange constant for surfactant

Together with the conditions of electroneutrality on both rock surfaces, $q^r = C_+^r + C_{++}^r$, and surfactant micelles, $C^s = C_+^s + C_{++}^s$, equations 4.7 and 4.8 are solved to obtain the equilibrium partitioning of monovalent and divalent cations.

The cation exchange model can be activated independently for rock exchanges and surfactant micelle exchanges using keywords [IONXROCK](#) and [IONXSURF](#) in the PROPS section, respectively. For rock ion exchanges the cation exchange constant, β^r , should be provided using the [IONXROCK](#) keyword. The rock exchange capacity must be entered for each grid block using keyword [IONROCK](#) in the GRID section. For surfactant micelle exchanges the surfactant option must first be activated using keyword [SURFACT](#) in the RUNSPEC section. The cation exchange constant for surfactant, β^s , should be provided using the [IONXSURF](#) keyword along with the equivalent molecular weight of the surfactant. Since the equations solve for the surfactant's mass concentration, an equivalent molecular weight is required to calculate the number of active sites available on surfactant micelles for ion exchange.

Note: The equivalent molecular weight refers to the weight of surfactant per mole of “active sites”. A surfactant molecule may have more than one active anionic site. For example, a divalent surfactant with a molecular weight of 500 kg/kg-M has an equivalent weight of 250 kg/kg-M.

When the flowing concentrations of cations are zero, the exchange equilibria are ignored and the rock and surfactant micelle associated concentrations are zero. This corresponds to ignoring the exchange capacity of the rock and surfactant along with the respective electroneutrality conditions. To ensure continuous behavior in the vicinity of zero cation concentrations, a damping function is imposed on the exchange capacities. The damping function satisfies $W(0) = 0$ and $W(\varepsilon) = 1$, where ε is a small threshold value. The damping is only imposed for $C_{++} \in [0, \varepsilon]$. The threshold value is set according to

$$\varepsilon = THIONX \times C_{++}^{ref} C_{++}^{ref} = \max(C_{++}^{inj}, 10^{-2}),$$

where

C_{++}^{inj} is the maximum injection concentration and

THIONX is a relative threshold concentration, which can be set using the **TUNING** keyword. The default value for *THIONX* is 10^{-2} .

The PVT properties of brine

The PVT properties of salt water solution are specified using the **PVTWSALT** and **BDENSITY** keywords in the PROPS section.

The **PVTWSALT** keyword replaces **PVTW** in brine models, and is used to specify the water properties at different values of the salt concentration C_s . The first record of each table contains two items of data: the reference pressure P_{ref} and the reference salt concentration C_{sref} . The reference salt concentration defines the formation volume factor $B_w(P_{ref}, C_{sref})$ for stock tank water. The reference salt concentration defaults to the minimum salt concentration in the table.

The **BDENSITY** keyword tabulates the brine surface density against the salt concentration (as specified in the **PVTWSALT** data). The **BDENSITY** keyword is optional. If the keyword is not used the brine surface density is taken as the water surface density (input using the **DENSITY** or **GRAVITY** keywords) for all salt concentrations.

The fluid properties of the salt water phase are computed from the information entered in the **PVTWSALT** keyword in different ways depending on whether Eclipse 100 or Eclipse 300 is used.

Eclipse 100

Eclipse 100 first calculates the values of formation volume factor B_w , compressibility C , viscosity μ_w and viscosibility C_v for the water phase as functions of the salt concentration at the previous timestep by linear interpolation of C and C_v and the **reciprocals** of B_w and $B_w\mu_w$. Then it calculates B_w and $B_w\mu_w$ at the current pressure P as follows:

$$B_w(P, C_s) = \frac{B_w(P_{ref}, C_s)}{1 + X + (X^2/2)} \quad \text{Eq. 4.9}$$

where $X = C(P - P_{ref})$

$$(B_w(P, C_s)\mu_w)(P, C_s) = \frac{B_w(P_{ref}, C_s)\mu_w(P_{ref}, C_s)}{1 + Y + (Y^2/2)} \quad \text{Eq. 4.10}$$

where $Y = (C - C_v)(P - P_{ref})$

Eclipse 300

Eclipse 300 also calculates B_w using equation 4.9. The water viscosity μ_w is then calculated similarly, using the expression:

$$\mu_w(P, C_s) = \frac{\mu_w(P_{ref}, C_s)}{1 + Y + (Y^2/2)} \quad \text{Eq. 4.11}$$

where $Y = -C_v(P - P_{ref})$

Low salinity option

This option provides a mechanism for modeling the salinity dependence of the oil and water relative permeabilities and the water-oil capillary pressure as functions of the salt concentration. The gas relative permeability and the gas-oil capillary pressure are assumed to be independent of salinity. This option is selected by specifying the **LOWSALT** keyword in the **RUNSPEC** section. The **SATNUM** and **LWSLTNUM** keywords are used to define the high and low salinity saturation regions respectively. (The **LSNUM** keyword may be used as an abbreviated alias for the **LWSLTNUM** keyword.)

The salinity dependence is modeled as a two stage process. Firstly, the oil and water saturation (horizontal) end-points are interpolated between the high and low table values. Secondly, this set of interpolated saturation end-points is used to look up the high and low salinity saturation functions in the high and low salinity saturation tables using the associated high and low relative permeability and capillary pressure (vertical) end-points. The resultant looked-up values of high and low salinity saturation functions are then combined using a further interpolation. Therefore, the saturation function table lookups are performed using scaled saturation end-points; that is, two-point, horizontal end-point scaling is applied even when end-point scaling has not been specified via the **ENDSCALE** keyword.

Given two sets of saturation functions, one for high salinity and one for low salinity, the oil and water table saturation end points are interpolated firstly according to:

$$\begin{aligned}
 S_{wco}^i &= F_1 S_{wco}^l + (1-F_1) S_{wco}^h \\
 S_{wcr}^i &= F_1 S_{wcr}^l + (1-F_1) S_{wcr}^h \\
 S_{wmax}^i &= F_1 S_{wmax}^l + (1-F_1) S_{wmax}^h \\
 S_{owcr}^i &= F_1 S_{owcr}^l + (1-F_1) S_{owcr}^h \\
 S_{ogcr}^i &= F_1 S_{ogcr}^l + (1-F_1) S_{ogcr}^h
 \end{aligned}
 \tag{Eq. 4.12}$$

where

F_1	is a weighting function which is tabulated as a function of the salt concentration and is defined in the second column of the LSALTFNC keyword.
S_{wco}	is the saturation table derived connate water saturation.
S_{wcr}	is the saturation table derived critical water saturation.
S_{wmax}	is the saturation table derived maximum water saturation.
S_{owcr}	is the saturation table derived critical oil saturation in water.
S_{ogcr}	is the saturation table derived critical oil saturation in gas.

superscripts h, l, i denote the high, low and interpolated table saturation end-points respectively.

Lower case subscripts and superscripts are used here to signify the saturation table derived end-points. Upper case versions are used subsequently to signify scaled end-points.

The high and low water-oil capillary pressures and water and oil relative permeabilities are looked up in the high and low salinity saturation tables by applying two-point saturation (horizontal) end-point scaling:

$$\begin{aligned}
P_{cow}^h &= f(S_w, S_{wco}^i, S_{wmax}^i, P_{cowmax}^h) \\
P_{cow}^l &= f(S_w, S_{wco}^i, S_{wmax}^i, P_{cowmax}^l) \\
k_{rw}^h &= f(S_w, S_{wcr}^i, S_{wmax}^i, k_{rwmax}^h) \\
k_{rw}^l &= f(S_w, S_{wcr}^i, S_{wmax}^i, k_{rwmax}^l) \\
k_{rog}^h &= f(S_{og}, S_{ogcr}^i, S_{omax}, k_{romax}^h) \\
k_{rog}^l &= f(S_{og}, S_{ogcr}^i, S_{omax}, k_{romax}^l) \\
k_{row}^h &= f(S_{ow}, S_{owcr}^i, S_{omax}, k_{romax}^h) \\
k_{row}^l &= f(S_{ow}, S_{owcr}^i, S_{omax}, k_{romax}^l)
\end{aligned}
\tag{Eq. 4.13}$$

where

f signifies the end-point scaling operation.

S_w is the grid block water saturation.

S_{ow} is the oil saturation for oil relative permeability to water.

S_{og} is the oil saturation for oil relative permeability to gas.

S_{omax} is the maximum oil saturation, $S_{omax} = (1 - S_{wco}^i - S_{gco}^h)$.

S_{gco} is the saturation table derived connate gas saturation.

k_{rwmax} is the saturation table derived maximum water relative permeability.

k_{romax} is the saturation table derived maximum oil relative permeability.

P_{cowmax} is the saturation table derived maximum oil-water capillary pressure.

For the default three-phase oil relative permeability model, the oil saturations used to look up the oil relative permeabilities are:

$$\begin{aligned}
S_{ow} &= (1 - S_w - S_g) \\
S_{og} &= (1 - S_w - S_g)
\end{aligned}
\tag{Eq. 4.14}$$

For the Stone 1 and Stone 2 three-phase oil relative permeability models, the oil saturations used to look up the oil relative permeabilities are:

$$\begin{aligned}
S_{ow} &= (1 - S_w) \\
S_{og} &= (1 - S_g - S_{wco}^i)
\end{aligned}
\tag{Eq. 4.15}$$

Finally, the relative permeabilities and capillary pressure are interpolated according to:

$$\begin{aligned}
k_{rw}^i &= F_1 k_{rw}^l + (1-F_1)k_{rw}^h \\
k_{row}^i &= F_1 k_{row}^l + (1-F_1)k_{row}^h \\
k_{rog}^i &= F_1 k_{rog}^l + (1-F_1)k_{rog}^h \\
P_{cow}^i &= F_2 P_{cow}^l + (1-F_2)P_{cow}^h
\end{aligned}
\tag{Eq. 4.16}$$

where

S_g is the grid block gas saturation.

F_2 is a weighting function which is tabulated as a function of the salt concentration and is defined in the third column of the `LSALTFNC` keyword. (Note that this defaults to the second column if the third column is defaulted.)

The gas relative permeability and gas-oil capillary pressure are assumed to be independent of salinity and are looked up directly in the high salinity gas saturation table. If the current grid block gas saturation exceeds the maximum gas saturation defined in the high salinity table, the values of gas relative permeability and gas-oil capillary pressure are assumed to remain constant at their corresponding maximum values.

Low salinity option with end-point scaling

End-point scaling (which is specified using the `ENDSCALE` keyword) may be applied to both the high and low salinity saturation functions.

For high salinity end-point scaling, the set of keywords `SWL`, `SWLPC`, `SWCR`, `SWU`, `SGL`, `SGLPC`, `SGCR`, `SGU`, `SOWCR`, `SOGCR`, `KRW`, `KRWR`, `KRG`, `KRGR`, `KRO`, `KRORW`, `KRORG`, `PCW` and `PCG` is available and applies to the tables defined by `SATNUM`. In addition, for the high salinity tables, the scaled end-points may also be specified as a function of depth using the keywords `ENPTVD`, `ENKRVD` and `ENPCVD`.

For the low salinity end-point scaling, the reduced set of keywords `LSWL`, `LSWLPC`, `LSWCR`, `LSWU`, `LSOWCR`, `LSOGCR`, `LKRW`, `LKRWR`, `LKRO`, `LKRORW`, `LKRORG` and `LPCW` is available and applies to the set of tables defined by `LWSLTNUM` (or `LSNUM`). The low salinity set corresponds to the high salinity set of keywords, except that the end-points associated with scaling the gas relative permeability and the gas-oil capillary pressure are omitted because these are assumed to be independent of salinity.

When end-point scaling has been specified, the salinity dependence is modeled in a similar manner to that described above except that the table derived end-points are replaced by the values from their corresponding end-point scaling arrays. (Where the end-point scaling arrays have been omitted from the input data, the end-point array values are defaulted to their corresponding table end-point values).

Given two sets of saturation functions, one for high salinity and one for low salinity, the scaled oil and water saturation end-points are interpolated firstly according to:

$$\begin{aligned}
S_{WL}^I &= F_1 S_{WL}^L + (1-F_1)S_{WL}^H \\
S_{WLPC}^I &= F_1 S_{WLPC}^L + (1-F_1)S_{WLPC}^H \\
S_{WCR}^I &= F_1 S_{WCR}^L + (1-F_1)S_{WCR}^H \\
S_{WU}^I &= F_1 S_{WU}^L + (1-F_1)S_{WU}^H \\
S_{OWCR}^I &= F_1 S_{OWCR}^L + (1-F_1)S_{OWCR}^H \\
S_{OGCR}^I &= F_1 S_{OGCR}^L + (1-F_1)S_{OGCR}^H
\end{aligned}
\tag{Eq. 4.17}$$

where

F_1	is a weighting function which is tabulated as a function of the salt concentration, and is defined in the second column of the <code>LSALTFNC</code> keyword.
S_{WL}	is the scaled minimum water saturation for relative permeability.
S_{WLPC}	is the scaled minimum water saturation for oil-water capillary pressure.
S_{WCR}	is the scaled critical water saturation.
S_{WU}	is the scaled maximum water saturation.
S_{OWCR}	is the scaled critical oil saturation in water.
S_{OGCR}	is the scaled critical oil saturation in gas.

superscripts H, L, I denote the high, low and interpolated scaled saturation end-points respectively.

Lower and upper case subscripts and superscripts are used here to signify the table derived and the scaled end-points respectively.

The capillary pressure and relative permeabilities are looked up in the high and low salinity saturation tables applying the set of scaled end-points shown below. For capillary pressure, only two-point scaling is employed:

$$\begin{aligned} P_{COW}^H &= f(S_W, S_{WLPC}^I, S_{WU}^I, P_{CW}^H) \\ P_{COW}^L &= f(S_W, S_{WLPC}^I, S_{WU}^I, P_{CW}^L) \end{aligned} \quad \text{Eq. 4.18}$$

For the relative permeabilities, two-point or three-point end-point scaling is employed according to whether item 1 of the `SCALECRS` keyword has been set to NO or YES respectively. (Two-point scaling is selected by default if this keyword is omitted.) For two-point scaling:

$$\begin{aligned} K_{RW}^H &= f(S_W, S_{WCR}^I, S_{owcr}^h, S_{WU}^I, K_{RW}^H, K_{RWR}^H) \\ K_{RW}^L &= f(S_W, S_{WCR}^I, S_{owcr}^l, S_{WU}^I, K_{RW}^L, K_{RWR}^L) \\ K_{ROG}^H &= f(S_{OG}, S_{OGCR}^I, S_{gcr}^h, S_{OMAX}, K_{RO}^H, K_{RORG}^H) \\ K_{ROG}^L &= f(S_{OG}, S_{OGCR}^I, S_{gcr}^h, S_{OMAX}, K_{RO}^L, K_{RORG}^L) \\ K_{ROW}^H &= f(S_{OW}, S_{OWCR}^I, S_{wcr}^h, S_{OMAX}, K_{RO}^H, K_{RORW}^H) \\ K_{ROW}^L &= f(S_{OW}, S_{OWCR}^I, S_{wcr}^l, S_{OMAX}, K_{RO}^L, K_{RORW}^L) \end{aligned} \quad \text{Eq. 4.19}$$

For three-point scaling, the scaled critical water saturation end-point is taken from the interpolated value or the respective high and low salinity values according to whether the default or the Stone 1 or 2 three-phase oil relative permeability models has been selected:

$$\begin{aligned}
K_{RW}^H &= f(S_W, S_{GL}^H, S_{WCR}^I, S_{OWCR}^I, S_{WU}^I, K_{RW}^H, K_{RWR}^H) \\
K_{RW}^L &= f(S_W, S_{GL}^H, S_{WCR}^I, S_{OWCR}^I, S_{WU}^I, K_{RW}^L, K_{RWR}^L) \\
K_{ROG}^H &= f(S_{OG}, S_{WL}^I, S_{OGCR}^I, S_{GCR}^H, S_{OMAX}, K_{RO}^H, K_{ROG}^H) \\
K_{ROG}^L &= f(S_{OG}, S_{WL}^I, S_{OGCR}^I, S_{GCR}^H, S_{OMAX}, K_{RO}^L, K_{ROG}^L)
\end{aligned}
\tag{Eq. 4.20}$$

Default three-phase oil relative permeability model:

$$\begin{aligned}
K_{ROW}^H &= f(S_{OW}, S_{GL}^H, S_{OWCR}^I, S_{WCR}^I, S_{OMAX}, K_{RO}^H, K_{ROW}^H) \\
K_{ROW}^L &= f(S_{OW}, S_{GL}^H, S_{OWCR}^I, S_{WCR}^I, S_{OMAX}, K_{RO}^L, K_{ROW}^L)
\end{aligned}
\tag{Eq. 4.21}$$

Stone 1 and 2 three-phase oil relative permeability models:

$$\begin{aligned}
K_{ROW}^H &= f(S_{OW}, S_{GL}^H, S_{OWCR}^I, S_{WCR}^H, S_{OMAX}, K_{RO}^H, K_{ROW}^H) \\
K_{ROW}^L &= f(S_{OW}, S_{GL}^H, S_{OWCR}^I, S_{WCR}^L, S_{OMAX}, K_{RO}^L, K_{ROW}^L)
\end{aligned}
\tag{Eq. 4.22}$$

where

f signifies the end-point scaling operation.

S_W is the grid block water saturation.

S_{OW} is the oil saturation for oil relative permeability to water.

S_{OG} is the oil saturation for oil relative permeability to gas.

S_{wcr} is the saturation table derived critical water saturation.

S_{owcr} is the saturation table derived critical oil in water saturation.

S_{WL} is the scaled minimum water saturation.

S_{OMAX} is the scaled maximum oil saturation, $S_{OMAX} = (1 - S_{WL}^I - S_{GL}^H)$.

S_{GL} is the scaled minimum gas saturation.

S_{GCR} is the scaled critical gas saturation.

S_{gcr} is the saturation table derived critical gas saturation.

K_{RW} is the scaled maximum water relative permeability.

K_{RWR} is the scaled water relative permeability at residual oil saturation.

K_{RO} is scaled maximum oil relative permeability.

K_{ROG} is the scaled oil relative permeability at residual gas saturation.

K_{ROW} is the scaled oil relative permeability at residual water saturation.

P_{CW} is the scaled maximum oil-water capillary pressure.

Noting that:

1. For the scaling of both the high and low salinity saturation tables, specifically, for the scaling of the relative permeabilities of oil to gas and water, only the high salinity minimum and critical scaled gas saturations are used.
2. For two-point scaling, the positioning of the relative permeability at residual saturation of the displacing phase is determined by the two-point scaled table value of the displacing critical phase saturation whereas, for three-point scaling, this is determined by the scaled end-point of the displacing phase critical saturation.
3. For the default three-phase oil relative permeability model, the critical water saturation used in the scaling of the high and low salinity oil relative permeabilities to water is based upon the interpolated value, whereas for the Stone 1 and 2 three-phase oil relative permeability models, the scaling is based upon the respective high and low salinity values.

For the default three-phase oil relative permeability model, the oil saturations used to look up the oil relative permeabilities are:

$$\begin{aligned} S_{OW} &= (1 - S_W - S_G) \\ S_{OG} &= (1 - S_W - S_G) \end{aligned} \quad \text{Eq. 4.23}$$

For the Stone 1 and Stone 2 three-phase oil relative permeability models, the oil saturations used to look up the oil relative permeabilities are:

$$\begin{aligned} S_{OW} &= (1 - S_W) \\ S_{OG} &= (1 - S_G - S_{WCO}^I) \end{aligned} \quad \text{Eq. 4.24}$$

Finally, the relative permeabilities and capillary pressure are interpolated according to:

$$\begin{aligned} K_{RW}^I &= F_1 K_{RW}^L + (1 - F_1) K_{RW}^H \\ K_{ROW}^I &= F_1 K_{ROW}^L + (1 - F_1) K_{ROW}^H \\ K_{ROG}^I &= F_1 K_{ROG}^L + (1 - F_1) K_{ROG}^H \\ P_{COW}^I &= F_2 P_{COW}^L + (1 - F_2) P_{COW}^H \end{aligned} \quad \text{Eq. 4.25}$$

where

F_1 is a weighting function which is tabulated as a function of the salt concentration, and is defined in the second column of the LSALTFNC keyword.

F_2 is a weighting function which is tabulated as a function of the salt concentration, and is defined in the third column of the LSALTFNC keyword. (Note that this defaults to the second column if not specified.)

The gas-oil capillary pressure and gas relative permeability are assumed to be independent of salinity. For capillary pressure two-point scaling is employed:

$$P_{COG} = f(S_G, S_{GLPC}^H, S_{GU}^H, P_{CG}^H) \quad \text{Eq. 4.26}$$

For gas relative permeability, depending upon the value of item 1 of the [SCALECRS](#) keyword, either two-point scaling:

$$K_{RG} = f(S_G, S_{GCR}^H, S_{ogcr}^h, S_{GU}^H, K_{RG}^H, K_{RGR}^H) \quad \text{Eq. 4.27}$$

or three-point scaling is employed:

$$K_{RG} = f(S_G, S_{WL}^H, S_{GCR}^H, S_{OGCR}^H, S_{GU}^H, K_{RG}^H, K_{RGR}^H) \quad \text{Eq. 4.28}$$

where

S_G is the grid block gas saturation.

S_{GLPC} is the scaled minimum gas saturation for capillary pressure.

S_{GU} is the scaled maximum gas saturation.

S_{ogcr} is the saturation table derived critical oil saturation in gas.

P_{CG} is the scaled maximum oil-gas capillary pressure.

K_{RG} is the scaled maximum gas relative permeability.

K_{RGR} is the scaled gas relative permeability at residual oil saturation.

Hence only the high salinity table and high salinity scaled end-points are used for modeling the gas saturation functions. If the current grid block gas saturation exceeds the maximum high salinity scaled gas saturation, the values of gas relative permeability and gas-oil capillary pressure are assumed to remain constant at their corresponding maximum values.

Low salinity and surfactant options

If, in addition to the [LWSALT](#) option, the [SURFACT](#) option has been specified, the high and low salinity immiscible saturation functions are combined in a similar manner to that described above and then combined with the miscible saturation functions. The miscible saturation functions are assumed to be independent of salinity. The [SURFNUM](#) keyword is used to define the miscible saturation regions and the [SATNUM](#) and [LWSLTNUM](#) (or [LSNUM](#)) keywords are used to define the immiscible high and low salinity saturation regions respectively. In this case, however, the initial interpolation of the saturation (horizontal) end-points and the subsequent interpolation of the vertical end-points following the table look-up involve a two stage process:

Given two sets of immiscible saturation functions, one for high salinity and one for low salinity, the immiscible oil and water table saturation end-points are interpolated firstly according to:

$$\begin{aligned} S_{wco}^{imm} &= F_1 S_{wco}^l + (1-F_1) S_{wco}^h \\ S_{wcr}^{imm} &= F_1 S_{wcr}^l + (1-F_1) S_{wcr}^h \\ S_{wmax}^{imm} &= F_1 S_{wmax}^l + (1-F_1) S_{wmax}^h \\ S_{owcr}^{imm} &= F_1 S_{owcr}^l + (1-F_1) S_{owcr}^h \\ S_{ogcr}^{imm} &= F_1 S_{ogcr}^l + (1-F_1) S_{ogcr}^h \end{aligned} \quad \text{Eq. 4.29}$$

and these are then combined with the miscible saturation end-points according to:

$$\begin{aligned}
S_{wco}^i &= F_3 S_{wco}^{mis} + (1-F_3) S_{wco}^{imm} \\
S_{wcr}^i &= F_3 S_{wcr}^{mis} + (1-F_3) S_{wcr}^{imm} \\
S_{wmax}^i &= F_3 S_{wmax}^{mis} + (1-F_3) S_{wmax}^{imm} \\
S_{owcr}^i &= F_3 S_{owcr}^{mis} + (1-F_3) S_{owcr}^{imm} \\
S_{ogcr}^i &= F_3 S_{ogcr}^{mis} + (1-F_3) S_{ogcr}^{imm}
\end{aligned} \tag{Eq. 4.30}$$

where

F_3 is tabulated as a function of the capillary number (expressed in terms of its logarithm base ten), and corresponds to the second column of the [SURFCAPD](#) keyword.

superscripts h, l denote the immiscible high and low table saturation end-points respectively.

superscripts imm, mis denote the immiscible interpolated and miscible table saturation end-points respectively.

superscript i denotes the interpolation between immiscible and miscible table saturation end-points.

The immiscible high and low salinity capillary pressures and relative permeabilities are looked up in the high and low salinity saturation tables by applying two-point saturation (horizontal) end-point scaling using the interpolated saturation end-points:

$$\begin{aligned}
P_{cow}^h &= f(S_w, S_{wco}^i, S_{wmax}^i, P_{cowmax}^h) \\
P_{cow}^l &= f(S_w, S_{wco}^i, S_{wmax}^i, P_{cowmax}^l) \\
k_{rw}^h &= f(S_w, S_{wcr}^i, S_{wmax}^i, k_{rwmax}^h) \\
k_{rw}^l &= f(S_w, S_{wcr}^i, S_{wmax}^i, k_{rwmax}^l) \\
k_{rog}^h &= f(S_{og}, S_{ogcr}^i, S_{omax}, k_{romax}^h) \\
k_{rog}^l &= f(S_{og}, S_{ogcr}^i, S_{omax}, k_{romax}^l) \\
k_{row}^h &= f(S_{ow}, S_{owcr}^i, S_{omax}, k_{romax}^h) \\
k_{row}^l &= f(S_{ow}, S_{owcr}^i, S_{omax}, k_{romax}^l)
\end{aligned} \tag{Eq. 4.31}$$

The miscible relative permeabilities are looked up in the miscible tables by applying two-point saturation (horizontal) end-point scaling using the interpolated saturation end-points:

$$\begin{aligned}
k_{rw}^{mis} &= f(S_w, S_{wcr}^i, S_{wmax}^i, k_{rwmax}^{mis}) \\
k_{rog}^{mis} &= f(S_{og}, S_{ogcr}^i, S_{omax}, k_{romax}^{mis}) \\
k_{row}^{mis} &= f(S_{ow}, S_{owcr}^i, S_{omax}, k_{romax}^{mis})
\end{aligned} \tag{Eq. 4.32}$$

The immiscible high and low salinity relative permeabilities and capillary pressure are interpolated according to:

$$\begin{aligned}
k_{rw}^{imm} &= F_1 k_{rw}^l + (1-F_1)k_{rw}^h \\
k_{row}^{imm} &= F_1 k_{row}^l + (1-F_1)k_{row}^h \\
k_{rog}^{imm} &= F_1 k_{rog}^l + (1-F_1)k_{rog}^h \\
P_{cow}^{imm} &= F_2 P_{cow}^l + (1-F_2)P_{cow}^h
\end{aligned}
\tag{Eq. 4.33}$$

The oil-water capillary pressure is taken directly from the interpolated immiscible term:

$$P_{cow}^i = P_{cow}^{imm} \tag{Eq. 4.34}$$

The miscible and immiscible water relative permeability and oil relative permeability to water are interpolated according to:

$$\begin{aligned}
k_{rw}^i &= F_3 k_{rw}^{mis} + (1-F_3)k_{rw}^{imm} \\
k_{row}^i &= F_3 k_{row}^{mis} + (1-F_3)k_{row}^{imm}
\end{aligned}
\tag{Eq. 4.35}$$

For the default three-phase oil relative permeability model, the oil relative permeability to gas is interpolated between the miscible and the immiscible values:

$$k_{rog}^i = F_3 k_{rog}^{mis} + (1-F_3)k_{rog}^{imm} \tag{Eq. 4.36}$$

whereas, for the Stone 1 and 2 three-phase oil relative permeability models, this is taken directly from the immiscible value:

$$k_{rog}^i = k_{rog}^{imm} \tag{Eq. 4.37}$$

The gas saturation functions are looked up in the high salinity immiscible table; that is, the gas relative permeability and gas-oil capillary pressure are assumed to exhibit no dependence upon salinity or surfactant.

When end-point scaling is specified via the [ENDSCALE](#) keyword, the interpolations follow a similar pattern to that described above in the absence of end-point scaling. Hence given two sets of immiscible saturation functions, one for low salinity and one for high salinity, the scaled saturation end points are interpolated firstly according to:

$$\begin{aligned}
S_{WL}^{IMM} &= F_1 S_{WL}^L + (1-F_1)S_{WL}^H \\
S_{WLPC}^{IMM} &= F_1 S_{WLPC}^L + (1-F_1)S_{WLPC}^H \\
S_{WCR}^{IMM} &= F_1 S_{WCR}^L + (1-F_1)S_{WCR}^H \\
S_{WU}^{IMM} &= F_1 S_{WU}^L + (1-F_1)S_{WU}^H \\
S_{OWCR}^{IMM} &= F_1 S_{OWCR}^L + (1-F_1)S_{OWCR}^H \\
S_{OGCR}^{IMM} &= F_1 S_{OGCR}^L + (1-F_1)S_{OGCR}^H
\end{aligned}
\tag{Eq. 4.38}$$

and these are then combined with the miscible scaled saturation end-points according to:

$$\begin{aligned}
S_{WL}^I &= F_3 S_{WL}^{MIS} + (1-F_3) S_{WL}^{IMM} \\
S_{WLPC}^I &= F_3 S_{WL}^{MIS} + (1-F_3) S_{WLPC}^{IMM} \\
S_{WCR}^I &= F_3 S_{WCR}^{MIS} + (1-F_3) S_{WCR}^{IMM} \\
S_{WU}^I &= F_3 S_{WU}^{MIS} + (1-F_3) S_{WU}^{IMM} \\
S_{OWCR}^I &= F_3 S_{OWCR}^{MIS} + (1-F_3) S_{OWCR}^{IMM} \\
S_{OGCR}^I &= F_3 S_{OGCR}^{MIS} + (1-F_3) S_{OGCR}^{IMM}
\end{aligned}
\tag{Eq. 4.39}$$

where

superscripts H, L denote the immiscible high and low scaled saturation end-points respectively.

superscripts IMM, MIS denote the immiscible interpolated and miscible scaled saturation end-points respectively.

superscript I denotes the interpolation between immiscible scaled and miscible saturation end-points.

The capillary pressures and relative permeabilities are looked up in the high and low salinity saturation tables applying the set of scaled end-points shown below. For capillary pressure, only two-point scaling is employed:

$$\begin{aligned}
P_{COW}^H &= f(S_W, S_{WLPC}^I, S_{WU}^I, P_{CW}^H) \\
P_{COW}^L &= f(S_W, S_{WLPC}^I, S_{WU}^I, P_{CW}^L)
\end{aligned}
\tag{Eq. 4.40}$$

For the immiscible relative permeabilities, either two-point or three-point end-point scaling is employed according to whether item 1 of the [SCALECRS](#) keyword has been set to NO or YES respectively. For two-point scaling the immiscible saturation functions are looked up in the high and low salinity saturation tables applying the set of scaled end-points shown below:

$$\begin{aligned}
K_{RW}^H &= f(S_W, S_{WCR}^I, S_{owcr}^h, S_{WU}^I, K_{RW}^H, K_{RWR}^H) \\
K_{RW}^L &= f(S_W, S_{WCR}^I, S_{owcr}^l, S_{WU}^I, K_{RW}^L, K_{RWR}^L) \\
K_{ROG}^H &= f(S_{OG}, S_{OGCR}^I, S_{gcr}^h, S_{OMAX}, K_{RO}^H, K_{RORG}^H) \\
K_{ROG}^L &= f(S_{OG}, S_{OGCR}^I, S_{gcr}^h, S_{OMAX}, K_{RO}^L, K_{RORG}^L) \\
K_{ROW}^H &= f(S_{OW}, S_{OWCR}^I, S_{wcr}^h, S_{OMAX}, K_{RO}^H, K_{RORW}^H) \\
K_{ROW}^L &= f(S_{OW}, S_{OWCR}^I, S_{wcr}^l, S_{OMAX}, K_{RO}^L, K_{RORW}^L)
\end{aligned}
\tag{Eq. 4.41}$$

For three-point scaling the immiscible water relative permeability and the oil relative permeability to gas are looked up in the high and low salinity saturation tables applying the set of scaled end-points shown below:

$$\begin{aligned}
K_{RW}^H &= f(S_W, S_{GL}^H, S_{WCR}^I, S_{OWCR}^I, S_{WU}^I, K_{RW}^H, K_{RWR}^H) \\
K_{RW}^L &= f(S_W, S_{GL}^H, S_{WCR}^I, S_{OWCR}^I, S_{WU}^I, K_{RW}^L, K_{RWR}^L) \\
K_{ROG}^H &= f(S_{OG}, S_{WL}^I, S_{OGCR}^I, S_{GCR}^H, S_{OMAX}, K_{RO}^H, K_{RORG}^H) \\
K_{ROG}^L &= f(S_{OG}, S_{WL}^I, S_{OGCR}^I, S_{GCR}^H, S_{OMAX}, K_{RO}^L, K_{RORW}^L)
\end{aligned}
\tag{Eq. 4.42}$$

For the immiscible oil relative permeability to water and for three-point scaling, the critical water saturation end-point is taken from the interpolated value for the default three-phase oil relative permeability model:

$$\begin{aligned}
K_{ROW}^H &= f(S_{OW}, S_{GL}^H, S_{OWCR}^I, S_{WCR}^I, S_{OMAX}, K_{RO}^H, K_{RORW}^H) \\
K_{ROW}^L &= f(S_{OW}, S_{GL}^H, S_{OWCR}^I, S_{WCR}^I, S_{OMAX}, K_{RO}^L, K_{RORW}^L)
\end{aligned}
\tag{Eq. 4.43}$$

but from the respective high and low salinity values for the Stone 1 and 2 three-phase oil relative permeability models:

$$\begin{aligned}
K_{ROW}^H &= f(S_{OW}, S_{GL}^H, S_{OWCR}^I, S_{WCR}^H, S_{OMAX}, K_{RO}^H, K_{RORW}^H) \\
K_{ROW}^L &= f(S_{OW}, S_{GL}^H, S_{OWCR}^I, S_{WCR}^L, S_{OMAX}, K_{RO}^L, K_{RORW}^L)
\end{aligned}
\tag{Eq. 4.44}$$

For the miscible relative permeabilities, either two-point or three-point end-point scaling is employed according to whether item 1 of the [SCALECRS](#) keyword has been set to NO or YES respectively. For two-point scaling the miscible saturation functions are looked up in the miscible saturation tables applying the set of scaled end-points shown below:

$$\begin{aligned}
K_{RW}^{MIS} &= f(S_W, S_{WCR}^I, S_{owcr}^{mis}, S_{WU}^I, K_{RW}^{MIS}, K_{RWR}^{MIS}) \\
K_{ROG}^{MIS} &= f(S_{OG}, S_{OGCR}^I, S_{gr}^{mis}, S_{OMAX}, K_{RO}^{MIS}, K_{RORG}^{MIS}) \\
K_{ROW}^{MIS} &= f(S_{OW}, S_{OWCR}^I, S_{wcr}^{mis}, S_{OMAX}, K_{RO}^{MIS}, K_{RORW}^{MIS})
\end{aligned}
\tag{Eq. 4.45}$$

For three-point scaling the miscible water relative permeability and the oil relative permeability to gas are looked up in the miscible tables applying the set of scaled end-points shown below:

$$\begin{aligned}
K_{RW}^{MIS} &= f(S_W, S_{GL}^{MIS}, S_{WCR}^I, S_{OWCR}^{MIS}, S_{WU}^I, K_{RW}^{MIS}, K_{RWR}^{MIS}) \\
K_{ROG}^{MIS} &= f(S_{OG}, S_{WL}^I, S_{OGCR}^I, S_{GCR}^{MIS}, S_{OMAX}, K_{RO}^{MIS}, K_{RORG}^{MIS})
\end{aligned}
\tag{Eq. 4.46}$$

For the miscible oil relative permeability to water and for three-point scaling, the critical water saturation end-point is taken from the interpolated value for the default three-phase oil relative permeability model:

$$K_{ROW}^{MIS} = f(S_{OW}, S_{GL}^{MIS}, S_{OWCR}^I, S_{WCR}^I, S_{OMAX}, K_{RO}^{MIS}, K_{RORW}^{MIS})
\tag{Eq. 4.47}$$

but from the miscible value for the Stone 1 and 2 three-phase oil relative permeability models:

$$K_{ROW}^{MIS} = f(S_{OW}, S_{GL}^{MIS}, S_{OWCR}^I, S_{WCR}^{MIS}, S_{OMAX}, K_{RO}^{MIS}, K_{RORW}^{MIS})
\tag{Eq. 4.48}$$

The immiscible high and low salinity relative permeabilities and capillary pressure are interpolated according to:

$$\begin{aligned}
K_{RW}^{IMM} &= F_1 K_{RW}^L + (1-F_1)K_{RW}^H \\
K_{ROW}^{IMM} &= F_1 K_{ROW}^L + (1-F_1)K_{ROW}^H \\
K_{ROG}^{IMM} &= F_1 K_{ROG}^L + (1-F_1)K_{ROG}^H \\
P_{COW}^{IMM} &= F_2 P_{COW}^L + (1-F_2)P_{COW}^H
\end{aligned}
\tag{Eq. 4.49}$$

The oil-water capillary pressure is taken directly from the interpolated immiscible term:

$$P_{COW}^I = P_{COW}^{IMM} \tag{Eq. 4.50}$$

The miscible and immiscible water relative permeability and oil relative permeability to water are interpolated according to:

$$\begin{aligned}
K_{RW}^I &= F_3 K_{RW}^{MIS} + (1-F_3)K_{RW}^{IMM} \\
K_{ROW}^I &= F_3 K_{ROW}^{MIS} + (1-F_3)K_{ROW}^{IMM}
\end{aligned}
\tag{Eq. 4.51}$$

For the default three-phase oil relative permeability model, the oil relative permeability to gas is interpolated between the miscible and the immiscible values:

$$K_{ROG}^I = F_3 K_{ROG}^{MIS} + (1-F_3)K_{ROG}^{IMM} \tag{Eq. 4.52}$$

whereas, for the Stone 1 and 2 three-phase oil relative permeability models, this is taken directly from the immiscible value:

$$K_{ROG}^I = K_{ROG}^{IMM} \tag{Eq. 4.53}$$

The gas-oil capillary pressure and gas relative permeability are assumed to be independent of salinity. For capillary pressure two-point scaling is employed:

$$P_{COG} = f(S_G, S_{GLPC}^H, S_{GU}^H, P_{CG}^H) \tag{Eq. 4.54}$$

For gas relative permeability, depending upon the value of item 1 of the [SCALECRS](#) keyword, either two-point scaling:

$$K_{RG} = f(S_G, S_{GCR}^H, S_{ogcr}^h, S_{GU}^H, K_{RG}^H, K_{RGR}^H) \tag{Eq. 4.55}$$

or three-point scaling is employed:

$$K_{RG} = f(S_G, S_{GCR}^H, S_{OGCR}^H, S_{GU}^H, K_{RG}^H, K_{RGR}^H) \tag{Eq. 4.56}$$

Hence only the high salinity table and high salinity scaled end-points are used for modeling the gas saturation functions.

Low salinity option and surfactant option with wettability dependence

If, in addition to the [LOWSALT](#) option, the [SURFACTW](#) option has been specified, two sets of high and low salinity functions are modeled, one corresponding to oil-wet and the other to water-wet. These are each processed and combined to form a composite immiscible set which is then combined with the miscible saturation functions. The miscible saturation functions are assumed to be independent of salinity. The [SURFNUM](#) keyword is used to define the miscible saturation regions, the [SATNUM](#) and [LWSLTNUM](#) (or [LSNUM](#)) keywords are used to define the immiscible high and low salinity oil-wet saturation regions respectively and the [SUREFNUM](#) (or [HWSNUM](#)) and [LSLTWNUM](#) (or [LWSNUM](#)) keywords are used to define the immiscible high and low salinity oil-wet saturation regions respectively.

The oil-wet and water-wet high and low salinity saturation functions are each combined in the manner described above for the `LOWSALT` plus `SURFACT` options. However, the interpolation process applied to the end-points involves an additional interpolation according to the adsorbed surfactant concentration. Using the saturation table derived critical water saturation as an example, the horizontal interpolation process prior to table lookup of the saturation functions consists of:

$$S_{wcr}^{imm-ow} = F_1 S_{wcr}^{ow-l} + (1-F_1) S_{wcr}^{ow-h} \quad \text{Eq. 4.57}$$

$$S_{wcr}^{imm-ww} = F_1 S_{wcr}^{ww-l} + (1-F_1) S_{wcr}^{ww-h}$$

$$S_{wcr}^{imm} = F_4 S_{wcr}^{imm-ow} + (1-F_4) S_{wcr}^{imm-ww}$$

$$S_{wcr}^i = F_3 S_{wcr}^{mis} + (1-F_3) S_{wcr}^{imm}$$

where

F_4 is tabulated as a function of the adsorbed surfactant concentration and corresponds to the second column of the `SURFADDW` keyword.

superscripts $ow-h$, $ow-l$ denote the oil-wet immiscible high and low table saturation end-points respectively.

superscripts $ww-h$, $ww-l$ denote the water-wet immiscible high and low table saturation end-points respectively.

superscripts $imm-ow$, $imm-ww$ denote the immiscible oil-wet and water-wet interpolated table saturation end-points respectively.

superscript i denotes the interpolation between immiscible and miscible table saturation end-points.

Following table look-up, the relative permeabilities and capillary pressures are combined in a similar manner to that described above. Hence using water relative permeability as an example, the vertical interpolation process following table lookup of the saturations functions consists of:

$$k_{rw}^{imm-ow} = F_1 k_{rw}^{ow-l} + (1-F_1) k_{rw}^{ow-h} \quad \text{Eq. 4.58}$$

$$k_{rw}^{imm-ww} = F_1 k_{rw}^{ww-l} + (1-F_1) k_{rw}^{ww-h}$$

$$k_{rw}^{imm} = F_4 k_{rw}^{imm-ow} + (1-F_4) k_{rw}^{imm-ww}$$

$$k_{rw}^i = F_3 k_{rw}^{mis} + (1-F_3) k_{rw}^{imm}$$

Hence apart from these additional interpolation steps, the interpolation process is similar to that for when only one pair of immiscible high and low salinity saturation functions are being modeled. Similar treatments apply for the oil-water capillary pressure and for the oil relative permeabilities, noting that for the Stone 1 and 2 three-phase oil relative permeability model, only the oil-wet immiscible oil relative permeability gas is employed. If end-point scaling is specified, the modeling is similar to that described for the low salinity option and surfactant option without wettability dependence (`LOWSALT` and `SURFACT`) except that two pairs of high and low salinity scaled end-points are used.

See "[Surfactant model](#)" for additional information.

Miscible and immiscible low and high salinity saturation regions and end-points

The following combinations of saturation regions are available:

1. If only the **LOWSALT** option is specified, the **SATNUM** and **LWSLTNUM** keywords are used to define the high and low salinity saturation function table numbers respectively.
2. If the **LOWSALT** and **SURFACT** options are specified, the **SATNUM** and **LWSLTNUM** keywords are used to define the high and low salinity saturation function table numbers respectively and the **SURFNUM** keyword is used to define the miscible saturation function table numbers.
3. If the **LOWSALT** and **SURFACTW** options are specified, the **SATNUM** and **LWSLTNUM** keywords are used to define the oil-wet high and low salinity saturation function table numbers respectively and the **SURFNUM** and **LSLTWNUM** keywords are used to define the water-wet high and low salinity saturation function table numbers respectively and the **SURFNUM** keyword is used to define the miscible saturation function table numbers.
4. If the only the **SURFACT** option is specified, the **SATNUM** and **SURFNUM** keywords are used to define the immiscible and miscible saturation function table numbers respectively. This does not involve the low salinity option but is included here for completeness. See "[Surfactant model](#)" for further information.
5. If the only the **SURFACTW** option is specified, the **SATNUM** and **SURFNUM** keywords are used to define the immiscible oil-wet and water-wet saturation function table numbers respectively and the **SURFNUM** keyword is used to define the miscible saturation function table numbers. This does not involve the low salinity option but is included here for completeness. See "[Surfactant model](#)" for further information.
6. The **LSNUM**, **LWSNUM** and **HWSNUM** keywords may be used as abbreviated aliases for the **LWSLTNUM**, **LSLTWNUM** and **SURFNUM** keywords respectively.

This is summarized below:

Options	Saturation Regions
LOWSALT	SATNUM and LWSLTNUM (or LSNUM)
LOWSALT and SUFRAC	SURFNUM, SATNUM and LWSLTNUM (or LSNUM)
LOWSALT and SUFRACW	SURFNUM, SATNUM and LWSLTNUM (or LSNUM) and SURFNUM (or HWSNUM) and LSLTWNUM (or LWSNUM)
SURFACT	SURFNUM and SATNUM
SURFACTW	SURFNUM, SATNUM and SURFNUM (or HWSNUM)

For the saturation regions, the following sets of end-point scaling arrays are available:

SATNUM	LWSLTNUM (or LSNUM)	SURFNUM (or HWSNUM)	LSLTWNUM (or LWSNUM)	SURFNUM
Immiscible High Salinity Oil-Wet	Immiscible Low Salinity Oil-Wet	Immiscible High Salinity Water-Wet	Immiscible Low Salinity Water-Wet	Miscible
SWL	LSWL	HWSWL	LWSWL	SSWL
SWLPC	LSWLPC	HWSWLPC	LWSWLPC	
SWCR	LSWCR	HWSWCR	LWSWCR	SSWCR

SATNUM	LWSLTNUM (or LSNUM)	SURFNUM (or HWSNUM)	LSLTWNUM (or LWSNUM)	SURFNUM
Immiscible High Salinity Oil-Wet	Immiscible Low Salinity Oil-Wet	Immiscible High Salinity Water-Wet	Immiscible Low Salinity Water-Wet	Miscible
SWU	LSWU	HWSWU	LWSWU	SSWU
SGL				SSGL
SGLPC				
SGCR				SSGCR
SSGU				
SOWCR	LSOWCR	HWSOWCR	LWSOWCR	SSOWCR
SOGCR	LSOGCR	HWSOGCR	LWSOGCR	SSOGCR
KRW	LKRW	HWKRW	LWKRW	SKRW
KRWR	LKRWR	HWKRWR	LWKRWR	SKRWR
KRG				
KRGR				
KRO	LKRO	HWKRO	LWKRO	SKRO
KRORW	LKRORW	HWKRORW	LWKRORW	SKRORW
KRORG	LKRORG	HWKRORG	LWKRORG	SKRORG
PCW	LPCW	HWPCW	LWPCW	
PCG				

Hysteresis

The low salinity option does not currently include relative permeability hysteretic effects in the water and oil phases where the dependence on salinity is being modeled; however, if hysteresis has been specified for the gas phase, this will be modeled according to the value of item 2 in the [EHYSTR](#) keyword.

Output of scaled end-point tables

If output of the scaled saturation functions to the DBG file is requested via the [EPSDEBUG](#) or [EPSDBGS](#) keywords, these will be reported for all the saturation tables involved in the high and low salinity calculations. However, with the exception of the high salinity set associated with the [SATNUM](#) keyword, the scaled gas saturation functions will be omitted because they are not used in these salinity and surfactant dependent calculations.

Two-phase water-oil models

Two phase water-oil models are modeled in a similar manner to that described above for three-phase models except that only the two-phase water relative permeability, two phase oil relative permeability to water and the water-oil capillary pressure are employed.

Version 2014.1 extensions

Prior to the 2014.1 version end-point scaling was only available for the high salinity saturation functions and both the oil-wet and water-wet sets employed the same set of end-points. For the 2014.1 version, the following changes have been made:

1. Only the high salinity (oil-wet) saturation tables are scaled using the set of end-point scaling keywords consisting of: [SWL](#), [SWLPC](#), [SWCR](#), [SWU](#), [SGL](#), [SGLPC](#), [SGCR](#), [SGU](#), [SOWCR](#), [SOGCR](#), [KRW](#), [KRWR](#), [KRG](#), [KRGR](#), [KRO](#), [KRORW](#), [KRORG](#), [PCW](#) and [PCG](#).
2. The low salinity (oil-wet) saturation tables are scaled using the set of end-point scaling keywords consisting of [LSWL](#), [LSWLPC](#), [LSWCR](#), [LSWU](#), [LSOWCR](#), [LSOGCR](#), [LKRW](#), [LKRWR](#), [LKRO](#), [LKRORW](#), [LKRORG](#) and [LPCW](#).
3. The high salinity (water-wet) saturation tables are scaled using the set of end-point scaling keywords consisting of [HWSWL](#), [HWSWLPC](#), [HWSWCR](#), [HWSWU](#), [HWSOWCR](#), [HWSOGCR](#), [HWKRW](#), [HWKRWR](#), [HWKRO](#), [HWKRORW](#), [HWKRORG](#) and [HWPCW](#).
4. The low salinity (water-wet) saturation tables are scaled using the set of end-point scaling keywords consisting of [LWSWL](#), [LWSWLPC](#), [LWSWCR](#), [LWSWU](#), [LWSOWCR](#), [LWSOGCR](#), [LWKRW](#), [LWKRWR](#), [LWKRO](#), [LWKRORW](#), [LWKRORG](#) and [LWPCW](#).
5. The [LSNUM](#), [LWSNUM](#) and [HWSNUM](#) keywords have been introduced as abbreviated aliases for the [LWSLTNUM](#), [LSLTWNUM](#) and [SURFNUM](#) keywords respectively.

In addition, for the 2014.1 version, changes of behavior may arise when compared to previous versions because of the following differences between this version and prior versions:

1. Prior to the 2014.1 version, the high and low scaling involving the gas saturation end-points utilizes the corresponding high and low gas saturation values. For the 2014.1 version, the scaling of the high and low salinity saturation functions involving the gas saturation end-points (both with and without end-point scaling being specified via the [ENDSCALE](#) keyword) utilizes only the high salinity values. This may lead to differences between the 2014.1 and prior versions in cases where the high and low minimum or critical gas saturation values are different, in particular, when three-point scaling has been specified.
2. Prior to the 2014.1 version, in some cases when three-point scaling has been specified, the low salinity scaling utilizes the minimum rather than the critical values of water saturation. This has been rectified for the 2014.1 version.
3. Prior to the 2014.1 version, if end-point scaling has been specified, three-point end-point scaling is disabled for the two-phase water-oil models whilst looking up the water and oil saturation functions

The version prior to 2014.1 may be selected by setting [item 225](#) of the [OPTIONS](#) keyword to 1.

Version 2015.1 extensions

For the 2015.1 version end-point scaling has been extended to the miscible saturation tables by adding the following end-point scaling keywords:

[SSWL](#), [SSWCR](#), [SSWU](#), [SSGL](#), [SSGCR](#), [SSOWCR](#), [SSOGCR](#), [SKRW](#), [SKRWR](#), [SKRO](#), [SKRORW](#) and [SKRORG](#).

Setting the initial conditions

The initial salt distribution throughout the reservoir is specified in the [SOLUTION](#) section. If the initial conditions are determined by **equilibration**, the salt concentration can vary with depth independently in each equilibration region. The [SALTVD](#) keyword supplies a table of salt concentrations versus depth for each equilibration region.

When setting the initial conditions by **enumeration** (keywords [PRESSURE](#) and [SWAT](#) for example), the [SALT](#) keyword is used to supply the initial salt concentration for each grid block.

Multi-component brine

Eclipse 100

When the multi-component brine model is activated the initial mass concentration of each salt specified in the [BRINE](#) keyword in the RUNSPEC section must be entered. For initialization by equilibration the [SALTVD](#) keyword supplies a table of salt mass concentrations versus depth with one column of data for each salt. For initialization by enumeration the [SALT](#) keyword supplies the initial salt concentration of each salt for every grid block.

For both initialization options the salt mass concentrations are converted into equivalent molar concentrations of anions, monovalent cations, and divalent cations, which are then tracked during simulation.

Wells, rivers and aquifers

The salinity of injected water can be specified independently for each water injector using the [WSALT](#) keyword (the argument(s) can be used as a user-defined argument). The default salt concentration is zero.

Eclipse 100

Brine economic limits can be specified for production connections, wells and groups by using the keywords [CECONT](#), [WECONT](#) and [GECONT](#) respectively. It is also possible to set brine cutback limits for wells and groups by using the keywords [WCUTBACT](#) and [GCUTBACT](#) respectively.

Note: The name 'BRI' should be used as the name of the tracer for these keywords when setting production and cutback limits for brine.

The salinity of water can be specified for individual river branches and reaches by using the [RIVSALT](#) keyword. The default salt concentration is zero.

The initial salinity of any analytic aquifers can be set in the keyword used to specify the aquifer ([AQUFET](#), [AQUFETP](#), [AQUCT](#), [AQUFLUX](#) or [AQUCHWAT](#)). The default salt concentration for all analytic aquifers is zero.

Tracer related options

Eclipse 100

As brine is modeled as a tracer it is possible to use some of the tracer related options for brine. It is possible to use a form of velocity and concentration dependent dispersion using the keywords [TRDIS](#), [DISPERSE](#) and [DIAGDISP](#). The tracer dispersion option is activated by specifying [DISPDIMS](#) in the RUNSPEC section.

For a brine tracer it is possible to replace the tracer concentration dispersion dependence by water density dispersion dependence by using the keyword [DSPDEINT](#).

The polymer flood option

Eclipse 100

The polymer flood model (see "[Polymer flood model](#)"), activated using the option [POLYMER](#) in the RUNSPEC section, allows you to investigate the effect of varying brine concentrations on the efficiency of the polymer flood. The study of the interaction of brine with the polymer solution is, however, optional within the model.

For models which have the polymer and brine options activated, the viscosity of a fully mixed polymer solution needs to be entered as a function of the salt concentration. The keywords [SALTNODE](#) and [PLYVISC](#) should be used to enter two-dimensional tables for the viscosity of the solution. The maximum polymer and salt concentrations to be used in calculating the effective fluid component viscosities are entered using the keyword [PLYMAX](#).

The polymer adsorption can be set to depend on the brine concentration and on the rock permeability using the [ADSORP](#) keyword with [POLYMER](#) as an argument. Polymer adsorption can also be set to depend on brine concentration using the keywords [PLYADSS](#) and [ADSALNOD](#).

The concentration of polymer and salt for the injection wells can be specified using the [WPOLYMER](#) keyword. The default polymer and salt concentrations are zero.

Multi-component brine and polymer

Eclipse 100

The multi-component brine model ([ECLMC](#) keyword in the [RUNSPEC](#) section) cannot currently be used with the polymer/brine flood model (that is, the combination of polymer and brine using the [SALTNODE](#) keyword; [ECLMC](#) can be used with the basic polymer flood model).

The surfactant flood option

Eclipse 100

The surfactant model (see "[Surfactant model](#)"), activated using the option [SURFACT](#) in the [RUNSPEC](#) section, allows you to investigate the effect of varying brine concentrations on the efficiency of the surfactant flood.

The study of the interaction of brine with the surfactant solution is, however, optional within the model. For models which have the surfactant and brine options activated, the oil-water surface tension of a fully mixed surfactant solution can be entered as a function of the salt concentration. For this, the keywords [ESSNODE](#) and [SURFSTES](#) can be used to enter two-dimensional tables for the oil-water surface tension of the solution as a function of surfactant and salt concentrations. The surfactant adsorption can be set to depend on the brine concentration and on the rock permeability using the [ADSORP](#) keyword with [SURFACT](#) as an argument.

Multi-component brine and surfactant

Eclipse 100

The multi-component brine model can be used to make the surfactant adsorption dependent on the salinity. Based on ion transport and exchanges an **effective salinity** is computed, and then used in the surfactant adsorption isotherm. The effective salinity is an empirical quantity incorporating the association between the surfactant and divalent cations, and calculated according to

$$C_{SE} = \frac{C_-}{1 - \alpha^s X^s}, \quad X^s = \frac{C_{++}^s}{C^s} \quad \text{Eq. 4.59}$$

where

C_{SE} denotes the surfactant effective salinity

X^s denotes the fraction of surfactant sites occupied by divalent cations

α^s denotes a user-input effective salinity parameter ($\alpha^s \in [0, 1]$) ([SURFESAL](#))

The effective salinity calculation is activated using keyword [SURFESAL](#) in the [PROPS](#) section, which also specifies the effective salinity parameter, α^s , in equation 4.59. This salinity is then used in the surfactant adsorption isotherms specified in the [ADSORP](#) keyword with [SURFACT](#) as an argument. Furthermore, this salinity as well as the surfactant concentration can be used to update the water-oil surface tension if such option is requested (using keywords [SURFSTES](#) and [ESSNODE](#)).

Solving the conservation equations

For the multi-component brine model the ion components are solved as active tracers. The ion concentrations determine the effective salinities, which in turn have a strong impact on surfactant-water-oil phase behavior and surfactant/polymer adsorption, and hence affect the phase flows. By default tracers are solved only at the end of a timestep once the equations for phases have converged. In this way the salinity is treated explicitly in the calculations which may lead to numerical instability if the timesteps are large and the dependence on salinity is strong. For this reason active tracers can now be solved in each Newton iteration so that the tracer concentrations are updated simultaneously with the phase equations.

When the [ECLMC](#) keyword is present in the RUNSPEC section the multi-component brine equations are solved in each Newton iteration. The standard behavior of solving the brine equations at the end of a timestep can be restored by setting [item 126](#) of the [OPTIONS](#) keyword to 2. When solved in each Newton iteration the brine concentration used when calculating effective salinities is computed as a weighted average of the concentration at time t and the concentration at time $t + \Delta t$ corresponding to the current iteration:

$$C = (1-\theta)C(t) + \theta C(t + \Delta t) \quad \text{Eq. 4.60}$$

where

θ is a weighting factor.

$\theta = 0$ corresponds to the standard, explicit treatment of brine concentration, and $\theta = 1$ corresponds to a fully coupled approach. The value of θ can be set in item 13 in the second record of the [TUNING](#) keyword (Default value: 1).

Summary of keywords

RUNSPEC

Restriction	Keyword	Description
	BRINE	activates the brine tracking option
Eclipse 100	LWSALT	activates the low salinity option
Eclipse 100	ECLMC	activates multi-component brine

GRID

Eclipse 100 keyword	Description
IONROCK	specify the rock ion exchange capacity
RPTGRID	controls output from the GRID section to the print file: Use argument IONROCK to output rock ion exchange capacity.

PROPS

Restriction	Keyword	Description
	BDENSITY	specify the brine density

Restriction	Keyword	Description
Eclipse 100	LSALTFNC	specify the low salinity fraction as function of the salt concentration in the grid block.
	PVTWSALT	specify water PVT functions with salt present
	RPTPROPS	controls output from the PROPS section to the print file: Use arguments BDENSITY and PVTWSALT to output brine properties to the print file.
Eclipse 100	IONXROCK	activate cation exchange on rock surfaces and specify cation exchange constant for rock.
Eclipse 100	IONXSURF	activate cation exchange on surfactant micelles and specify cation exchange constant for surfactant and equivalent molecular weight of surfactant.
Eclipse 100	SURFESAL	specify parameter for effective salinity calculation with respect to surfactant phase behavior.
Eclipse 100	PLYESAL	specify parameter for effective salinity calculation with respect to polymer properties.
Eclipse 100	DSPDEINT	interpolation of dispersion tables by water density.
Eclipse 100	SWL, SWLPC, SWCR, SWU, SGL, SGLPC, SGCR, SGU, SOWCR, SOGCR, KRW, KRWR, KRG, KRGR, KRO, KRORW, KRORG, PCW and PCG	Immiscible (oil-wet) high salinity scaled end-points.
Eclipse 100	LSWL, LSWLPC, LSWCR, LSWU, LSOWCR, LSOGCR, LKRW, LKRWR, LKRO, LKRORW, LKRORG and LPCW	Immiscible (oil-wet) low salinity scaled end-points.
Eclipse 100	HWSWL, HWSWLPC, HWSWCR, HWSWU, HWSOWCR, HWSOGCR, HWKRW, HWKRWR, HWKRO, HWKRORW, HWKRORG and HWPCW	Immiscible water-wet (high salinity) scaled end-points.
Eclipse 100	LWSWL, LWSWLPC, LWSWCR, LWSWU, LWSOWCR, LWSOGCR, LWKRW, LWKRWR, LWKRO, LWKRORW, LWKRORG and LWPCW	Immiscible water wet low salinity scaled end-points.
Eclipse 100	SSWL, SSWCR, SSWU, SSGL, SSGCR, SSOWCR, SSOGCR, SKRW, SKRWR, SKRO, SKRORW and SKRORG	Miscible scaled end-points.

REGIONS

Restriction	Keyword	Description
	SATNUM	specify the high salinity table number for each block. For the Surfactant option, this corresponds to the immiscible high salinity oil-wet tables.
Eclipse 100	LWSLTNUM (or LSNUM)	specify the low salinity table number for each block. For the Surfactant option, this corresponds to the immiscible low salinity oil-wet tables.
Eclipse 100	SURFNUM (or HSWNUM)	For the Surfactant option, this corresponds to the immiscible high salinity water-wet tables.
Eclipse 100	LSLTWNUM (or LWSNUM)	For the Surfactant option, this corresponds to the immiscible low salinity water-wet tables.

SOLUTION

Restriction	Keyword	Description
Eclipse 100	RPTSOL	Controls output from the SOLUTION section to the print file: Use SALT to output initial grid block salt concentration values. Use SALTVD to output tables of initial salt concentration. For the Multi-Component Brine model use SALTVD to output tables of initial anion and divalent cation concentrations.
	SALT	specify initial salt concentration(s)
	SALTVD	specify salt concentration(s) versus depth for equilibration

SUMMARY

Refer to [Brine option quantities](#) in the *Eclipse Reference Manual*.

SCHEDULE

Description	Keyword	Description
Eclipse 100	RIVSALT	Sets the salt concentrations in the upstream flow of river branches.
	RPTSCHED	Controls output from the SCHEDULE section to the print file: Use SALT to output grid block salt concentration values. Eclipse 100: FIPSALT outputs salt fluid in place reports Eclipse 100: ANIONS and CATIONS output grid block anion and divalent cations concentrations for the Multi-Component Brine model. Eclipse 100: ESALSUR and ESALPLY output grid block effective salinities for surfactant and polymer, respectively, in the Multi-Component Brine model.
	RPTRST	VISC allows to output extra viscosity when or Brine is active: EWV_SAL: effective water viscosity due to salt concentration

Description	Keyword	Description
	WSALT	Set salt concentration for injection wells (the argument can be set as a user-defined argument)
Eclipse 100	TUNING	Item 12 is the second record sets THIONX, the relative threshold for damping in the ion exchange calculation

Examples

This example shows only sections where data related to the brine option are mentioned.

RUNSPEC

```

RUNSPEC
TITLE
Example Brine Model
-- Phases present
OIL
WATER
BRINE
FIELD
DIMENS
15 1 10 /
START
1 'JAN' 1983 /
-- Well dimensions
WELLDIMS
2 10 1 2 /
-- Aquifer dimensions
AQUDIMS
0 0 1 36 2 10 /

```

PROPS

```

PROPS

-- Rock properties
ROCK
4000.0 0.000008 /

-- Fluid densities at surface conditions
DENSITY
45 64 0.09 /

-- PVT data for oil
PVDO
400 1.012 1.160
1200 1.0040 1.164
2000 0.9960 1.167
2800 0.9880 1.172
3600 0.9802 1.177
4400 0.9724 1.181
5200 0.9646 1.185
5600 0.9607 1.190
/

-- PVT data for water with salt present
PVTWSALT
3000.0 /
0.0 1.000 3.0E-6 1.00 1.0E-6
10.0 0.8 3.0E-6 1.00 1.0E-6 /

-- Saturation functions
SWFN
0.20 0.0 0.0

```



```

0.40 0.1 0.0
0.80 0.5 0.0
1.0 1.0 0.0
/

```

```

SOF2
0.2 0.0
0.4 0.12
0.6 0.35
0.7 0.68
0.8 1.0
/

```

SOLUTION

```

SOLUTION

-- Equilibration data
EQUIL
5270 4000.0 5270.0 0.0 400.0 0.0 0 0 5 /

-- Salt concentration versus depth table
SALTVD
5000.0 9.0
5500.0 9.0 /

-- Aquifer data
AQUFETP
1 5400 1* 2.0E9 3.0E-5 540.96 1 10.0 /
/

AQUCT
2 5400.0 1* 20.0 0.10 3.0E-5 2400.0 140.0 4.8 1 1 9.5 /
/

AQUANCON
1 15 15 1 1 1 5 'I+' /
2 15 15 1 1 6 10 'I+' /
/

```

SUMMARY

```

SUMMARY

-- Field salt production rates and totals
FSPR
FSPT

-- Field salt injection rates and totals
FSIR
FSIT

-- Field salt production and injection concentrations
FSPC
FSIC

-- Field salt in place
FSIP

-- Group salt production and injection rates
GSPR

```

```

'G' /
GSIR
'G' /

-- Well salt production and injection rates
WSPR
'PRODUCER' /
WSIR
'INJECTOR' /

-- Block salt concentration
BSCN
1 1 1 /
/

```

SCHEDULE Example

SCHEDULE section example

```

SCHEDULE

-- Well specification data
WELSPECS
  'PRODUCER' 'G' 1 1 5200 'OIL' /
  'INJECTOR' 'G' 14 1 5400 'WAT' /
/

COMPDAT
  'PRODUCER' 1 1 1 10 'OPEN' 1 -1 0.5 /
  'INJECTOR' 14 1 5 10 'OPEN' 1 -1 0.5 /
/

WCONPROD
  'PRODUCER' 'OPEN' 'ORAT' 200.0 4* 1500.0 /
/

GCONINJE
  'FIELD' 'WAT' 'VREP' 3* 0.7 /
/

WCONINJE
  'INJECTOR' 'WAT' 'OPEN' 'GRUP' /
/

-- Salt concentration for injection well
WSALT
  'INJECTOR' 1.0 /
/

-- Timestep data
TSTEP
1.0 9.0 90.0 100.0 100.0 100.0 100.0 250.0 250.0 1000.0 1000.0 /

END

```

Gi pseudo-compositional model

x	Eclipse 100
	Eclipse 300

This chapter is concerned with the Gi pseudo-compositional model for gas condensate/volatile oil systems in Eclipse 100.

The scope and accuracy of gas condensate modeling using black oil reservoir simulators such as Eclipse is now established. Both depletion and gas cycling above the dew point can be modeled and yield an adequate match with the results from multi-component compositional simulators. The main inadequacy with the existing Eclipse approach lies with the treatment of gas injection below the dew point where the primary compositional effect - the stripping of liquid components in inverse proportion to their molecular weights - is completely ignored.

The standard black oil model assumes that the saturated hydrocarbon fluid properties are functions of pressure only and disregards any compositional dependence in the saturated fluid PVT properties. As a consequence, when dry gas is injected into a condensate below its dew point the gas continues to re-vaporize liquid at a rate governed only by the ambient pressure. The vapor saturates over a zone whose thickness is of the order of one grid block - in particular all the liquid in the vicinity of the injectors evaporates rapidly. Results obtained with fully compositional simulation models suggest that liquid saturation profiles would vary more slowly with increasing distance from gas injectors.

The simulation of low pressure gas injection into gas condensate reservoirs can be improved in one of two ways. One is to use a multi-component compositional model to track the movement of the individual hydrocarbon components through the grid mesh. At each timestep the thermodynamic equilibrium between all the components present in each grid block is computed using an equation-of-state flash calculation or K-value correlations which define the vapor- liquid mole fraction ratios for each component over a wide range of conditions in the simulation.

An alternative approach to modeling low pressure gas injection is to extend the black oil, two pseudo-component model to take account of fluid property changes occurring during gas injection. A number of extensions of the black oil model to treat compositional effects arising during gas injection have been reported in the literature, and all methods hinge on extending the fluid property treatment so that the saturated fluid properties depend not only on pressure but also on an additional parameter which characterizes compositional changes in the reservoir liquid and vapor phases at constant pressure.

The model described here is based on the method proposed by [Cook et al \(1974\)](#). The extra parameter introduced into the model to represent the light-end stripping effects is a cumulative injection gas flow parameter (Gi) and is analogous to tracking the flow of the dry injection gas around the reservoir grid.

Description of the model

The model requires an extra parameter to be introduced in order to characterize the compositional changes taking place in the reservoir fluids due to gas injection - this is cumulative volume of injection gas which has passed over the reservoir liquid in each grid cell.

A simple active gas tracking calculation is performed to determine the cumulative amount of dry injection gas (Gi), which has contacted the oil in each grid cell. Gi can then be used as the correlating parameter for determining the fluid property changes at constant pressure. An extra variable Gi has therefore to be stored in each grid cell.

The tracking calculation is performed in a novel way. If f_{dr} denotes the fraction of invading gas (with $R_v = R_{v, inj}$), then the effective R_v of the invading gas is:

$$R_{v, inv} = f_{dr} \cdot R_{v, inj} + (1 - f_{dr})R_{v, cell} \quad \text{Eq. 4.61}$$

which yields the following expression for the fractional injection gas flux into each grid cell:

$$f_{dr} = \frac{R_{v, inv} - R_{v, cell}}{R_{v, inj} - R_{v, cell}} \quad \text{Eq. 4.62}$$

If the well injects dry gas ($R_{v, inj} = 0$), this expression reduces to $f_{dr} = 1 - R_{v, inv}/R_{v, cell}$.

The cumulative injection gas flowing into each grid cell (G_i) can then be defined as:

$$G_i = \int \frac{f_{dr} F_g}{HCPV} dt \quad \text{Eq. 4.63}$$

where

F_g is the surface flow rate of gas,

$HCPV$ is the hydrocarbon pore volume of the cell.

The saturated PVT properties (principally R_s , R_v , B_o , and B_g) should now contain a dependence on both P_o and G_i . In practice it is usually the variation of $R_{v, sat}$ and $R_{s, sat}$ with G_i that is the most significant and should be ignored. For example it is likely that $R_{v, sat}$ could be reduced by an order of magnitude before complete vaporization of the liquid in a cell finally occurred. More significantly, the difference in the fluid property variations with G_i from cell to cell in a one dimensional compositional case is usually found to be small – this tends to suggest that G_i is a good correlating parameter for the purposes of modeling gas injection effects.

The differential equation to be solved for G_i can be easily derived. The gas entering each grid cell can be regarded as being composed of two components. Firstly, a component which has the same oil-gas ratio as the gas already present in the cell; this component is regarded as being in equilibrium with the gas already present in the cell. The second component in the gas is dry ($R_v = 0$) and can cause liquid vaporization.

The inflow of the free gas which is dry is then:

$$f_{dr} \cdot F_g = \left(1 - \frac{R_{v, inv}}{R_{v, cell}}\right) F_g \quad \text{Eq. 4.64}$$

If the grid cell contains a well completion injecting dry gas, there is an additional term to represent the dry gas inflow from the well:

$$f_{drw} \cdot Q_g = \left(1 - \frac{R_{v, inj}}{R_{v, cell}}\right) Q_g \quad \text{Eq. 4.65}$$

where

f_{drw} is the dry gas fraction of the well flow,

Q_g is the well gas flow rate.

Now the change in G_i from the definition is:

$$\delta G_i = \frac{f_{dr} F_g + f_{drw} Q_g}{HCPV} \delta t \quad \text{Eq. 4.66}$$

$$\delta G_i = \frac{(R_v - R_{v, up}) F_g \delta t}{R_v \cdot \phi \cdot V \cdot (1 - S_w)} + \frac{(R_v - R_{v, inj}) Q_g \delta t}{R_v \cdot \phi \cdot V \cdot (1 - S_w)} \quad \text{Eq. 4.67}$$

where

R_v is the vaporized oil-gas ratio,

$R_{v, up}$ is the R_v in the upstream cell,

$R_{v, inj}$ is the R_v of the injection fluid,

V is the cell volume

φ is the cell porosity

S_w is the water saturation.

In the continuum limit $VOL \rightarrow 0$, $\delta t \rightarrow 0$ the differential equation for Gi in one dimension becomes:

$$\frac{d}{dt}Gi = \frac{F_g}{\varphi \cdot R_v \cdot A \cdot (1 - S_w)} \frac{dR_v}{dx} \quad \text{Eq. 4.68}$$

where

A is the flow area.

Data generation procedure

The model requires two-dimensional tables (P, Gi) to be entered for the saturated R_s , R_v , B_o and B_g data tables spanning the range of pressures and Gi parameter values encountered in the simulation. Dependence of the phase viscosities on Gi is assumed to be negligible. Variations of the surface oil and gas gravities with Gi are also ignored.

The new PVT tables are generated by an extension to the standard constant volume depletion procedure usually used to produce black oil PVT data. Instead of simply generating black oil tables by depleting the PVT cell, the simulator needs to deplete the PVT cell to a pressure below the dew point and then inject dry gas into the cell in a series of steps. At each step the mixture is flashed to thermodynamic equilibrium and the excess gas withdrawn until the mixture occupies the same volume before the gas injection step. The new values for R_v , R_s , B_o , and B_g are calculated to correspond with the updated value of Gi. The procedure is repeated injecting gas at a lower pressure — in each case the liquid should completely evaporate to yield the full saturated PVT curves.

It is assumed in the model that the data generated by the above method is sufficiently path independent to adequately represent the compositional effects occurring during gas injection (that is, no matter how a given point is reached in (P, Gi) space, the R_v , R_s , B_o , and B_g values will be the same as that used in the generating procedure).

Note: The Gi model with Gi=0 will not always behave as a black oil model without the GI option in it. According to [EQ 31.3], in a Gi Model, there will be a finite amount of Gi generation whenever there exists the combination of a finite fractional gas flux into a grid cell and a finite gas flow rate in the grid cell. Furthermore, Gi accumulates over time. In a black oil model without GI option, this does not happen.

Using the Gi pseudo-compositional model

The model should only be used in oil-water-gas systems. The keyword `GI`MODEL in the RUNSPEC section is used to activate the option. Gas condensate and live oil must also be declared active phases using keywords `VAPOIL` and `DISGAS` in the RUNSPEC section. The `IMPES` solution method should not be used when the Gi model is active.

The oil and gas PVT properties are specified using the `PVTO` and `PVTG` keywords in the usual way. When the Gi model is active, the data specified using the `PVTO` and `PVTG` keywords refer to fluid conditions at $G_i=0$. A two-dimensional variation of saturated fluid properties is required for the option to specify the full (P, G_i) dependence of $R_{s\text{ sat}}$, $R_{v\text{ sat}}$, $B_{g\text{ sat}}$ and $B_{o\text{ sat}}$.

Either the keywords `RVGI`, `RSGI`, `BGGI`, `BOGI` or the single keyword `GIAL`L are used to specify these variations, and should be defined in the PROPS section. Note that the function values entered under these keywords refer to **multiplier** values on the saturated data contained in the `PVTO` and `PVTG` keywords (that is, at $G_i=0$). The corresponding Gi-nodal values are specified using the `GINODE` keyword in the PROPS section.

The PVT properties specific to the Gi Model can be echoed using the 'GINODE' mnemonic in the `RPTPROPS` keyword. Output of the values of G_i in each cell at each report time can be produced using the 'GI' mnemonic in the `RPTSCHED` keyword. Debug from the option can be produced using integer 30 of the `DEBUG` keyword.

Summary of keywords

RUNSPEC

The keywords are:

- `GI`MODEL Activates the Gi pseudo-compositional model.

PROPS section

Keyword	Description
<code>BGGI</code>	Variation of saturated gas FVF with pressure and G_i .
<code>BOGI</code>	Variation of saturated oil FVF with pressure and G_i .
<code>GIAL</code> L	Variation of saturated properties with pressure and G_i .
<code>GINODE</code>	Gi nodal values.
<code>RPTPROPS</code>	Controls output from the PROPS section. Arguments <code>GINODE</code> and so on output saturated fluid properties.
<code>RSGI</code>	Variation of saturated GOR with pressure and G_i .
<code>RVGI</code>	Variation of saturated OGR with pressure and G_i .

SOLUTION section

- `GI` Initial G_i values for enumeration.

SUMMARY section

- BGI Block Gi value

SCHEDULE section

Keyword	Description
RPTSCHED	Controls output from the SCHEDULE section. Argument GI outputs the current Gi values.

Example problem

This problem illustrates the use of the option. The grid is one dimensional and the initial fluids consist of connate water and a volatile undersaturated oil. Injection of dry gas begins at t=0 in grid cell (1,1). The production well completed in cell (15,1) operates at constant BHP.

RUNSPEC

```
RUNSPEC
TITLE
  1D Test Case - Volatile Oil Model - Gi Model Activated

DIMENS
  15    1    1  /

NONNC
OIL
WATER
GAS
DISGAS
VAPOIL
GIMODEL
METRIC

EQLDIMS
  1  100  20    1    1  /

ENDSCALE
  'NODIR'      'REVERS'      1    1  /

TABDIMS
  1    1  20  30    1  40  /

WELLDIMS
  2    4    1    2  /

START
  1 'JAN' 1987  /
```

GRID

```
GRID =====
NOGGF

EQUALS
  'DX' 100.0  /
  'DY' 100.0  /
  'DZ' 10.0   /
  'TOPS' 3995.0 /
  'PORO' 0.25  /
  'PERMX' 200.0 /
/

RPTGRID
```

```
'DX' 'DY' 'DZ' 'PERMX' 'PERMY' 'PERMZ' 'MULTX'
'MULTY' 'MULTZ' 'PORO' 'NTG' 'TOPS' 'PORV'
'DEPTH' 'TRANX' 'TRANX' 'TRANZ' 'COORDSYS' /
```

PROPS

```
PROPS =====
PVTW
408.70000 1.05200 5.00E-5 0.26000 0.00E+00 /
```

The following [PVTG](#) table describes the wet gas properties corresponding to $G_i=0$. The saturated gas properties depend only on pressure in the usual way.

```
PVTG
11.40 0.00057098 0.134826 0.01316 /
49.20 0.00018704 0.028395 0.01534 /
107.50 0.00015314 0.012619 0.01765 /
154.00 0.00018098 0.008819 0.01994 /
207.30 0.00024415 0.006699 0.02343 /
248.00 0.00031469 0.005778 0.02685 /
275.00 0.00037429 0.005357 0.02957 /
304.30 0.00045362 0.005019 0.03306 /
330.00 0.00053982 0.004806 0.03675 /
344.09 0.00058662 0.004713 0.03890 /
358.09 0.00062914 0.004632 0.04096 /
371.27 0.00067040 0.004566 0.04301 /

383.69 0.00071060 0.004514 0.04507 /
395.41 0.00075159 0.004472 0.04717 /
406.48 0.00079051 0.004439 0.04925 /
416.96 0.00082863 0.004413 0.05134 /
426.88 0.00086703 0.004393 0.05348 /
436.29 0.00090351 0.004378 0.05560 /
445.21 0.00093977 0.004366 0.05775 /
453.68 0.00097372 0.004358 0.05986
0.00064914 0.004119 0.04974
0.00032457 0.003904 0.04034
0.00000000 0.003718 0.03230 /
/
```

The following [PVTO](#) table describes the live oil properties corresponding to $G_i=0$. The saturated oil properties depend only on pressure in the usual way.

```
PVTO
3.02 11.40 1.095 0.71545 /
24.84 49.20 1.179 0.57933 /
56.57 107.50 1.284 0.44463 /
83.57 154.00 1.367 0.36695 /
120.00 207.30 1.476 0.29689 /
153.30 248.00 1.575 0.25293 /
179.05 275.00 1.652 0.22707 /
211.49 304.30 1.749 0.20134 /
245.23 330.00 1.850 0.18027 /
266.64 344.09 1.915 0.16908
445.41 1.840 0.18829
546.74 1.785 0.2062 /
284.74 358.09 1.966 0.16218 /
302.84 371.27 2.017 0.15597 /
320.95 383.69 2.068 0.15036 /
339.05 395.41 2.119 0.14527 /
357.15 406.48 2.170 0.14063 /
375.25 416.96 2.220 0.13638 /
393.35 426.88 2.271 0.13248 /
411.45 436.29 2.322 0.12888 /
429.55 445.21 2.373 0.12555 /
447.65 453.68 2.423 0.12246
555.01 2.322 0.13498
```



```
656.33 2.246 0.14656 /
/
```

The Gi node values to be used in the specifying the data points in the [RSGI](#), [RVGI](#), [BOGI](#) and [BGGI](#) keywords are defined using the [GINODE](#) keyword

```
GINODE
0.0
10.0
20.0
40.0
120.0
200.0
300.0
500.0
800.0
/
```

RVGI defines the variation of saturated oil-gas values with pressure and Gi. Note that the entries in this table represent multiplier values for the pressure dependent saturated OGR data entered in the [PVTG](#) table.

```
RVGI
248.0 1.0
0.95
0.90
0.85
0.80
0.72
0.60
0.50
0.40
0.30 /
416.96 1.0
0.95
0.90
0.85
0.80
0.72
0.60
0.50
0.40
0.30 /
/
```

RSGI defines the variation of saturated gas-oil values with pressure and Gi. Note that the entries in this table represent multiplier values for the pressure dependent saturated GOR data entered in the [PVTG](#) table.

```
RSGI
248.0 1.0
0.98
0.95
0.91
0.86
0.82
0.78
0.75
0.72
0.70 /
416.96 1.0
0.98
0.95
0.91
0.86
0.82
0.78
0.75
0.72
```

```
0.70 /
/
```

BGGI defines the variation of saturated gas FVF values with pressure and Gi. Note that the entries in this table represent multiplier values for the pressure dependent saturated FVF data entered in the PVTG table.

```
BGGI
248.0 1.0
      0.998
      0.996
      0.994
      0.992
      0.990
      0.988
      0.986
      0.984
      0.982 /
416.96 1.0
      1.0
      1.0
      1.0
      1.0
      1.0
      1.0
      1.0
      1.0
      1.0 /
/
```

BOGI defines the variation of saturated oil FVF values with pressure and Gi. Note that the entries in this table represent multiplier values for the pressure dependent saturated FVF data entered in the PVTO table.

```
BOGI
248.0 1.0
      0.995
      0.990
      0.985
      0.980
      0.975
      0.970
      0.965
      0.960
      0.955 /
416.96 1.0
      0.99
      0.98
      0.96
      0.94
      0.92
      0.90
      0.88
      0.86
      0.84 /
/

DENSITY
732.000 1000.05 1.10900 /

SGFN
0.00 0.0000 0.0
0.03 0.0000 0.0
0.110 0.0246 0.0
0.307 0.1212 0.0
0.435 0.2684 0.0
0.491 0.3701 0.0
/

SWFN
0.221 0.00 0.0
0.25 0.010 0.0
```

```

0.400 0.070 0.0
0.547 0.163 0.0
0.582 0.179 0.0
0.670 0.260 0.0
0.731 0.360 0.0
/
SOF3
0.269 0.000000 0.000000
0.288 0.000038 0.000000
0.294 0.000050 0.000380
0.300 0.000065 0.000740
0.320 0.000170 0.001400
0.500 0.035000 0.034000
0.750 0.470000 0.760000
0.779 1.000000 1.000000
/
ROCK
407.800 7.00E-5 /
RPTPROPS
'GINODE' /

```

SOLUTION

```

SOLUTION =====
PRESSURE
15*370.0 /
SWAT
15*0.2 /
SGAS
15*0.0 /
RS
15*266.64 /
RV
15*0.0006 /
RPTSOL
'PRES' 'SOIL' 'SWAT' 'SGAS' 'RS' 'RV' 'FIP'
'EQUIL' 'RSVD' 'GI' /

```

SCHEDULE

```

SCHEDULE =====
WELSPECS
'GINJ' 'G' 1 1 4000.00 'GAS' 1* 'NO' 'SHUT' 'YES' 1* 'SEG' /
'PROD' 'G' 15 1 4000.00 'OIL' 0.0 'NO' 'SHUT' 'YES' 1* 'SEG' /
/
COMPDAT
'GINJ' 1 1 1 1 'OPEN' 0 1* 0.4 /
'PROD' 15 1 1 1 'OPEN' 0 1* 0.4 /
/
WCONPROD
'PROD' 'OPEN' 'BHP' 1* 1* 1* 1* 1* 370.0 /
/
WCONINJE
'GINJ' 'GAS' 'OPEN' 'RATE' 150000.0 /
/
RPTSCHED
'PRES' 'SOIL' 'SWAT' 'SGAS' 'RS' 'RV' 'RESTART=2'
'FIP' 'WELLS=2' 'SUMMARY=2' 'CPU=2'

```

```
'NEWTON=2'  /  
TSTEP  
10*20.0  /  
END
```

GASWAT option

	Eclipse 100
x	Eclipse 300

The GASWAT option in Eclipse 300 provides a method of modeling gas phase/aqueous phase equilibria using an equation of state. The Peng Robinson equation of state is modified following the suggestions of [Soreide and Whitson \(1992\)](#) to obtain accurate gas solubilities in the aqueous phase.

Note: The GASWAT option is two-phase only. By using it, the gas and water phases are implicitly requested. It is not possible to model the oil phase while using this option. If the [GASWAT](#) keyword is present, then the [OIL](#), [WATER](#) and [GAS](#) keywords are not necessary and will be ignored.

The temperature of the reservoir may vary with depth, and this is reflected in the equation of state treatment, in which temperature coefficients are stored to be combined with the temperature of each cell. A global salinity value may be entered to correct the default equation of state modifications for brine concentration. You can override equation of state coefficients.

The GASWAT option also allows molar calorific values to be entered under the [CALVAL](#) or [CALVALR](#) keywords. Gas qualities, calorific production rates and calorific production totals for wells and groups are calculated and reported. A group can be given a quality target using the [GQUALITY](#) keyword - quality is here defined as calorific value per mole of gas. The group control algorithm will attempt to meet this quality target by distributing production between wells of better and worse quality.

The reservoir may be equilibrated with a single phase present or with an initial gas-water contact. In the latter case the condition that the two phases are in equilibrium at the contact is honored - you specify the composition of one of the phases and the program obtains the composition of the other from the equilibrium condition.

A number of new summary items have been added so that calorific production rates, calorific production totals and quality values may be obtained for wells, groups and the field.

The water analytic aquifers are dealt with in the same way as in the CO2STORE option. Refer to "[Analytic water aquifers](#)".

The determination of single phase as gas or water use a simplified criteria. If the temperature is less than the pseudo critical temperature, as computed by the Li criteria, see keyword [FACTLI](#), and the phase contains mostly H₂O, the phase is labeled water otherwise gas. It is possible to activate the pre-2008.1 behavior by setting [item 146](#) to 6 in the [OPTIONS3](#) keyword. Also the flash will treat H₂O as the heaviest component unless [item 125](#) in [OPTIONS3](#) is set to -1.

Note that the default density and viscosity calculations for water with solutes may be replaced by Ezrokhi's method, the latter also used in conjunction with [CO2STORE](#) and [GASSOL](#). This is done by adding the keywords [DENAQA](#) and [VISCAQA](#) respectively for density and viscosity, and specifying the Ezrokhi temperature coefficients for each component.

Equation of state modifications

The conventional Peng-Robinson equation of state

The A -coefficient in the Peng-Robinson equation of state is given by:

$$A = \sum_{jk} x_j x_k A_{jk} \quad \text{Eq. 4.69}$$

where x are component mole fractions and

$$A_{jk} = (1 - k_{jk})(A_j A_k)^{\frac{1}{2}} \quad \text{Eq. 4.70}$$

k_{jk} are binary interaction coefficients. A_{jk} is defined as:

$$A_j = \frac{\Omega_j P_{rj}}{T_{rj}^2} \quad \text{Eq. 4.71}$$

The reduced critical temperatures and pressures are $T_r = T/T_c$ and $P_r = P/P_c$. The reduced A-coefficients are defined by extracting this temperature and pressure dependence:

$$A_j = \frac{A_j^r \cdot P}{T^2} \quad \text{Eq. 4.72}$$

The Peng-Robinson equation of state then defines A_j^r as:

$$A_j^r = \frac{\Omega_{oj} T_{cj}^2 \alpha_j}{P_{cj}} \quad \text{Eq. 4.73}$$

with

$$\alpha_j^{\frac{1}{2}} = 1 + f(w_j) \left(1 - T_{rj}^{\frac{1}{2}} \right) \quad \text{Eq. 4.74}$$

$f(w_j)$ is a polynomial in the acentric factor, given by:

$$f(w_j) = 0.37464 + 1.54226w_j - 0.26996w_j^2 \quad \text{Eq. 4.75}$$

Rewriting to account for the temperature dependence in this expression gives:

$$A_j^r = \frac{\Omega_{oj} T_{cj}^2 \left(p_j + q_j T^{\frac{1}{2}} \right)^2}{P_{cj}} \quad \text{Eq. 4.76}$$

where $p_j = 1 + f(w_j)$ and $q_j = -f(w_j)/T_c^{\frac{1}{2}}$.

Then:

$$A_{rjk} = [1 - k_{jk}] \left[\frac{\Omega_{oj} T_{cj}^2}{P_{cj}} \right]^{\frac{1}{2}} \left[\frac{\Omega_{ok} T_{ck}^2}{P_{ck}} \right]^{\frac{1}{2}} \left[p_j p_k + (p_j q_k + p_k q_j) T^{\frac{1}{2}} + q_j q_k T \right] \quad \text{Eq. 4.77}$$

Eclipse 300 stores the three coefficients A_{1jk} , A_{2jk} and A_{3jk} , where:

$$A_{1jk} = (1 - k_{jk}) \left(\frac{\Omega_{oj} T_{cj}^2}{P_{cj}} \right)^{\frac{1}{2}} \left(\frac{\Omega_{ok} T_{ck}^2}{P_{ck}} \right)^{\frac{1}{2}} (p_j p_k) \quad \text{Eq. 4.78}$$

$$A_{2jk} = (1 - k_{jk}) \left(\frac{\Omega_{oj} T_{cj}^2}{P_{cj}} \right)^{\frac{1}{2}} \left(\frac{\Omega_{ok} T_{ck}^2}{P_{ck}} \right)^{\frac{1}{2}} (p_j q_k + p_k q_j) \quad \text{Eq. 4.79}$$

$$A_{3jk} = (1 - k_{jk}) \left(\frac{\Omega_{oj} T_{cj}^2}{P_{cj}} \right)^{\frac{1}{2}} \left(\frac{\Omega_{ok} T_{ck}^2}{P_{ck}} \right)^{\frac{1}{2}} [q_j q_k] \quad \text{Eq. 4.80}$$

This enables the equation of state coefficient for a cell of temperature T to be simply assembled as:

$$A_{jk} = \left(A_{1jk} + A_{2jk} T^{\frac{1}{2}} + A_{3jk} T \right) \frac{P}{T^2} \quad \text{Eq. 4.81}$$

Modifications for the vapor-aqueous system

If the Peng-Robinson equation is used in the above form for a gas-aqueous system the predicted saturated mole fractions of gases dissolved in water are hopelessly wrong - predicted solubilities are typically 10^{-7} mole/mole, whilst experimental values are nearer 10^{-3} mole/mole.

Soreide and Whitson propose two changes:

- The expression for $\alpha^{\frac{1}{2}}$ above is modified for the water component. It is replaced by an expression with a more complex temperature dependence:

$$\alpha^{\frac{1}{2}} = 1 + aq_1(1 - aq_2 T_r) + aq_3(T_r^{-3} - 1) \quad \text{Eq. 4.82}$$

with default values:

$$aq_1 = 0.4530$$

$$aq_2 = 1 - 0.0103 c_s^{1.1}$$

$$aq_3 = 0.0034$$

c_s is the brine concentration expressed as a molality. (Molality is given by $[1000 \times (\text{moles of salt}) / (\text{mass of water})]$, where moles and mass are expressed in consistent units such as kg-M and kg respectively. This is discussed further below).

In Eclipse 300 you may overwrite the defaults for aq_1 , aq_2 and aq_3 by entering the [AQCOEF](#) keyword.

This modification is only done for the water component, and the remainder take the usual form. A general form for all the components is thus:

$$\alpha^{\frac{1}{2}} = p + q T_r^{\frac{1}{2}} + r T_r + s T_r^{(-3)} \quad \text{Eq. 4.83}$$

- Different binary interaction coefficients are used for the gas and aqueous phases. This would cause problems near the critical point, but for a water-dominated system the critical temperature is in the region of $T_{cw} = 647\text{K}$, which should be well above the area of interest.

Soreide and Whitson propose adding a temperature dependence to the aqueous phase binary interaction coefficients:

$$k_{jw}^a = bq_1 + bq_2 T_{rj} + bq_3 T_{rj}^2 \quad \text{Eq. 4.84}$$

with default values:

$$bq_1 = A_0(1 + S_0 c_s)$$

$$bq_2 = A_1(1 + S_1 c_s)$$

$$bq_3 = A_2(1 + S_2c_s)$$

where:

$$A_0 = 1.112 - 1.7369w_j^{-0.1}$$

$$A_1 = 1.1001 + 0.836w_j$$

$$A_2 = -0.15742 - 1.0988w_j$$

and:

$$S_0 = 0.017407$$

$$S_1 = 0.033516$$

$$S_2 = 0.011478$$

Special values are used for binary interaction coefficients between the aqueous phase and N₂, CO₂ and H₂S:

$$k_{N_2w}^a = -1.70235 \left(1 + 0.025587c_s^{\frac{3}{4}} \right) + 0.44338 \left(1 + 0.08126c_s^{\frac{3}{4}} \right) T_{rN_2} \quad \text{Eq. 4.85}$$

$$k_{CO_2w}^a = -0.31092(1 + 0.15587c_s^{0.7505}) + (0.23580) \left(1 + 0.17837c_s^{0.979} \right) T_{rCO_2} - 21.2566e^{(-6.7222T_{rCO_2} - c_s)} \quad \text{Eq. 4.86}$$

$$K_{H_2Sw}^a = -0.20441 + 0.23426T_{rH_2S} \quad \text{Eq. 4.87}$$

These fit into the general quadratic form, apart from the exponential term in the CO₂ case: this is evaluated at the reservoir temperature or surface standard conditions and added to the bq_1 term.

In Eclipse 300 you may overwrite these defaults for bq_1 , bq_2 and bq_3 using the [BICAQ1](#), [BICAQ2](#) and [BICAQ3](#) keywords.

The above forms for α and k_{ajk} may then be simply assembled with the cell temperature to build up the equation of state A-coefficient.

The program automatically uses both these modifications in GASWAT mode. The water component is identified as that with name 'H2O' - this must be present. Nitrogen, carbon dioxide and hydrogen sulfide, if present, should be identified with the names 'N2', 'CO2' and 'H2S' respectively. Keywords such as [SALINITY](#), [SALINITR](#), [AQCOEF](#), [BICAQ1](#), [BICAQ2](#) and [BICAQ3](#) are entered once, and need not be specified for each equation of state region or separately for reservoir and surface conditions. The gas binary interaction coefficients are obtained from the usual user-input data.

Calorific values

Units

These are entered in the PROPS section under the [CALVAL](#) or [CALVALR](#) keywords. The metric units are kJ/kg-M. For conversion purposes 1 m³=0.04165935 kg-M at normal standard conditions (compare with ["Conversion for an ideal gas \(Z=1\)"](#)). Well and group quality is reported in kJ/kg-M. Calorific rates and totals are in kJ/day and kJ respectively. The equivalent field units are Btu/lb-M etc.

Calorific well and group production rates and totals are reported in the PRINT file under the REPORT, FIELD and WELLS headers. The gas quality is also reported - this is the mixture calorific value per mole.

Specifying group production quality

Where a group target is set - for example a required group gas production rate - the group distributes this target amongst the wells in the group using guide rates. Modifying these guide rates can weight the production between low and high quality wells so that a required group quality is obtained. This required quality can be set using the [GQUALITY](#) keyword.

This option applies only if:

- The group has more than one child under group control
- At least one of the group controlled children has a quality better than the required target and at least one has a quality worse than the required target.

Generally a point is ultimately reached when the required calorific value can no longer be met (for example if all the wells in the group are producing gas of too low a quality). The quality does not then honor the required target.

Salinity

In the above, salinity is defined by molality: $c_s = 1000 \cdot n_s / m_w$, where n_s is the number of moles of salt, for instance in kg-M, and m_w the mass of the water, for instance in kg. This may be related to the usual definition in terms of parts per million by mass:

$$c_s^{ppm} = A \cdot c_s / (1 + B \cdot c_s) \quad \text{Eq. 4.88}$$

$$c_s = c_s^{ppm} / (A - B \cdot c_s^{ppm}) \quad \text{Eq. 4.89}$$

with $A = 10^3 M_{\text{NaCl}}$, $B = 10^{-3} M_{\text{NaCl}}$, $M_{\text{NaCl}} = 58.44 \text{ kg/kg-M}$

The salinity variation of the equation of state coefficients is not treated on a cell by cell basis - a single value is entered using the [SALINITY](#) keyword in the PROPS section. If it is not entered the salinity value is assumed to be zero.

Relative permeability tables

The water relative permeability is input using the [WSF](#) keyword, while the gas relative permeability and the gas-water capillary pressure are entered using the [GSF](#) keyword. These keywords are aliases for the [SOF2](#) and [SGFN](#) keywords, which can also be used (with all references to oil treated as references to water).

The saturation tables can be output to the PRINT file by entering the WSF or GSF mnemonics in the [RPTPROPS](#) keyword.

Initialization

The equilibration of the reservoir is straightforward if only a single phase is present: the composition may be entered using [ZI](#) or with respect to depth using [ZMFVD](#).

The situation is slightly more complicated if an initial gas-water contact exists within the reservoir. The need for equilibrium at the contact means that the two phase compositions cannot be specified independently. You may specify either the vapor or aqueous phase composition at the contact ([ZMFVD](#) or

COMPVD may be used). The ninth argument of the [EQUIL](#) keyword specifies which is the phase specified. The other phase composition is then defined by the bubble or dew point condition. Strictly, this also defines the pressure and by default this is reset to the equilibrium value. However, this resetting of the pressure may be overridden using the tenth argument of [EQUIL](#).

Note: The phase equilibrium is only required at the contact: COMPVD allows arbitrary aqueous phase compositions below the water-gas contact and arbitrary vapor compositions above it.

Summary items

Refer to [Gas calorific value control option](#) in the *Eclipse Reference Manual*.

DENSITY values are not used in the GASWAT option: both gas and water density values are obtained from the equation of state.

Note as well that the water phase is considered as “hydrocarbon” when calculating the "hydrocarbon weighted" field pressure reported by FPR and FPRH. For a comparison with a simulation involving an oil phase such as [CO2SOL](#), the "pore volume weighted" pressure returned by FPRP can be used. Additionally, setting [item 151](#) in [OPTIONS3](#) to 1 ensures that only the gas phase is treated as "hydrocarbon".

Equations of state

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Two-parameter equations of state

Eclipse 300 incorporates four equations of state and two additional variations to the Peng-Robinson equation.

When an equation of state is selected, this is used to obtain Z-factors and phase fugacities, to be used to define inter-phase equilibrium and fluid densities.

The equations of state are:

- PR: Peng-Robinson
- RK: Redlich-Kwong
- SRK: Soave-Redlich-Kwong
- ZJ: Zudkevitch-Joffe-Redlich-Kwong

These equations of state are implemented in generalized form using Martin's equation ([Martin, 1973](#)) and [Coats \(1982\)](#).

The generalized form of such an equation of state is:

$$Z^3 + E_2 Z^2 + E_1 Z + E_0 = 0 \quad \text{Eq. 4.90}$$

with:

$$E_2 = (m_1 + m_2 - 1)B - 1 \quad \text{Eq. 4.91}$$

$$E_1 = A - (m_1 + m_2 - m_1 m_2)B^2 - (m_1 + m_2)B$$

$$E_0 = -[AB + m_1 m_2 B^2 (B + 1)] \quad \text{Eq. 4.92}$$

The coefficients m_1 and m_2 depend upon the equation used. Refer to table 4.2 for details.

Equation of State	m1	m2
Redlich-Kwong	0	1
Soave-Redlich-Kwong	0	1
Zudkevitch-Joffe	0	1
Peng-Robinson	$1 + \sqrt{2}$	$1 - \sqrt{2}$

Table 4.2: The dependence of m_1 and m_2 on the equation of state

The cubic equation for the Z-factor may be solved to obtain Z-factors for the liquid and vapor phases. In general, three solutions are obtained.

The distinction between the liquid and the vapor phase is made by choosing the smallest root as the Z-factor for the liquid phase, and the largest root as the Z-factor for the vapor phase.

Fugacity coefficients are calculated using:

$$\ln(f_i/(px_i)) = -\ln(Z-B) + \frac{A}{(m_1-m_2)B} \left(\frac{2\Sigma_i}{A} - \frac{B_i}{B} \right) \ln \left(\frac{Z+m_2B}{Z+m_1B} \right) + \frac{B_i}{B}(Z-1) \quad \text{Eq. 4.93}$$

where:

$$\Sigma_i = \sum_j A_{ij} x_j \quad \text{Eq. 4.94}$$

$$A = \sum_{j=1}^n \sum_{k=1}^n x_j x_k A_{jk} \quad \text{Eq. 4.95}$$

$$B = \sum_{j=1}^n x_j B_j \quad \text{Eq. 4.96}$$

$$A_{jk} = (1 - \delta_{jk}) (A_j A_k)^{\frac{1}{2}} \quad \text{Eq. 4.97}$$

and δ_{jk} are binary interaction coefficients, normally between hydrocarbons and non-hydrocarbons.

These four equations express the mixing laws used in all the equations of state.

The variables are defined by the following equations:

$$A_j = \Omega_a(T, j) \frac{P_{rj}}{T_{rj}^2} \quad \text{Eq. 4.98}$$

$$B_j = \Omega_b(T, j) \frac{P_{rj}}{T_{rj}} \quad \text{Eq. 4.99}$$

$\Omega_a(T, j)$ and $\Omega_b(T, j)$ in equations 4.98 and 4.99 respectively are functions of the acentric factor w_j and the reduced temperature T_{rj} .

For Redlich-Kwong:

$$\Omega_a(T, j) = \Omega_{a0} T_{rj}^{-1/2}$$

$$\Omega_b(T, j) = \Omega_{b0}$$

For Soave-Redlich-Kwong:

$$\Omega_a(T, j) = \Omega_{a0} [1 + (0.48 + 1.574w_j - 0.176w_j^2)(1 - T_{rj}^{1/2})]^2$$

$$\Omega_b(T, j) = \Omega_{b0}$$

For Zudkevitch-Joffe:

$$\Omega_a(T, j) = \Omega_{a0} F_{aj}(T) T_{rj}^{-1/2}$$

$$\Omega_b(T, j) = \Omega_{b0} F_{bj}(T)$$

For Peng-Robinson:

$$\Omega_a(T, j) = \Omega_{a0} [1 + (0.37464 + 1.54226w_j - 0.26992w_j^2)(1 - T_{rj}^{1/2})]^2$$

$$\Omega_b(T, j) = \Omega_{b0}$$

As $T_r = T/T_c$, the SRK and PR forms for Ω_a may be expanded as a polynomial in $T^{1/2}$,

$$\Omega_a(T, j) = A + B \cdot T^{1/2} + CT \quad \text{Eq. 4.100}$$

This expansion is used if [TEMPVD](#) is specified, so that each cell and well has its own temperature.

The normal PR form can be optionally modified for a large acentric factor, using the factor $(0.379642 + 1.48503w_j - 0.164423w_j^2 + 0.016666w_j^3)$ rather than $(0.37464 + 1.53226w_j - 0.2699w_j^2)$ for $w_j > 0.49$. This correction is invoked by use of the [PRCORR](#) keyword.

Ω_{a0} and Ω_{b0} are constants depending upon the equation of state, as shown in table 4.3.

Equation of state	Ω_{a0}	Ω_{b0}
RK, SRK, ZJ	0.4274802	0.08664035
PR	0.457235529	0.077796074

Table 4.3: The dependence of PR constants on the equation of state

These default values may be overwritten if required, by using the [OMEGAA](#) and [OMEGAB](#) keywords.

The Zudkevitch-Joffe equation contains additional temperature factors, denoted $F_{aj}(T)$ and $F_{bj}(T)$ multiplying the usual RK temperature dependence. These are adjusted to match the pure-component fugacity values along the vapor pressure line, and to observe the correct component liquid density.

Eclipse 300 obtains the required variations of component saturation pressure and liquid density with temperature by using the correlations of Reidel, and Gunn and Yamada. These correlations require the input of the normal boiling point and the reference liquid density at a specified reference temperature.

Flash calculations

For a thermodynamic system to be in equilibrium, it is further required that the fugacities in the liquid and vapor phases must be equal for each component:

$$f_{iL} = f_{iV} \quad \text{Eq. 4.101}$$

As described above, the fugacities are functions of temperature, pressure and composition:

$$f_i = f_i(T, p, x_i) \quad \text{Eq. 4.102}$$

The fugacities can be calculated directly from an equation of state.

Equilibrium constants (otherwise known as K-values) for each component can be defined as:

$$K_i = \frac{y_i}{x_i} \quad \text{Eq. 4.103}$$

The mole fractions of each component in the liquid and vapor phases are given as:

$$x_i = \frac{z_i}{1 + (K_i - 1)V} \quad \text{Eq. 4.104}$$

$$y_i = \frac{K_i z_i}{1 + (K_i - 1)V} \quad \text{Eq. 4.105}$$

If K-values are used to specify inter-phase equilibria, these two equations are used to specify the liquid and vapor compositions directly, with K-values obtained by table look-up as functions of pressure.

Viscosity evaluation

In compositional mode the phase viscosity values are obtained using either the Lorentz-Bray-Clark method or the Pedersen et al. method.

The Lorentz-Bray-Clark method

By default the Lorentz-Bray-Clark method is used ([Lorentz and Bray, 1964](#)). The viscosity μ for each phase is given by

$$[(\mu - \mu^o) \cdot \xi + 0.0001]^{1/4} = \sum_{i=1}^5 a_i \cdot b_r^{i-1} \quad \text{Eq. 4.106}$$

μ^o and ξ are functions of the composition x_i , the component molecular weights, critical pressures and critical temperatures.

The coefficients a_i are set equal to the values given in [Lorentz and Bray \(1964\)](#). These values can be modified by using the `LBCCOE` keyword.

The reduced molar densities b_r are given by

$$b_r = \frac{b_p}{b_c} \quad \text{Eq. 4.107}$$

where the phase volumes b_p are calculated from the pressure and compressibility (and volume shift if used), and the critical molar density b_c for each phase. The critical molar density is defined by:

$$\frac{1}{b_c} = V_c = \sum x_i V_{c,i} \quad \text{Eq. 4.108}$$

The critical molar volumes for each component $V_{c,i}$ can be specified directly by either of the keywords `VCRIT` or `VCRITVIS`, or can be calculated from critical z-factors entered by the keywords `ZCRIT` or `ZCRITVIS`.

The modified Lorentz-Bray-Clark method

A modified form of the Lorentz-Bray-Clark method can also be used:

$$[(\mu - \mu^o) \cdot \xi + 0.0001] = \left(\sum_{i=1}^5 a_i \cdot b_r^{i-1} \right)^4 \cdot e\left(\frac{b_r}{b_{max,r} - b_r}\right) \quad \text{Eq. 4.109}$$

This modified equation is intended to improve the modeling of the behavior of heavy oils. It was requested by a client and has not been published.

In this equation the viscosity increases exponentially as the reduced molar density b_r approaches a maximum $b_{max,r}$. For small values of b_r , such that $b_r \ll b_{max,r}$, the modified Lorentz-Bray-Clark equation has the same form as the original equation, Equation 4.106. The maximum reduced molar density can be expressed in terms of a maximum molar density b_{max}

$$b_{max,r} = \frac{b_{max}}{b_c}$$

which in turn can be expressed in terms of the minimum component volumes $V_{min,i}$

$$\frac{1}{b_{\max}} = V_{\min} = \sum x_i V_{\min,i} \quad \text{Eq. 4.110}$$

The minimum component volumes $V_{\min,i}$ are entered using the keyword **VCOMPACT**. If the **VCOMPACT** keyword is found the modified Lorentz-Bray-Clark method will be used instead of the Lorentz-Bray-Clark method.

When entering $V_{\min,i}$ data it is important to ensure that the actual molar density never exceeds the maximum, so that

$$b_p < b_{\max} \quad \text{Eq. 4.111}$$

If the molar densities do exceed this value then a large constant value is used for the viscosity.

A simplified version of the modified Lorentz-Bray-Clark method can be used if the minimum component volumes $V_{\min,i}$ are set proportional to the critical component volumes

$$V_{\min,i} = \frac{1}{b_0} \cdot V_{c,i} \quad \text{Eq. 4.112}$$

In this case the maximum reduced molar density in equation 4.109 will be a constant:

$$b_{\max,r} = b_0 \quad \text{Eq. 4.113}$$

Example

In this example of the simplified version of the modified Lorentz-Bray-Clark method, the reduced molar density $b_{\max,r}$ is set to a constant value of 4.776. Given values of **VCRIT** for each of the three components, the three **VCOMPACT** values are obtained by dividing each **VCRIT** value by 4.776.

```
VCRIT
 99.20  93.90 148.30 /
VCOMPACT
 20.77  19.66 31.05 /
```

The Pedersen method

If the Pedersen et al viscosity calculation (*Pedersen et al, 1989*) is required instead of the default method, then the **PEDERSEN** keyword should be added to the **PROPS** section. The calculation is based on a modification of the corresponding states principle in which the mixture viscosity μ is related to a reference viscosity μ_o by terms involving properties of the mixture (denoted with a subscript m) and the reference material (denoted with a subscript o). The properties are the critical temperature T_c , critical pressure P_c , molecular weight M_w and rotational coupling factor α :

$$\mu(P, T) = \left(\frac{T_{cm}}{T_{co}}\right)^{-1/6} \cdot \left(\frac{P_{cm}}{P_{co}}\right)^{2/3} \cdot \left(\frac{M_{wm}}{M_{wo}}\right)^{1/2} \cdot \left(\frac{\alpha_m}{\alpha_o}\right) \cdot \mu_o(P_o, T_o)$$

The reference material is taken to be methane. Pedersen (*Pedersen et al, 1989*) gives formulas for the mixture properties, and the reference pressure P_o and reference temperature T_o . The mixture molecular weight is given by:

$$M_{wm} = c_1 \cdot 1.304 \times 10^{-4} \cdot \left(\overline{M}_w^{c_2 \cdot 2.303} - \overline{M}_n^{c_2 \cdot 2.303}\right) + \overline{M}_n$$

where $\overline{M}_w$ and $\overline{M}_n$ are the weight average and number average mole weights. The rotational coupling factors are given in terms of the reduced density ρ_{rm} by:

$$\alpha_m = 1 + c_5 \cdot \rho_{rm}^{c_3} \cdot M_{wm}^{c_4}$$

$$\alpha_o = 1 + c_6 \cdot \rho_{rm}^{c_3}$$

The coefficients $c_1, \dots, c_6$ can be defined by the [PEDTUNE](#) and [PEDTUNER](#) keywords. Note that the default value of c_6 is related to the default values of c_4 and c_5 by

$$c_6 = c_5 \cdot M_{wo}^{c_4}$$

Three parameter equations of state

The traditional weakness of the two-parameter equations of state (EoS), such as the Peng-Robinson, Redlich-Kwong for example, is their poor prediction of liquid properties, especially liquid densities and saturations.

The critical compressibilities predicted by the Van der Waals, Redlich-Kwong and Peng-Robinson equations of state are $Z_c^{VdW} = 0.375$, $Z_c^{RK} = 0.333$ and $Z_c^{PR} = 0.307$. But hydrocarbons are known to have $Z_c < 0.29$.

Peneloux *et al.* proposed a molar volume correction, referred to as volume shift, which adds a third parameter to the EoS, and, in turn, greatly improves liquid property estimation.

For a mixture of N components, the phase molar volume $V_{mol, p}$ is given by:

$$V_{mol, p} = V_{mol, p}^{EoS} - \sum_{i=1}^N z_i c_i \quad \text{Eq. 4.114}$$

where

p = (liquid, vapor) represents the phase of the system

$V_{mol, p}^{EoS}$ is the molar volume of the phase predicted by the traditional 2-parameter EoS

$z_i = (x_i, y_i)$ are the liquid and vapor mole compositions

c_i constitute a set of volume corrections.

The component corrections are usually related to a set of dimensionless shift parameters, s_i , by:

$$s_i = \frac{c_i}{b_i} \quad \text{Eq. 4.115}$$

where:

$$b_i = \frac{\Omega_{b,i} RT_{ci}}{p_{ci}} \quad \text{Eq. 4.116}$$

The shift parameters are entered in the dimensionless form using the keyword [SSHIFT](#).

Use of the equation of state for hydrocarbon mixtures

A typical oil is composed of many millions of components. The equation of state has one value of A and one value of B to model the phase behavior of this mixture.

For example, the PR EoS is defined as follows:

$$P = \frac{RT}{V-B} - \frac{A}{V(V+B) + B(V-B)} \quad \text{Eq. 4.117}$$

In a typical reservoir simulation model, 4-8 components and pseudo-components are used to define the mixture. Each component has a value of T_c , P_c , V_c or Z_c , ω , Ω_a , Ω_b and binary interaction coefficients, δ_{ij} (that increase or decrease the forces between component pairs), that must be specified in the fluid characterization process.

These parameters are for pure components, such as Methane or Nitrogen, and are usually considered fixed. Thus, you must alter the remaining component parameters so that the resulting calculated phase behavior matches the experimental data. This phase behavior matching process is carried out with the PVT i program.

Calculation process

This section outlines the calculation process used by Eclipse 300 to determine phase compositions, and the amount of liquid and vapor present in a grid block, at equilibrium.

The Eclipse 300 calculation process utilizes linear, non-linear (Newton) and flash iterations to obtain a converged solution. The final solution at the end of a timestep for each grid block consists of: a pressure; water, oil and gas saturations; liquid and vapor molar fractions; and mole fractions of each component in the liquid and vapor phase.

The remainder of this section describes the iteration process used to flash the mixture in each grid block, to determine the molar volumes and the phase compositions at equilibrium.

After the linear problem converges, there are new pressure, water and molar densities for each grid block.

A stability test tells us if there are one or two hydrocarbon phases. If there are two hydrocarbon phases then K-values, K_i , are assigned to each component, i . These values can be estimated from the values used in the previous iteration, using equation 4.126, or, when no previous values exist, from Wilson's formula:

$$K_i = \frac{e^{\left[5.37(1 + \omega_i)\left(1 - \frac{1}{T_{ri}}\right)\right]}}{P_{ri}} \quad \text{Eq. 4.118}$$

Then, given z_i and K_i , Eclipse 300 solves the Flash Equation to obtain the molar fraction of vapor, V . The flash equation solved by Eclipse 300 is:

$$g(V) = \sum_{i=1}^{N_{\text{comps}}} \frac{z_i(K_i - 1)}{1 + V(K_i - 1)} = 0 \quad \text{Eq. 4.119}$$

Eclipse 300 solves this equation to obtain the mole fractions of each component in the liquid and vapor phases:

$$x_i = z_i / [1 + V(K_i - 1)], \quad i = 1, 2, \dots, N_{\text{comps}} \quad \text{Eq. 4.120}$$

$$y_i = (K_i z_i) / [1 + V(K_i - 1)], \quad i = 1, 2, \dots, N_{\text{comps}} \quad \text{Eq. 4.121}$$

Using these mole fractions, the Peng-Robinson equation of state can then be expressed as a cubic equation of the compressibility factor, Z :

$$Z^3 - Z^2(1-b) + Z(a-3b^2-2b) - (ab-b^2-b^3) = 0 \quad \text{Eq. 4.122}$$

where

Z is defined by $Z = \frac{PV}{RT}$ and the two remaining parameters are defined by:

$$a = \frac{AP}{R^2T^2} \quad \text{Eq. 4.123}$$

$$b = \frac{BP}{RT} \quad \text{Eq. 4.124}$$

The solution of this cubic equation gives, in a two phase region, three real roots. The largest root is the compressibility factor of the vapor phase, and the smallest positive root is the compressibility factor of the liquid phase.

With this information, Eclipse 300 calculates the fugacity of each component in the liquid and vapor phases.

When a grid block is in equilibrium at a given pressure and temperature, the fugacities of all components must be the same in the liquid and the vapor. In other words:

$$f_{Li} = f_{Vi}$$

A convergence criterion is used to determine if a grid block is in equilibrium (if ε is very small),

$$\sum_{i=1}^{N_{\text{comps}}} \left(\frac{f_{iL}}{f_{iV}} - 1 \right)^2 < \varepsilon \quad \text{Eq. 4.125}$$

If equilibrium is not achieved, then new estimates for the K-values are made and the iteration process continues with a new solution of the flash equation, etc. Using successive substitution, the new estimate of the K-value is:

$$K_i^{\text{NEW}} = K_i^{\text{OLD}} \cdot \left[\frac{f_{iL}}{f_{iV}} \right] \quad \text{Eq. 4.126}$$

After equilibrium is achieved, Eclipse 300 begins the next non-linear iteration.

Calculation of phase states

A cell in a compositional simulation may be either water-filled, or in a single-phase or two-phase hydrocarbon state. The single-phase states may possess a saturation pressure, usually lower than the current pressure, at which a second hydrocarbon phase appears. For temperatures above the cricondentherm however, such a pressure does not exist, and the cell composition leads to a single-phase hydrocarbon at all pressures.

The appearance of a two-phase state is detected by application of the Michelsen stability test ([Michelsen, 1982](#)). This is carried out anew at each timestep, and does not involve assumptions concerning the state of the cell carried over from previous steps.

By default Eclipse 300 uses the stability test for all cells, avoiding saturation pressure tracking altogether.

The Michelsen stability test involves the minimization of a function G^* , as a function of a trial phase composition Y , starting from both liquid and vapor-like initial estimates. If a non-trivial minimum can be found with $G^* < 0$ the hydrocarbon mixture is unstable, and a flash calculation may be performed to establish the phase split.

It is possible to cast the Michelsen stability test in a residual form and use a Newtonian method, but this is not entirely suitable, as:

- There are generally two minima to G^* , and the Newton method can converge towards a local maximum as easily as a minimum.
- Newton's method involves order N^3 operations per iteration, due to the matrix operations involved. As a number of iterations are required from each initial Y , this becomes expensive.

A better method, which also yields quadratic convergence near the solution, is the BFGS iteration ([Dennis and Schnabel, 1996](#)). This is a minimization method which uses an approximate Hessian that is constrained to be symmetric and positive definite. In the early stages of the iteration an accelerated successive substitution method, which uses a two-point GDEM technique ([Crowe and Nishio, 1975](#)), is effective.

Fast flash option

In some cases a rigorous stability test may not be necessary. In the fast flash option, the simulator checks for single phase cells in the middle of single phase regions. The stability calculation is not performed for these cells. Instead, the state is assumed to remain the same, as long as the state of the neighboring cells is unchanged. This can lead to a considerable improvement in simulator speed - depending on the proportion of single phase cells in the simulation.

The fast flash option can be invoked using the [SKIPSTAB](#) keyword. It can be switched off and on using [item 60](#) of the [OPTIONS3](#) keyword. [Item 109](#) of the [DEBUG3](#) keyword can be used to gauge the usefulness and accuracy of the method for a particular data set.

Equations of state regions

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x	Eclipse 300

It is possible to set up more than one equation of state description, and associate these with sets of cells using the [EOSNUM](#) keyword in the REGIONS section. The most common application of this is a study containing two or more disconnected reservoirs, coupled through the production system.

The number of reservoir equations of state is set by [NMEOSR](#) in the TABDIMS keyword, and the number of surface equations of state is set by [NMEOSS](#), also in TABDIMS. Both default to 1. If a value is entered for NMEOSR it becomes the default for NMEOSS. There is a limit of 10 on both settings.

Reservoir equation of state

For each reservoir equation of state (EoS), Eclipse 300 expects information corresponding to the following keywords:

Restriction	Keyword
	RTEMP or TEMPVD
	EOS
	TCRIT
	PCRIT
	VCRIT or ZCRIT
optional - default VCRIT or ZCRIT	VCRITVIS or ZCRITVIS
	MW
	ACF
	OMEGAA
	OMEGAB
optional - default 0	SSHIFT
ZJ equation only	TBOIL
ZJ equation or Thermal	TREF
ZJ equation or Thermal	DREF or GREF
Thermal only	CREF
Thermal only	PREF
Thermal only	THERMEX1
optional - miscible only	PARACHOR
	BIC
Thermal only	OILVISC
Thermal only	OILVISC
Thermal only	GASVISC
Thermal only	HEATVAP
Thermal only	HEATVAPE

Restriction	Keyword
Thermal only	SPECHA
Thermal only	SPECHG
Thermal only	KVCR
Thermal only	ZFACTOR

This information is provided by entering the required data [NMEOSR](#) times. In the case of [NMEOSR](#)=2, for example:

```
TCRIT
548.4 227.16 343.0 549.7 665.64 806.54 838.1 1058.0 1291.8 /
548.4 227.16 343.0 549.7 665.64 808.3 860.6 1107.5 1402.0 /
```

The same number of components must exist in all equation of state regions, and they must have the same names. The most common practice is to alter the properties of the heavy components when, for example, performing regressions to a number of reservoir fluids.

Having defined the EoS properties to be associated with the EoS regions, they are associated with cells using [EOSNUM](#), which acts like [SATNUM](#). For example, in the case of a 35 x 14 x 12 study, with the top 4 layers using EoS1 and the bottom 8 using EoS2:

```
EOSNUM
490*1 490*1 490*1 490*1
490*2 490*2 490*2 490*2 490*2 490*2 490*2 490*2 /
```

Alternatively, the [EQUALS](#) keyword may be used in the usual manner:

```
EQUALS
EOSNUM 1 4* 1 4 /
EOSNUM 2 4* 5 12 /
/
```

If cells with different EoS region numbers are in the same equilibration region, the initial state is not generally in hydrostatic equilibrium, as the EoS determines the phase densities. The code issues a warning if this occurs.

Surface equation of state

Turning to the surface facilities, the default value for [NMEOSS](#) in keyword [TABDIMS](#) is the same as [NMEOSR](#); that is, there are the same number of surface equations of state as there are reservoir equations of state.

The properties of each surface EoS default to those of the corresponding reservoir equation of state, but any or all of these may be overwritten using the following keywords:

Restriction	Keyword
	EOSS
	TCRITS
	PCRITS
	VCRITS or ZCRITS
	EOSS

Restriction	Keyword
	MWS
	ACFS
	OMEGAAS
	OMEGABS
optional - default 0	SSHIFTS
ZJ equation only	TBOILS
ZJ equation or Thermal	TREFS
ZJ equation or Thermal	DREFS or GREFS
Thermal only	PREFS
	BICS
parachors and viscosity values are not employed in surface calculations	RTEMP

Items in these keywords default to the reservoir equation of state values. For example, to modify just the last two [TCRIT](#) values, with NMEOSS=2, it is possible to enter:

```
TCRITS
7* 1078.0 1345.0 /
7* 1067.0 1356.0 /
```

If NMEOSR is set to a value less than NMEOSS, then the last (NMEOSS-NMEOSR) surface EoS values are taken from the last (NMEOSRth) EoS.

When performing separator calculations, a choice of surface equation of state must be made. By default, the flash calculations for a well use the EoS number corresponding to the cells in which that well is completed. For example, a well completed in cells in reservoir EoS region 2, uses surface EoS2. If a well is completed in cells in more than one reservoir EoS region, a warning is issued, and the first value encountered is used.

It is possible to over-ride this choice of surface EoS using the keywords [FIELDSEP](#) and [SEPCOND](#). The eighth argument of [FIELDSEP](#) and the tenth argument of [SEPCOND](#) may be used to specify a non-default EoS number to be used in the separator fluid-in-place and surface rate calculations, respectively.

By default, [FIELDSEP](#) obtains separator volumes in place for the whole field and for fluid-in-place regions using the equation of state specified for the cell, as specified with [EOSNUM](#).

Asphaltene option

	Eclipse 100
x	Eclipse 300

The purpose of this option is to model asphaltene formation and its impact on reservoir performance. There are a number of asphaltene models currently in existence, but there is still no consensus about the characterization of asphaltene behavior. The asphaltene option implements a simplified but practical model that provides sufficient flexibility to simulate a wide range of processes associated with asphaltene behavior.

The asphaltene modeling option is activated with the [ASPHALTE](#) keyword in the RUNSPEC section.

Asphaltene formation

The following terms describe asphaltene formation and may be different to those in the literature:

- precipitation: where asphaltene particles (“fines”) appear in the fluid
- dissolution: the reverse of precipitation, where fines dissolve back into the oil
- flocculation: where the precipitated fines aggregate to form bigger particles, (“flocs”)
- dissociation: the reverse of flocculation; where flocs split up into fines
- deposition: where flocs leave the fluid and attach to the reservoir rock
- entrainment: the reverse of deposition, where deposited asphaltene becomes suspended in the fluid.

Asphaltene damage

Asphaltene precipitation and deposition can alter the reservoir flow in a number of ways. It is possible to model the effects of asphaltene on the following reservoir and fluid properties:

- porosity
- permeability
- viscosity
- wettability

Pre-2011.1 model

A modified asphaltene model was introduced in the 2011.1 version, called the three-component model, since it uses separate components to represent the precipitation, flocculation and deposition states of the asphaltene. The pre-2011.1 model is still available, and both are described in this chapter. The main differences are:

- The three-component model simplifies the precipitation model by allowing only one component to represent the asphaltene in the oil phase (precipitate).
- The three-component model uses an extra solid component to model deposited asphaltene. This leads to some differences in the way the deposition terms are calculated. There are also slight differences in the way asphaltene deposition data is entered in keywords [ASPDEPO](#), [ASPLCRT](#) and [ASPWETF](#). The solid (deposition) component does not take part in the flash, and the solid properties need to be specified.

The value of the solid density in the three-component model is important as it affects the reservoir pressure. If it is less than or greater than the equivalent density of the precipitate in the oil phase, then as asphaltene is deposited the reservoir pressure will increase or decrease to preserve the fluid volume.

In the pre-2011.1 model, deposited asphaltene is still considered as part of the oil phase, so the oil properties do not change as asphaltene is deposited. In the three-component model, deposited asphaltene is removed from the oil phase, leading to a lighter oil and increased volatility. This in turn will affect the reservoir pressure if gas comes out of solution.

- The three-component model no longer uses the pre-2011.1 asphaltene permeability damage model, keyword [ASPKDAM](#). Instead a solid-dependent mobility multiplier can be used.

Asphaltene formation: three-component model

Characterization

Three special components can be defined using the [CATYPE](#) keyword:

- A precipitation component. This component represents the oil components that can become asphaltene. A fraction of the component concentration is identified as being precipitated fines.
- A floc component.
- A deposited asphaltene component. This component must also be a solid component, see ["Solid Phase"](#). This component is only required if asphaltene deposition is modeled.

Precipitation - dissolution

The asphaltene precipitation component can be either dissolved in oil or precipitated as a fine. The [ASPREWG](#) or [ASPPW2D](#) keyword can be used to define the maximum amount of the asphaltene that can be dissolved W_{lim} , which is expressed as the weight of the precipitation component in oil as a percentage of the total weight of oil. This can be expressed as a limit on the mole fraction of the precipitation component dissolved in oil:

$$x_{lim} = \frac{W_{lim}}{100} \cdot \frac{m_{woil}}{m_{wp}} \quad \text{Eq. 4.127}$$

where $m_{woil} = \sum m_{wi} \cdot x_i$ is the molecular weight of the oil.

If the mole fraction of the precipitation component is less than this value, $x_p < x_{lim}$, the precipitation component will be totally dissolved and there will be no fines. If the mole fraction of the precipitation component exceeds this value, $x_p > x_{lim}$, the dissolved and fine fractions of the precipitation component are given by:

$$x_{pd} = x_{lim}$$

$$x_{pf} = x_p - x_{lim}$$

The fines take part in the flocculation process.

Flocculation - dissociation

Flocculation and dissociation are modeled by two reactions. In the flocculation reaction the fines are converted into the flocculation component, with a rate given by:

$$R_f = V_f \cdot r_f \cdot \frac{m_{wp}}{m_{woil}} \cdot C_{pf} \quad \text{Eq. 4.128}$$

where

$V_f = (1 - S_s) \cdot V_p$ is the fluid volume

V_p is the pore volume

S_s is the solid saturation, representing the fraction of the pore volume filled with asphaltene deposit

$C_{pf} = S_o \cdot B_o \cdot x_{pf}$ is the concentration of the fines in the oil phase

S_o is the oil saturation

B_o is the oil molar density

r_f is the flocculation rate specified by the [ASPFLRT](#) keyword.

In the dissociation reaction, the flocculation component is converted into fines, with a rate given by:

$$R_{dis} = V_f \cdot r_{dis} \cdot C_f \quad \text{Eq. 4.129}$$

where

$C_f = S_o \cdot B_o \cdot x_f$ is the concentration of the flocs in the oil phase

r_{dis} is the dissociation rate specified by the [ASPFLRT](#) keyword.

Deposition - entrainment

Deposited asphaltene is modeled by a solid component. Deposition is modeled by a pair of reactions between the asphaltene floc component and the deposited asphaltene component.

Deposition

The first (forward) reaction models adsorption and plugging. The rate at which asphaltene floc is converted to deposit is given by:

$$R_d = \alpha \cdot V_f \cdot C_f + \gamma_2 \cdot \overline{F_o} \cdot L \cdot (C_f - C_{fcr})^+ \quad \text{Eq. 4.130}$$

where

α is the adsorption or static deposition coefficient, specified by the [ASPDEPO](#) keyword

γ_2 is the plugging coefficient, specified by the fifth item of the [ASPDEPO](#) keyword

$C_f = S_o \cdot B_o \cdot x_f$ is the concentration of the flocs in the oil phase (flowing flocs)

C_{fcr} is the critical concentration of flocs for plugging to occur, specified by the [ASPLCRT](#) keyword

$\overline{F}_o$ is the average oil flow rate

L is the length of the flow path

and the equation uses the notation

$$(x)^+ = \max(x, 0)$$

Equation 4.130 can be written in terms of $\overline{U}_o$, the oil phase average velocity:

$$R_d = V_f \cdot [\alpha \cdot C_f + \gamma_2 \cdot \overline{U}_o \cdot (C_f - C_{fcr})^+] \quad \text{Eq. 4.131}$$

where

$$\overline{F}_o \cdot L = \overline{U}_o \cdot V_f$$

Plugging only occurs if the floc concentration exceeds the critical floc concentration $C_f > C_{fcr}$ and if the volume fraction of deposited asphaltene, represented by the solid saturation S_s , exceeds a critical value $S_s \geq S_{cr}$, where C_{fcr} and S_{cr} are specified by the [ASPLCRT](#) keyword. The half-life of the reaction gives an indication of how large a timestep the simulator can take. Ignoring the C_{fcr} term in equation 4.131, the half-life of the deposition reaction is

$$T_{1/2} = \ln 2 / [\alpha + \gamma_2 \cdot \overline{U}_o] \quad \text{Eq. 4.132}$$

Entrainment

The second (backward) reaction models entrainment. The rate at which the deposited asphaltene is converted to the floc component is given by:

$$R_e = V_f \cdot \beta \cdot (U_o - U_{cr})^+ \cdot S_o \cdot S_s \quad \text{Eq. 4.133}$$

where

β is the entrainment coefficient, specified by the [ASPDEPO](#) keyword

U_{cr} is the critical velocity for entrainment, specified by the [ASPDEPO](#) keyword.

The average oil flow rate $\overline{F}_o$ and average oil velocity $\overline{U}_o$ are both calculated at the previous timestep. First the total oil flow rates in each direction (F_{ox} , F_{oy} and F_{oz}) and flow areas (A_x , A_y and A_z) are calculated by summing the flows and areas on opposite faces, and including any non-neighbor connection contributions:

$$F_{on} = \sum \left| T \cdot \frac{k_{ro}}{\mu_o} \cdot \Delta P \right| \quad \text{Eq. 4.134}$$

$$A_n = \sum A \quad \text{Eq. 4.135}$$

Here the flow is calculated from the inter-cell transmissibility T and pressure difference ΔP , and the cell oil relative permeability k_{ro} and viscosity μ_o . The average velocity in each direction is given by:

$$U_{on} = F_{on} / A_n \quad \text{Eq. 4.136}$$

The average flow is determined by taking the average over the N dimensions:

$$\overline{F}_o = (F_{ox} + F_{oy} + F_{oz}) / (2N) \quad \text{Eq. 4.137}$$

The phase average velocity $\overline{U}_o$ is calculated using the root mean square of the direction velocities:

$$\overline{U}_o = \sqrt{(U_{ox}^2 + U_{oy}^2 + U_{oz}^2)} \cdot \frac{C}{\phi \cdot (1 - S_s) \cdot S_o} \quad \text{Eq. 4.138}$$

where

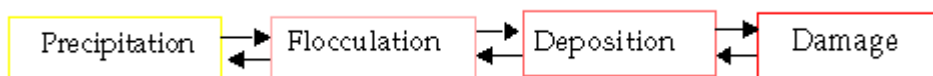
ϕ is the porosity

C is a conversion factor which is used to convert volumes from reservoir barrels to cubic feet if the simulation is in field units.

These flow and velocity calculations only take into account the flows between grid blocks. Well terms are not included - except through their effect on the flow into the grid block. This is discussed further in ["Velocity calculations"](#).

Asphaltene formation: Pre-2011.1 model

Asphaltene modeling is decomposed into different stages:



The double arrow indicates reversibility (partial or total). Precipitation triggers the sequence of flocculation deposition and damage, including a viscosity effect, as detailed below.

Asphaltene characterization and precipitation

The precipitation, flocculation, deposition and damage processes are modeled using separate keywords. The basis of this model is that asphaltene is characterized by the weight percentage of component(s) making up the asphaltene that can remain dissolved in the oil as a function of pressure.

Asphaltene as weight percentage

In this approach the asphaltene is defined as a set of component(s) that can precipitate depending on their percentage molar weight in the solution. You define the percentage limit as a function of pressure using the [ASPREWG](#) keyword.

This limit can be set against the pressure or temperature or molar fraction of a specified component using the [ASPP1P](#) keyword. The default setting is against pressure.

This limit can also be set as a function of two variables chosen from pressure (P), temperature (T) and molar fraction (Z). This is specified using keywords [ASPP2P](#) and [ASPPW2D](#). Keyword [ASPP2P](#) sets the variables (P,T) or (P,Z) or (T,Z). If the molar fraction Z is chosen the 3rd argument of [ASPP2P](#) must be the component number. Keyword [ASPPW2D](#) takes a 2D table where the first row represents the temperature if (P,T) is chosen, and the molar fraction if (P,Z) or (T,Z) is chosen. The first column of the 2D table represents the pressure if (P,T) or (P,Z) is chosen and the temperature if (T,Z) is chosen.

Note that the temperature can be used as a precipitation triggering variable only in thermal runs.

The amount of precipitate corresponds to the excess of a specified component in the oil phase with respect to the limit that you define.

The ASPREWG keyword like the ASPPW2D keyword is used in conjunction with the ASPFLOC keyword, which is used to specify the range of components that can precipitate. You can use this approach to model some aspects of the polydispersity that have been reported in the literature. You can also interpret it to some extent as a dissolution model, where the oil phase can only dissolve a certain amount of asphaltene component(s) at a given pressure, temperature or molar fraction.

Note: The precipitation process does not change the fluid composition. It only flags the fraction of a precipitate component that will contribute to the flocculation process.

The list of components is enumerated as $1, 2, \dots, i, i+1, \dots, j, \dots, k, \dots, N$. If the first component number to precipitate is i , the last component is j , and k is the flocs component then the dissolved fraction of asphaltene is simply:

$$w_{adis} = \frac{\sum_{l=i}^j m_{wl} x_l}{\sum_{l=1}^N m_{wl} x_l} \quad \text{Eq. 4.139}$$

and the total fraction of asphaltene (dissolved and flocs) is calculated as:

$$w_{apre} = \frac{m_{wk} x_k + \sum_{l=i}^j m_{wl} x_l}{\sum_{l=1}^N m_{wl} x_l} \quad \text{Eq. 4.140}$$

where

m_{wl} is the molecular weight of component l

x_l is the liquid fraction of component l

At a given pressure, temperature or concentration the precipitate fraction coming out from the dissolved asphaltene gives the excess of w_{adis} with respect to user limit input entered using ASPREWG keyword, as:

$$E = w_{adis} - asp_{lim}(P)$$

The $asp_{lim}(P)$ limit is interpreted as a fraction of asphaltene component(s):

$$asp_{lim}(P) = \frac{\sum_{l=i}^j m_{wl} x_l^a}{\sum_{l=1}^N m_{wl} x_l} \quad \text{Eq. 4.141}$$

and the composition of the precipitate is calculated by assuming that x_l^a is the same for all the asphaltene component(s), which means that

$$x_l^a = asp_{lim}(P) \frac{\sum_{l=1}^N m_{wl} x_l}{\sum_{l=i}^j m_{wl}} \quad \text{Eq. 4.142}$$

Hence the fraction of precipitate from the dissolved asphaltene writes:

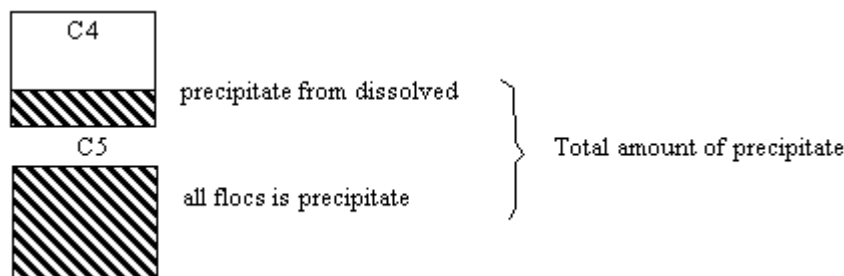
$$E = \frac{\sum_{l=i}^j m_{wl} (x_l - x_l^a)}{\sum_{l=1}^N m_{wl} x_l} \quad \text{Eq. 4.143}$$

The second argument of ASPREWG can take values from 0 (no dissolved asphaltene component(s)) to 100 (no precipitate), however the actual upper can be estimated from the expression of w_{adis} if the molar fraction(s) of the precipitating component(s) is known. The $asp_{lim}(P)$ limit and w_{adis} can be output to

check out the evolution of precipitate fraction. A list of outputs for the precipitation process is provided in ["Outputs"](#).

An example

Let consider a case with 5 components, and assume component 4 is precipitating and component 5 is the flocs component, then the amount of precipitate is calculated as depicted below:



Asphaltene flocculation-dissociation

Flocculation is the process whereby fines obtained from the precipitation process aggregate to form bigger particles referred to as flocs. This process is modeled by a set of two kinetic reactions to allow reversibility (partial or total) between aggregation of the fines into flocs and dissociation of the flocs into fines.

Let C_i denote the concentration of the fines coming from component i and C_a the concentration of the flocs. Once precipitation occurs, the aggregation rate of the fines i into flocs a is modeled by:

$$R_a = \frac{\partial C_a}{\partial t} = r_{ia}C_i - r_{ai}C_a \quad \text{Eq. 4.144}$$

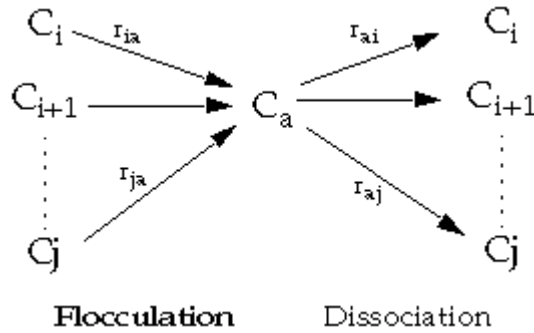
where

r_{ia} is the flocculation rate coefficient of the fines

and r_{ai} is the dissociation rate coefficient of the flocs.

The range of components that precipitate and the flocs component a are specified by the [ASPFLOC](#) keyword. This keyword is used in conjunction with the [ASPFLRT](#) keyword that specifies rates r_{ia} and r_{ai} . This process is depicted in the figure below.

In the case where the asphaltene is seen as single pure component, this flocculation reduces to two kinetic reactions only.



Data input

The floculation-Dissociation process can generate significant changes in the composition, which may result in un-physical behavior if data input is not carefully set up. Given that the flocs component is a result of the floculation process, it is expected that the flocs properties (molar weight, critical properties, and other EOS parameters) to be some average values of the components that precipitate. These average values should also take into account the floculation rates to have the right weight contribution of the different components that precipitate.

You should provide floculation properties that are consistent with the components that precipitate. A helpful tool to provide an estimate of the floculation properties would be a PVT package with a grouping capability that derives the properties of a pseudo-component from its parts.

As the composition of the oil can be changed by the floculation and/or deposition process(es), the oil/gas densities will change, and therefore the pressure too.

Asphaltene deposition

Deposition represents the process whereby the flocs are exchanged between the solution (oil phase) and the rock surface. The flocs can adsorb on the rock surface, can be trapped within the porous media because of their size (plugging) or can be entrained and returned to the oil phase because of high, local velocity (shear).

Note: Changes in the rock density are not taken into account as they are not thought to contribute to the process at this stage.

The deposition in the flow direction i is modeled as follows

$$\frac{\partial \varepsilon_i}{\partial t} = \frac{\alpha}{d} \phi C_a + \gamma_1 |F_{oi}| C_a - \beta (|U_{oi}| - U_{cr})^+ \varepsilon_i \quad \text{Eq. 4.145}$$

where

ε_i is the volume fraction of deposit in the i direction of the flow

α is the adsorption or static deposition coefficient

d is the dimension of the problem (1, 2 or 3)

ϕ is the current porosity (time t)

C_a is the volumetric concentration of the flocs in the oil phase (flowing flocs)

γ_1 is the plugging coefficient

F_{oi} is the oil Darcy flux

β is the entrainment coefficient

U_{oi} is the oil phase velocity ($F_{oi}/(A\varphi)$), A is the section area between connecting cells

U_{cr} is the user-input critical velocity.

The overall volume fraction deposit is the sum of the deposits in each direction i :

$$\varepsilon = \sum_{i=1}^d \varepsilon_i \quad \text{Eq. 4.146}$$

Assuming a very small binary interaction between the flocs component and the other components, the volumetric flocs concentration is taken as:

$$C_a = Z_a = n_a / \left(\sum_{i=1}^{N_{\text{comps}}} n_i \right) \quad \text{Eq. 4.147}$$

In the pre-2010.2 version, the flocs mole fraction that deposits is taken as: $n_a^{dp} = \varepsilon n_a^f$, and the flocs mole fraction that remains in solution is corrected to: $n_a^f = n_a^f - n_a^{dp}$.

The concentration of the flowing part of the flocs is calculated as: $C_a = \frac{n_a^f S_o x_a}{b_o}$

where

b_o is the oil molar density,

S_o the oil saturation and

x_a the molar fraction of the flocs in the liquid phase.

In some cases, this approach tends to overestimate the deposition level. [Item 242](#) of the `OPTIONS3` keyword can be used to revert to this pre-2010.2 behavior. Note also that [item 231](#) of `OPTIONS3` can be used to revert to the pre-2010.1 deposition behavior.

Input parameters α , γ , β and U_{cr} are user input, which you specify using the `ASPDEPO` keyword in the `PROPS` section. You should get these parameters from core flood experiments.

Plugging control

Plugging can be controlled by using the `ASPLCRT` keyword to specify a critical volume fraction of asphaltene deposit ε_{cr} and a critical concentration of flocs C_{acr} above which the plugging happens. If the `ASPLCRT` keyword is absent the pre-2008.1 modeling is recovered. It actually corresponds to using the defaults values of $\varepsilon_{cr} = 0$ and/or $C_{acr} = 0$. The transport equation is then slightly altered as

$$\frac{\partial \varepsilon_i}{\partial t} = \frac{\alpha}{d} \varphi C_a + \gamma_1 |F_{oi}| (C_a - C_{acr})^+ - \beta (|U_{oi}| - U_{cr})^+ \varepsilon_i \quad \text{Eq. 4.148}$$

Basically the coefficient γ is set as:

$$\gamma_1 = \begin{cases} \gamma_1 & \text{if } \varepsilon > \varepsilon_{cr} \text{ and/or } C_a > C_{acr} \\ 0 & \text{if not} \end{cases} \quad \text{Eq. 4.149}$$

This means that plugging will occur only if the flocs concentration gets above the critical value and/or the volume fraction deposit gets bigger than the critical value. Note that when the concentration gets above critical only the concentration difference will contribute to the plugging.

The critical volume fraction of asphaltene deposit can be interpreted as a reduction in the average pore radius. Using the empirical correlation and a critical radius can be estimated as

$$r \sim \sqrt{\frac{K}{\phi}} \rightarrow r_{cr} \sim r_0 \left(1 - \frac{\varepsilon_{cr}}{\phi_0}\right)^{\frac{\delta-1}{2}} \quad \text{Eq. 4.150}$$

so the remain flowing radius is $r = r_0 - r_{cr}$. Exponent δ is the same as in below. The critical floc concentration can be interpreted as a likelihood that above that value the size of some of the flowing flocs is big enough to plug some pore throats.

The `ASPLCRT` keyword takes three arguments: the critical volume fraction of asphaltene deposit ε_{cr} , the critical concentration of the flocs C_{acr} and a third argument equal to “AND” or “OR” for the combination of the two first arguments.

Asphaltene damage (both deposition models)

Asphaltene damage represents the effects of asphaltene on the reservoir and reservoir flow.

Porosity

Asphaltene deposition reduces the fluid volume in the reservoir. In the solid deposition model this effect is modeled by the solid saturation. In the pre-2011.1 model the porosity is reduced by asphaltene deposition:

$$\varphi = \varphi_0 - \int_0^t \frac{\partial \varepsilon}{\partial t} dt \quad \text{Eq. 4.151}$$

where φ_0 is the initial porosity, is ε the volume fraction of asphaltene deposit.

Permeability

In the three-component deposition model, solid is deposited and the effects of deposition on permeability can be modeled by using a solid-dependent mobility multiplier. See keyword `SOLIDMMS` or `SOLIDMMC`.

In the pre-2011.1 model the effects of deposition on permeability can be modeled in one of two ways, using the `ASPKDAM` keyword.

- If the rock permeability is correlated to the porosity, a reduction in permeability can also be taken into account using a parametrized power law relationship given the ratio of the permeability K at time t with respect to the initial permeability K_0 :

$$\frac{K}{K_0} = \left(1 - \frac{\varepsilon}{\varphi_0}\right)^\delta \quad \text{Eq. 4.152}$$

where δ is a user input (based on core experiment data).

To use this model you need to specify PORO in the second argument of [ASPHALTE](#) keyword, and provide the exponent with the [ASPKDAM](#) keyword.

- Alternatively, if the rock permeability is independent of the porosity, or data giving a relationship between the permeability and the amount of asphaltene deposit are available, this can be directly used within a lookup table. In that case the second argument of the [ASPHALTE](#) keyword should be set to “TAB” and the [ASPKDAM](#) keyword takes a table as an argument where the first column represents the volume fraction of asphaltene deposit and the second column is the corresponding permeability multiplier.

Oil viscosity

When precipitation occurs, asphaltene fines can be considered as colloids in the oil phase, which can alter the viscosity of the transporting phase, that is the oil phase.

Using the [ASPVISO](#) keyword in conjunction with item 3 of the [ASPHALTE](#) keyword, you can model the viscosity alteration as a function of the mass fraction of precipitate in the oil C_p . This includes both the fines and the floc component:

$$C_p = \frac{m_{wp} \cdot x_{pf} + m_{wf} \cdot x_f}{m_{woil}}$$

Three different models are available:

- Generalized Einstein model (one parameter)

$$\frac{\mu}{\mu_0} = 1 + aC_p \quad \text{Eq. 4.153}$$

where the default for a is $a = 2.5$, and μ_0 is the oil viscosity at $C_p = 0$. You can specify the coefficient a using [ASPVISO](#).

- Krieger and Dougherty model (two parameter)

$$\frac{\mu}{\mu_0} = \left(1 - \frac{C_p}{C_{p0}}\right)^{-\eta C_{p0}} \quad \text{Eq. 4.154}$$

where:

η is the intrinsic viscosity, $\eta = 2.5$ for spherical colloids

C_{p0} is the concentration for maximum packing.

You can specify the coefficient η and the maximum concentration C_{p0} using [ASPVISO](#).

- Lookup table that gives the oil viscosity multiplier as a function of the mass fraction of the asphaltene precipitate. The first column is the volume fraction and the second column is the viscosity multiplier. You can set the maximum number of rows with [MXRASO](#) in the [TABDIMS](#) keyword.

Depending on the model chosen, the [ASPVISO](#) keyword takes the appropriate number of entries.

Wettability change

Asphaltene deposition can trigger a wettability change. The exact physics on how this alteration occurs is still a research topic, however it has been reported that some of its effect can be captured by a relative permeability shift from a water-wet system to an oil-wet system.

It is assumed that the wettability can be determined from the water-oil capillary pressure values: if the water-oil capillary pressure is zero or negative, the grid cell is assumed to be oil-wet and there will be no wettability change.

If the grid cell is water-wet, that is it has a positive water/oil capillary pressure, the relative permeabilities (entered using, for example, [SWFN](#) and [SOF3](#)) are combined with the oil-wet saturation functions entered using the [ASPKROW](#) keyword as follows:

1. The weighting factor F is determined as a function of the volume fraction of asphaltene deposit (S_s or ϵ), provided using the [ASPWETF](#) keyword.
2. The critical oil in water and water saturations are scaled

$$S_{ocr} = FS_{ocra} + (1-F)S_{ocri}$$

$$S_{wcr} = FS_{wcra} + (1-F)S_{wcric}$$

where S_{ocra} and S_{ocri} are the critical oil in water saturations from the [ASPKROW](#) table and [SOF3](#) table respectively, and S_{wcra} and S_{wcric} are the critical water saturations from the [ASPKROW](#) table and [SWFN](#) table respectively.

3. Then a lookup for relative permeability is carried out on the scaled saturations from step 1, followed by a linear interpolation between the resulting water-wet and oil-wet relative permeabilities as:

$$k_{rw} = Fk_{rwa} + (1-F)k_{rwi}$$

$$k_{row} = Fk_{rowa} + (1-F)k_{rowi}$$

where k_{rwa} and k_{rwi} are the oil-wet and water-wet relative permeabilities to water respectively (from [ASPKROW](#) and [SWFN](#)), and k_{rowa} and k_{rowi} are the oil-wet and the water-wet relative permeabilities to oil respectively (from [ASPKROW](#) and [SOF3](#)).

The number of tables in the [ASPKROW](#) keyword is the same as the number of saturation tables (set by [NTSFUN](#) in the [TABDIMS](#) keyword), and the maximum number of rows in each table is also the same as for the saturation tables (set by [NSSFUN](#) in the [TABDIMS](#) keyword).

The number of tables in the [ASPWETF](#) keyword is the same as the number of saturation tables, and the maximum number of rows in each table is set by [MXRATF](#) in the [TABDIMS](#) keyword.

General comments on viscosity behavior

Viscosity behavior

Oil viscosity changes are due to two competing effects:

- the precipitation, as modeled above, tends to increase the oil viscosity
- the deposition of heavier component(s) tends to make the oil lighter and therefore reduces its viscosity.

This can be checked by running a case with deposition parameters set to zero, and running the same case with non-zero deposition parameters.

List of keywords

RUNSPEC section

- [ASPHALTE](#) activates the asphaltene option. It takes three arguments that select the precipitation model, the permeability damage model and the oil viscosity model.

PROPS section

Restriction	Keyword	Description
	ASPP1P	sets the variable asphaltene precipitation is tabulated against: pressure P, or molar fraction Z, or temperature T.
	ASPREWG	sets the table for the dissolution-precipitation asphaltene model based on percentage molar weight of asphaltene content as a function of one variable: P or T or Z.
	ASPP2P	sets the variables asphaltene precipitation is tabulated against: (P,T) or (P,Z) or (T,Z).
	ASPPW2D	sets the 2D table for the dissolution-precipitation asphaltene model based on percentage molar weight of asphaltene content as a function of two variables: (P,T) or (P,Z) or (T,Z).
Pre-2011.1 model	ASPFLOC	sets the precipitating components range and the flocs component number.
	ASPFLRT	sets the flocculation-dissociation rates.
	ASPDEPO	sets the deposition parameters.
Pre-2011.1 model	ASPKDAM	sets the permeability damage model: either an exponent if the permeability is correlated to the porosity or a table of permeability multiplier as a function of volume fraction of asphaltene deposit
	ASPVISO	sets the oil viscosity model parameters
	ASPLCRT	sets the critical values for the volume fraction of asphaltene deposit and the critical value for flocs concentration for plugging control, and how to use these critical values (AND or OR).
	ASPKROW	sets the water and oil wet relative permeability as a function of water saturation for the wettability change due to asphaltene deposition.
	ASPWETF	sets the wettability weighting factor between the water-wet and oil-wet relative permeability as a function of the volume fraction of asphaltene deposit.
	CATYPE	identifies the components for precipitation, flocculation and deposition. It is a replacement for ASPFLOC

Outputs

The following arguments are associated with the [RPTRST](#) / [RPTSCHED](#) keywords for the ASPHALTENE option.

Restriction	Keyword	Description
	AREAC	Asphaltene reaction rates.

Restriction	Keyword	Description
Pre-2011.1 model	ASPADS	Asphaltene cumulative adsorption volume fraction. The volume in these reported values is considered to be the grid block volume (bulk volume).
Pre-2011.1 model	ASPDOT	Asphaltene cumulative net deposition volume fraction.
Pre-2011.1 model	ASPENT	Asphaltene instantaneous entrainment volume fraction.
Pre-2011.1 model	ASPFLO	Oil phase velocities used in the entrainment calculations. The mnemonics output to the restart file are VELOILI+, VELOILJ+, and VELOILK+ depending on the dimension of the problem.
Three-component model	ASPFLT	Average oil flow rate at the previous timestep.
	ASPFDR	Asphaltene fraction precipitated and dissolved.
Pre-2011.1 model	ASPKDM	Asphaltene permeability damage multiplier.
	ASPLIM	Asphaltene precipitate fraction limit.
Pre-2011.1 model	ASPLUG	Asphaltene instantaneous plugging volume fraction
	ASPRET	Asphaltene precipitate fraction including flocs
	ASPREW	Asphaltene precipitate fraction
Three-component model	ASPVEL	Average oil velocity at the previous timestep.
	ASPVOM	Asphaltene oil viscosity multiplier

For information on the SUMMARY keywords refer to [Asphaltene SUMMARY output controls](#) in the *Eclipse Reference Manual*.

In pre-2011.1 versions of the simulator, the asphaltene flocculation rates could be reported using the ASPFL/BASPFL keywords. These keywords are no longer available - instead use the AREAC/BAREAC keywords.

Asphaltene reaction number

The BAREAC summary keyword is followed by a number of records, each containing four values, the I, J, K values of the cell block and the reaction number:

```
BAREAC
1 1 1 1 /
1 1 1 2 /
/
```

Pre-2011.1 model

In the pre-2011.1 model, pairs of reactions are used to model flocculation and dissociation. The reaction numbers have the following meanings:

2N-1 the rate of flocculation for the Nth precipitating component

2N the rate of dissociation for the Nth precipitating component

Three-component model

In the three-component model, pairs of reactions are used to model flocculation and dissociation and optionally deposition and entrainment. Only one component is allowed to precipitate so the reaction numbers have the following meanings:

1. the rate of flocculation

2. the rate of dissociation
3. the rate of deposition (if used)
4. the rate of entrainment (if used)

Example

In the RUNSPEC section, switch on asphaltene deposition model and the solid model (if required):

```

ASPHALTE
  WEIGHT  /
SOLID

```

In the PROPS section, specify the hydrocarbon and asphaltene components. For example if components 6, 7 and 8 are the precipitation, floc and deposition components:

```

CNAMES
  C1    C3    C6    C10   C15   C20   ASPH   DEPO /

-- Asphaltene component type
CATYPE
  5*    PREC   FLOC   DEP  /
-- Make deposition component a solid
CVTYPE
  7*    SOLID /
-- ... define asphaltene concentration limits
ASPREWG
--   pres   % wt
  1000.0   30.0
  10000.0  100.0 /
-- ... asphaltene floc rates
ASPFLRT
-- C20 <-> ASPH
  0.001
  0.0001 /
-- ... asphaltene deposition
ASPDEPO
-- adsorp plug entrain Vcr
  5.0E-6 0.0 0.00 1.0E6 /
-- ... asphaltene damage ratio
SOLIDMMS
-- deposit mult
  0.0    1.0
  1.0    0.5 /
-- ... asphaltene viscosity change
ASPVISO
-- vfrac mult
  0.00 1.00
  1.00 1.00 /

```

Converting a pre-2011.1 model

Pre-2011.1 asphaltene data sets can be converted to the three-component solid deposition model. Assuming the pre-2011.1 model has only one precipitating component, the conversion is straightforward:

1. Add the [SOLID](#) keyword in the RUNSPEC section.
2. Increase the number of components by one in the [COMPS](#) keyword.
3. Add the [CVTYPE](#) keyword in the PROPS section and define the new last component as a solid.

4. Add the **CATYPE** keyword in the PROPS section and define the precipitation, flocculation and deposition components. The precipitation and flocculation components are the same as those defined in the **ASPFLOC** keyword, and the deposition component is the new solid component. Remove the **ASPFLOC** keyword.
5. Add properties for the new solid component. The fluid properties can be copied from the flocculation component, although many of these will not be used since the solid component will not be considered in the flash. The deposition component molecular weight **MW** should be the same as the flocculation component molecular weight if mass is to be conserved. Solid properties for the deposition component can be defined using the appropriate keywords. See "**Solid keywords**". The solid component density **SDREF** is important as it will affect the volume balance and hence the cell pressure as asphaltene is deposited.
6. The two models use different plugging coefficients. The pre-2011.1 model uses γ_1 , specified by item 2 of the **ASPDEPO** keyword, while the three-component model uses γ_2 , specified by item 5 of the same keyword. The coefficient can be converted using $\gamma_2 = A \cdot \gamma_1$ where A is an average cell face area.
7. The two models use different measures for asphaltene deposit. The pre-2011.1 model uses the volume fraction in the cell ε , whereas the solid model uses the solid saturation S_s , that is the volume fraction in the pore $S_s = \varepsilon / \phi$. The ε values used in keywords must be converted to solid saturations:
 - replace ε_{cr} with S_{cr} in argument 1 of keyword **ASPLCRT**.
 - replace ε values with S_s values in the first column of **ASPWETF**, if the keyword is used.
8. Replace the **ASPKDAM** keyword, if used, with the **SOLIDMMS** keyword. If the **ASPKDAM** data is supplied as a table then the same data can be used for **SOLIDMMS**, except the first column must be converted from ε values to S_s values. If the **ASPKDAM** data is supplied as an exponent δ then an equivalent **SOLIDMMS** table must be created with columns S_s and $(1-S_s)^\delta$.
9. Replace the pre-2011.1 summary keywords.

Pre-2011.1 model	Three-component model
ASPADS/ BASPADSASPLUG/ BASPLUG	Adsorption and plugging are considered together as deposition. The volume fraction of adsorbed and plugging asphaltene are not calculated. Instead the rate of deposition can be found from the third reaction reported by AREAC/ BAREAC.
ASPDOT/BASPDOT	Use the solid keywords to report the amount of deposition SSOLID/ BSSOLID.
ASPENT /BASPENT	The volume fraction of entrained asphaltene is not calculated. Instead the rate of deposition can be found from the fourth reaction reported by AREAC/ BAREAC.
ASPFLO	Use keyword ASPFLT to report the flow rate and ASPVEL to report the average velocity. In addition the summary vectors BASPFLT and BASPVEL can be used.
ASPKDM/BASPKDM	Use the solid keywords to report the change in mobility that is due to asphaltene damage: SMMULT/BSMMULT.

Table 4.4: Summary keywords

Water components

	Eclipse 100
x	Eclipse 300

In Eclipse, water is traditionally modeled using a single component. Density and viscosity variations caused by salinity changes can be modeled using the Brine Tracking option, see "[Brine tracking](#)". The brine tracking option uses a tracer to track the salt movement in the reservoir.

This single-component model can be applied to many practical cases. However, there are some situations where more than one water component is required:

- as an alternative brine model. When density variations are large, the tracer method may become unstable, since the tracer equations are solved after the flow equations.
- to model brine in multisegment wells, see "[Multisegment wells](#)". At present the brine tracking option cannot be used at with the multisegment well option.
- to model reactions involving water borne components such as bacteria or dissolved solids, refer to "[Chemical reactions](#)".
- to model a surfactant component for the foam model, see "[Foam model](#)".
- to model a surfactant component for surface tension effects, see "[Surfactant model](#)".
- to model a polymer component for water phase viscosity effects, see "[Polymer model](#)".

Water component properties

Density

The molar density of water is given by

$$b_w = \frac{1}{V_w} \quad \text{Eq. 4.155}$$

where V_w is the specific volume given by:

$$V_w = \sum_c w_c \cdot \frac{MW_c}{\rho_{refc} \cdot (1 + X_c + 0.5X_c^2)} \quad \text{Eq. 4.156}$$

In equation 4.156 the sum is over the water components where:

w_c is the water component fraction in the water phase

ρ_{refc} is the reference density, specified with keyword [DREFW](#)

MW_c is the water component molecular weight, specified by the keyword [MWW](#)

The compressibility terms X_c in equation 4.156 are given by

$$X_c = c_{Pc} \cdot (P - P_{refc})$$

where

c_{Pc} is the water component compressibility, specified with keyword [CREFW](#)

P_{ref_c} is the reference pressure, specified with keyword [PREFW](#)

Viscosity

The viscosity of water is given by

$$\mu_w = \frac{\sum w_c \cdot \mu_{ref_c}}{(1 + Y_c + 0.5Y_c^2)} \quad \text{Eq. 4.157}$$

where

w_c is the water component fraction in the water phase

μ_{ref_c} is the reference viscosity, specified with keyword [VREFW](#)

The “viscosibility” terms Y_c in equation 4.157 are given by

$$Y_c = -c_{V_c} \cdot (P - P_{ref_c})$$

where

c_{V_c} is the “viscosibility”, specified with keyword [VREFW](#)

P_{ref_c} is the reference pressure, specified with keyword [PREFW](#)

Component names

Water components can be named using the [WNAMEs](#) keyword. If this keyword is not specified then the components take the default names. If there is only one water component, the default name is WATER. However, if there are more than one water components they are named WAT01, WAT02, and so on. The names are used in reporting and for associating a [TRACER](#) with a component.

Surface properties

For surface calculations, the optional keywords [MWWS](#), [DREFWS](#), [CREFWs](#) and [PREFWS](#) can be used to supply the values used in equations 4.155 and 4.156. The default, if these are not supplied, is to use the reservoir values given by [MWW](#), [DREFW](#), [CREFW](#) and [PREFW](#) (or [STCOND](#)).

Aquifer properties

The [AQSTREAW](#) keyword must be used to specify the composition of each analytic aquifer defined by [AQUCHWAT](#), [AQUCT](#), [AQUFETP](#) or [AQUFLUX](#). If this is not provided an error message will be emitted to the PRINT file and the water composition of an analytic water aquifer will be made up entirely of the last water component. However if the last water component is a surfactant, polymer or alkaline component (see [CWTYPE](#)), then it will not be used; instead the preceding water component will be used for the default aquifer water composition.

For a numerical aquifer defined by [AQUNUM](#), the aquifer composition is the water composition of the cell representing the aquifer, which is set in the usual way (see [SOLUTION section keywords](#)).

Any water component is allowed to leave the reservoir or enter it, depending on the aquifer composition. If [item 131](#) of the `OPTIONS3` is set to 1, this reverts to the pre-2008.1 behavior where the water entering / leaving the reservoir is made up entirely of the last water component. Additionally, the water properties used in the analytic aquifers are calculated as described above. If [item 119](#) of `OPTIONS3` is set to 1, this reverts to the pre-2008.1 behavior where the PVT tables were used. See "[Aquifer Modeling Facilities](#)".

Converting PVTW data to water component data

In standard simulations, water is modeled as a single component, with properties specified by the `PVTW` and `DENSITY` keywords. This case can also be modeled using the multiple component water option, with a single water component. Table 4.5 gives values for the water component keywords, derived from the `PVTW` and `DENSITY` data.

Keyword	Value
<code>MWW</code>	18.015
<code>PREFW</code>	Item 1 of <code>PVTW</code>
<code>DREFW</code>	$DENS_w / BW_{ref}$ where $DENS_w$ is Item 2 of <code>DENSITY</code> BW_{ref} is Item 2 of <code>PVTW</code>
<code>CREFW</code>	Item 3 of <code>PVTW</code>
<code>VREFW</code>	Items 4 and 5 of <code>PVTW</code>

Table 4.5: Water component values

In addition, in standard simulations the water moles are converted to surface volumes using a factor based on the water molecular weight, 18.015, and the water density as specified by `DENSITY` item 2. In a water component model, this factor is calculated based on the water composition as defined by equations 4.155 and 4.156 where surface properties should be specified as in Table 4.6.

Keyword	Value
<code>MWWS</code>	18.015
<code>PREFWS</code>	Item 2 of <code>STCOND</code>
<code>DREFWS</code>	Item 2 of <code>DENSITY</code>

Table 4.6: Surface water component values

Notes

- `PVTW` and `DENSITY` data are specified for each PVT region, while water component data is specified for each EOS region, so the two models can produce different answers if the regions do not match.
- If more than one water component is specified in the model but only one is actually present in the reservoir, use the last water component in the model to represent the one in the reservoir. This improves the numerical stability when solving the equations. See "[Formulation of the equations](#)". Failing to do so may introduce some noise in very sensitive decks.

Converting PVTWSALT data to water component data

Water with varying salinity can also be modeled using the Brine Tracking Option, see "Brine tracking". The Multiple Component Water provides an alternative mechanism, with one component used to model freshwater and another component used to represent brine. In the Brine Tracking Option, water properties are defined using the `PVTWSALT` keyword. This can be converted to component data in a similar manner to the `PVTW` data, described above, except that the density term must now include the weight of salt

$$DREFW = (DENS_w + SAL_w) / BW_{ref}$$

where SAL_w is the salt concentration.

Note that if the optional keyword `BDENSITY` is used, then the density conversion is simply

$$DREFW = DENS_w / BW_{ref}$$

where $DENS_w$ is specified by `BDENSITY`.

Effective salinity calculations

Effective salinity calculations are triggered when the multi-component water option is activated (`COMPW` in the `RUNSPEC` section) and effective salinity parameters are specified using the `SALTEFF` keyword in the `PROPS` section. In the current release, the keyword `SALTEFF` can only be used with the salinity-dependent Surfactant model (see "Surfactant model") or the Polymer model (see "Polymer flood model").

Effective salinity calculations can be defined by one of two formulae:

Formula 1: valid for the Surfactant and Polymer models	$E_s = \frac{a_0 + \sum_{i=1}^{N_w} a_i C_i}{b_0 + \sum_{i=1}^{N_w} b_i C_i}$
Formula 2: valid for the Surfactant model only	$E_s = \frac{a_0 + \sum_{i=1}^{N_w} a_i C_i}{b_0 + \frac{1}{C_s} \sum_{i=1}^{N_w} b_i C_i}$

where

C_i is the concentration of the i^{th} water component, and

a_i and b_i , $i = 0, \dots, N_w$ denote the effective salinity parameters specified by the keyword `SALTEFF`.

Note: The implementation of Formula 2 includes a smoothing factor that improves numerical stability when the surfactant concentration in the water phase is negligibly small. Rather than dividing directly by C_s the term $C_s + \delta$ is used, where by default δ is set to 1 E-6. You can change the value of δ using item 2 of `SURFOPTS`.

The keyword `SALTEFF` allows the input of any values of coefficients a_i and b_i . This allows you to define many salinity models (see for example *Chapter 3: "Salinity Effect and Ion Exchange"* in *Sheng (2011)*). The examples below show how these formulae can be used to represent different effective salinity models including those described in the Eclipse 100 simulator.

Example 1

The multi-component water option in Eclipse 300 can be used as an alternative to the brine model. For example, you could set three water components: one being fresh water, another representing brine and a final water component representing polymer.

```
COMPW
3 /

WNAMEs
WATER SALT POLY /

CWTYPE
1* 1* POLY /
```

The properties of each water component must then be specified with keywords [MWW](#), [PREFW](#), [DREFW](#), [CREFW](#) and [VREFW](#). Information on converting [PVTW](#) and [PVTWSALT](#) data into water component data can be found in sections “[Converting PVTW data to water component data](#)” and “[Converting PVTWSALT data to water component data](#)”.

If you want the effective salinity to represent salt concentration, then the following parameters in [SALTEFF](#) should be specified:

```
SALTEFF
-- Type  Eq a0 a1 a2 a3 b0 b1 b2 b3
POLY    1  0  0  1  0  1  0  0  0  /
```

According to Formula 1, this choice of coefficients in [SALTEFF](#) yields $E_s = C_2$, where C_2 is the salt concentration as desired.

Example 2

Let us now consider the following set of water components: one water component to represent fresh water, one to represent anions in the water phase, another water component to represent divalent cations associated with polymer and finally one water component to represent polymer solution:

```
COMPW
4 /

WNAMEs
WATER ANIONS CATIONS POLY /

CWTYPE
1* 1* 1* POLY /
```

If you want the effective salinity to be calculated using formula triggered by keyword [PLYESAL](#) in Eclipse 100, then the following parameters should be chosen in [SALTEFF](#)

```
SALTEFF
-- Type  Eq a0 a1 a2 a3  a4 b0 b1 b2 b3 b4
POLY    1  0  0  1  α -1  0  1  0  0  0  0  /
```

where

α is the effective salinity parameter that would be specified in [PLYESAL](#) item 1.

Example 3

Let us now consider a similar set of water components that now contain surfactant: one water component to represent fresh water, one to represent anions in the water phase, another water component to represent divalent cations associated with surfactant and finally one water component to represent Surfactant solution:

```
COMPW
4 /

WNNAMES
WATER ANIONS CATIONS SURF /

CWTYPE
1* 1* 1* SURF /
```

If you want the effective salinity to be calculated using the formula triggered by keyword [SURFESAL](#) in Eclipse 100, then the following parameters should be chosen in [SALTEFF](#)

```
SALTEFF
-- Type  Eq a0 a1 a2 a3 a4 b0 b1 b2 b3 b4
SURF    2  0  0  1  0  0  1  0  0 -  $\alpha$  0 /
```

where

α is the effective salinity parameter that would be specified in [SURFESAL](#) item 1.

Units

The extra flexibility in the choice of parameters in [SALTEFF](#) means that the units of the resulting effective salinity may vary from one combination of parameters to another:

- In Example 1 above, the coefficients in [SALTEFF](#) were chosen in such a way that the resulting effective salinity, according to Formula 1, has the following units: kg/sm³ ([METRIC](#)), lb/stb ([FIELD](#)), gm/sec ([LAB](#)). Considering that the effective salinity in this example equals to the concentration of salt (or the concentration of the second water component), the resulting units are as expected. In general, if all the coefficients b_i , $i = 1, \dots, N_w$ are zero, then the resulting effective salinity will have these units.
- On the other hand in Example 2 above, the resulting effective salinity is, according to Formula 1, dimensionless. This is consistent with the units of effective salinity in Eclipse 100 computed according to parameters specified in [PLYESAL](#). In general, units of effective salinity computed using Formula 1 will be dimensionless, with exception of cases described earlier.

Flux boundary conditions

The water components option can be used for two different scenarios when combined with the "[Flux Boundary Conditions](#)":

1. The full field model has the Water Component option active, and the reduced run also has the Water Components option active.

This is a standard use of the Flux Boundary option to allow runs to be performed on a small section of a field. It is required that the same number of water components ([COMPW](#)) is present in the full field model and in the reduced run.

- The full field model has the standard water model active (water properties input with [PVTW](#)), and the reduced run has the Water Components option active.

This set up allows the simulation of the injection of waterborne components in a given section of the reservoir without need to run the full field model, for example to investigate the effects of surfactant or polymer flooding on a sector of the reservoir. The water related properties read from the `FLUX` file are all assumed to be from the first water component defined in the reduced run and set to zero for any other water component. It is important that the properties of the first water component input in the reduced run match the properties of the water in the full field model as described in “[Converting PVTW data to water component data](#)”. In addition, you should ensure that the newly injected fluids do not change the flow conditions on the flux boundary; this might be achieved by selecting a flux boundary large enough so that the newly injected fluids do not cross the flux boundary.

Limitations

The multi-component water model can be used in place of the single component water model in most standard simulations (black oil, EoS or K-value). However, it cannot be used at present in certain simulations where water has special properties. Therefore, options that cannot be used with the multi-component water model include:

- The Thermal option, where water is allowed in the gas phase (see “[Thermal option](#)”).
- The GASWAT option, where water is modeled using an equation of state (see “[GASWAT option](#)”).
- The GASSOL option, where compositional gas is allowed to dissolve in aqueous phase (see “[GASSOL option](#)”).
- The CO2SOL option, where CO₂ dissolves in water (see “[CO2/H2S solution in the aqueous phase](#)”).
- Satellite groups (see [GSATPROD](#)).
- [IMPSAT](#)

In addition, there are currently some limitations when specifying water source terms:

- The water entering a multisegment well using the source / sink term ([WSEGEXSS](#)) is made up entirely of the last water component.

Summary of keywords

RUNSPEC

Keyword	Description
COMPW	Enables the water component model and specifies the number of water components

PROPS

Keyword	Description
CREFW	Compressibility of water components at the reference pressure
CREFWS	Compressibility of water components for surface calculations
CWTYPE	Water component type
DREFW	Density of water components at the reference pressure

Keyword	Description
DREFWS	Density of water components for surface calculations
MWW	Molecular weight of water components
MWWS	Molecular weight of water components for surface calculations
PREFW	Reference pressure for water component density and viscosity calculations
PREFWS	Reference pressure for water component density calculations at surface conditions
VREFW	Viscosity and viscosibility for water components at the reference pressure
WNAMEs	Water component names

SOLUTION

Keyword	Description
AQSTREAW	Sets composition of water analytic aquifer (can also be used in the SCHEDULE section)
WI	Specifies overall initial water composition when setting the initial conditions by equilibration (can also be used in the PROPS section)
WMF	Specifies initial water composition explicitly when setting the initial conditions by enumeration
WMFVD	Specifies initial water composition with respect to depth when setting the initial conditions by equilibration (can also be used in the PROPS section)

SUMMARY

The SUMMARY keywords that can be used to obtain water component production and injection data are displayed below.

See the following in the *Eclipse Reference Manual*..

- [Analytic aquifer quantities](#)
- ["Format of SUMMARY keywords with a component index"](#)

Field	Group	Well	Connection	Description
FAMF	GAMF	WAMF	CAMF	Water component mole fractions
FWMPR	GWMPR	WWMMPR		Water component molar production rate
FWMPT	GWMPPT	WWMPT		Water component molar production total
FWMIR	GWMIR	WWMIR		Water component molar injection rate
FWMIT	GWMIT	WWMIT		Water component molar injection total
FCWM				Component in water phase
			CKFR	Component flow rate
			CKFT	Component flow total

Block	Region	Segment	Description
BMLSC			Component moles / reservoir volume
BAMF			Water component mole fractions

Block	Region	Segment	Description
	RKFR		Component inter-region flow rate
	RKFT		Component inter-region flow total
BCFRI			Component inter-block flow in the I-direction
BCFRJ			Component inter-block flow in the J-direction
BCFRK			Component inter-block flow in the K-direction
		SKFR	Segment component flow rate

For water component summary keywords, the water component index starts at $N_{\text{comps}} + 1$ and ends at $N_{\text{comps}} + N_w$ where N_{comps} is the number of components given by **COMPS** and N_w is the number of water components given by the **COMPW** keyword.

SCHEDULE

Keyword	Description
WELLSTRW	Sets composition of the injection water stream
WINJW	Specifies the nature of the injection water

OUTPUT keywords

The arguments AMF and MWAT can be used with the keywords **RPTRST**, **RPTSCHED** and **RPTSOL** to output water mole fraction and molar concentrations.

For summary block quantities, such as BMLSC as the water component index starts at $N_{\text{comps}} + 1$ and ends at $N_{\text{comps}} + N_w$ where N_{comps} is the number of components given by **COMPS** and N_w is the number of water components given by the **COMPW** keyword.

Examples

The following data sets come with the Eclipse installation:

File	Description
MCW0 . DATA	Brine model (using tracers, for comparison)
MCW1 . DATA	Brine model
MCW2 . DATA	MSW Brine model
MCW3 . DATA	Brine + solids + reactions
MCW . GRF	GRF to compare runs

Table 4.7: Multi-component water example files

Chemical reactions

	Eclipse 100
x	Eclipse 300

This chapter describes how chemical reactions are modeled in the Eclipse 300 simulator. Chemical reactions can only be used in fully implicit runs. They can be used in black-oil runs and compositional runs (EoS, K-value and thermal).

The keywords for chemical reactions can be used to model a number of other processes, for example:

- combustion
- biodegradation
- decay of radioactive tracers
- non-equilibrium reactions
- foam decay and adsorption.

Formulation

The implementation of chemical reactions in Eclipse follows *Coats (1980)*.

The number of chemical reactions is specified by the **REACTION** keyword. Each reaction is defined by a stoichiometry equation and reaction rate.

Stoichiometry equation

The stoichiometry equation for reaction r is given by

$$\Sigma(S_{Rri} \cdot C_i) \rightarrow \Sigma(S_{Pri} \cdot C_i) \quad \text{Eq. 4.158}$$

where:

S_{Rri} is the stoichiometric coefficient for the reactant C_i ,

S_{Pri} is the stoichiometric coefficient for the product C_i .

The stoichiometric coefficients are defined by the **STOREAC** and **STOPROD** keywords. The phase of the reactants can be specified by the **REACPHA** keyword.

Reaction rate

The reaction rate R_r for reaction r in a grid cell is given by an Arrhenius equation:

$$R_r = V_b \cdot A_r \cdot e^{(-E_r/(RT))} \cdot \Pi c_{ri}^{n_{ri}} \quad \text{Eq. 4.159}$$

where:

v_b is the bulk volume (rock volume plus pore volume),

A_r is a reaction rate constant, defined by **REACRATE**,

E_r is the activation molar energy, defined by **REACTACT**,

R is the gas constant,

T is the temperature which may be limited using [REACLIMS](#),

n_{ri} is the index or order of the component dependent term, defined by [REACCORD](#).

Component concentration

The terms c_{ri} represent the component concentration in the reacting phase ([REACPHA](#)). If the reactant is in the oil phase:

$$c_{ri} = \theta \cdot b_o \cdot S_o \cdot x_i \quad \text{Eq. 4.160}$$

If the reactant is in the gas phase:

$$c_{ri} = \theta \cdot b_g \cdot S_g \cdot y_i \quad \text{Eq. 4.161}$$

If the reactant is water in the water phase:

$$c_{ri} = \theta \cdot b_w \cdot S_w \cdot a_i \quad \text{Eq. 4.162}$$

If the reactant is in all phases:

$$c_{ri} = \theta \cdot m_i \quad \text{Eq. 4.163}$$

Here

θ is the porosity,

b_o , b_g and b_w are the molar densities of oil, gas and water;

S_o , S_g and S_w are the oil, gas and water saturations;

x_i , y_i , a_i and m_i are the mole fractions in the oil, gas and water phases and the molar density of component i .

For reactants in the gas phase, the reaction rate can be made to depend on the partial pressure. Instead of equation [4.161](#):

$$c_{ri} = P \cdot y_i \quad \text{Eq. 4.164}$$

where P is the pressure and can be limited using the [REACLIMS](#) keyword.

If the reactant is in the solid phase, equation [4.163](#) can be used - since at present a solid component cannot appear in any other phase.

Component concentration order

The order n_{ri} of the component term in equation [4.159](#) should be non-negative for all reactants. The order for a reactant can be set to zero, in which case the reaction rate is independent of the concentration of that component; however, this is not recommended, unless keyword [EQLDREAC](#) is also used. If the order of a component dependent term is non-zero, and that component is not a reactant, then it acts as a catalyst or inhibitor.

To improve stability when simulating zero-order reactions, the reaction rate is multiplied by the scaling factor $m/(m + s)$ where m is the molar density of the reactant and s is given by item 4 of keyword

REACOPS. Please note that the default value of $s(1)$ is an experimental value and may be adjusted to ensure that the scaling factor $m/(m + s)$ tends to zero with the concentration of the reactant.

Porosity order

In equations 4.160 - 4.163, the component terms c_{ri} represent the concentrations per cell volume. In Eclipse it is more natural to consider the concentrations per pore volume:

$$c'_{ri} = c_{ri}/\theta$$

The reaction rate, equation 4.159 can then be rewritten as

$$R_r = V_p \cdot \theta^{n_{rp}} \cdot A_r \cdot e^{(-E_r/(RT))} \cdot \prod c'_{ri}{}^{n_{ri}} \quad \text{Eq. 4.165}$$

where:

V_p is the pore volume, $V_p = \theta \cdot V_b$

n_{rp} is the index or order of the porosity term given by $n_{rp} = \sum_i n_{ri} - 1$

Here the sum is over all reaction components **except** those where the rate is defined by the partial pressure 4.164. If necessary, the porosity order can be modified using the **REACPORD** keyword.

Scaling chemical reaction speed based on water volumes

The default formulation for chemical reactions can be expressed as $R_r = V_b f(c_{ri,b})$

where:

R_r is the reaction rate in a grid cell

V_b is the bulk volume

$c_{ri,b} = \frac{n_{ri}}{V_b}$ are concentrations expressed as number of moles in a cell per cell bulk volume

The change of moles in a cell relies on a function using concentrations per bulk volume scaled by the cell overall bulk volume.

Using **REACPORD**, it is possible to reformulate the chemical reaction using the following form:

$$R_r = V_p f(c_{ri,pv})$$

where:

V_p is the pore volume

$c_{ri,pv} = \frac{n_{ri}}{V_p}$ are concentrations expressed as number of moles in a cell per cell pore volume

Both formulations have their advantages.

The chemical reaction feature allows us to model many phenomena involving species dissolved in water, from low salinity effects to Co2 dissolution and acidity, barite deposition, and other water-based chemical

reactions. For these types of reactions, if WATERV is selected for item 5 of REACOPS, the formulation can be simplified to $R_r = V_p f(c_{ri,w})$

where:

V_w is the cell water reservoir volume

$c_{ri,w} = \frac{n_{ri}}{V_w}$ are concentrations expressed as number of moles in a cell per cell water volume

Developed in more detail, the reaction speed can be expressed as $R_r = V_b \theta S_w A_r e^{(-E_r/(RT))} \prod (c'_{wri})^{n_{ri}}$ where $c'_{wri} = b_w a_i$

The change of reaction speed is only applied to reactions involving water components as either reactant or products.

Handling of mixed kinetics reaction

The previous formulations imply that the reaction speed depends on the water volume in place. Some reactions may involve a mix of gas and water components interacting together, leading to a mixed kinetic model where speed depends on concentration of both a gas component and a water component. If the new model is selected for that kind of situation, a mixed approach is used for the reaction speed. The mixing of bulk volume concentration and water volume concentration is not recommended, but the approach is allowed to offer flexibility.

New model

$$R_r = V_b \theta S_w A_r e^{(-E_r/(RT))} \prod_{o,g} c_{ri}^{n_{ri}} \prod (b_w a_i)^{n_{ri}} \quad \text{Eq. 4.166}$$

Reactions involving solid components

In simulations with solid components the fluid saturations are modified to include the solid saturation $\hat{S}_s$:

$$\hat{S}_o + \hat{S}_g + \hat{S}_w + \hat{S}_s = 1$$

Normalized fluid saturations can then be defined

$$S_o = \hat{S}_o / (1 - \hat{S}_s)$$

$$S_g = \hat{S}_g / (1 - \hat{S}_s)$$

$$S_w = \hat{S}_w / (1 - \hat{S}_s)$$

Equations 4.160 - 4.162 are modified, for example 4.160 becomes:

$$c_{ri} = \theta \cdot b_o \cdot \hat{S}_o x_i = \theta \cdot b_o \cdot S_o \cdot x_i \cdot (1 - \hat{S}_s)$$

Equation 4.165 then becomes

$$R_r = V_p \cdot \theta^{n_{rp}} \cdot \left(1 - \hat{S}_s\right)^{n_{rs}} \cdot A_r \cdot e^{(-E_r/(RT))} \cdot \prod c_{ri}^{n_{ri}}$$

where:

$$n_{rs} = \sum_i n_{ri}$$

and the sum is over all reaction components **except** those where the rate is defined by the partial pressure, equation 4.164 or where the reactant is in all phases, equation 4.163. If necessary, the values of n_{rs} , can be modified using the REACSORD keyword.

Problems may occur in the matrix inversion, if the solid saturation is close to 1. To prevent this, the solid dependent term in the reaction rate is replaced by

$$\left(1 - \frac{\hat{S}_s}{\hat{S}_{smax}}\right)^{n_{rs}}$$

The value of $\hat{S}_{smax} \leq 1$ can be defined using the REACSMAX keyword. As the solid saturation approaches the limit $\hat{S}_{smax}$, the reaction rate tends to zero.

If the WATERV option is selected in item 5 of the REACOPS keyword, the treatments of oil and gas concentrations and water concentration are different, analogous to the WATERV mixed model without

$$\text{solids. } R_r = V_p \theta^{n_{rp}} S_w \left(1 - \hat{S}_s\right)^{n_{rp} + 1} A_r e^{(-E_r/(RT))} \prod_{o,g} (c_{ri}')^{n_{ri}} \prod (b_w a_i)^{n_{ri}}$$

$$\text{where } n_{rp} = \sum_{o,g} n_{ri}$$

The order of $(1 - \hat{S}_s)$ can still be modified with REACSORD.

Equilibrium deviation reaction rates

The reaction rate can contain extra terms that depend on the deviation from an equilibrium value. Equation 4.159 is replaced with:

$$R_r = V_b \cdot A_r \cdot e^{(-E_r/(RT))} \cdot \prod c_{ri}^{n_{ri}} \cdot \prod D_{mijk} \quad \text{Eq. 4.167}$$

The additional term $\prod D_{mijk}$ is the product of one or more terms of the form:

$$D_{mijk} = \text{sign}(d_{mijk}) \cdot |d_{mijk}|^n \quad \text{Eq. 4.168}$$

The equilibrium deviation terms d_{mijk} are given by one of the following models:

$$d_{1ijk} = \theta \cdot b_o \cdot S_o \cdot (x_i - y_j / K_k) \quad \text{Eq. 4.169}$$

$$d_{2ijk} = \theta \cdot b_g \cdot S_g \cdot (K_k \cdot x_i - y_j) \quad \text{Eq. 4.170}$$

$$d_{3ijk} = \theta \cdot b_w \cdot S_w \cdot (a_i - 1 / K_k) \quad \text{Eq. 4.171}$$

$$d_{4ijk} = \theta \cdot (F_k(a_i) - C_a) \quad \text{Eq. 4.172}$$

In these equations x_i is a component oil mole fraction, y_j is a component gas mole fraction, a_i is a water component mole fraction and C_a is the adsorbed solid concentration. The [EQLDREAC](#) keyword should be used to define the values of m , i , j , k and n , where:

m determines which of equations 4.169 - 4.172 is used,

i is the first component number,

j is the second component number,

k is either

- the number of the equilibrium k-value, K_k , defined by the [EQLDKVCR](#) keyword
- the number of the equilibrium table F_k , defined by the [EQLDTAB](#) keyword

n is the order of the deviation term in equation 4.168.

The equilibrium position is given by $x_i = y_j/K_k$, $a_i = 1/K_k$ or $C_a = F_k(a_i)$. The equilibrium rate term D_{mijk} is zero when the components i and j are in equilibrium, and can be positive or negative or both. The reaction stoichiometry should be set so that component i reacts to form component j for positive reaction rates. The direction of the reaction can be restricted to forwards, in which case equation 4.168 is replaced by:

$$D_{mijk} = (\max(0, d_{mijk}))^n \quad \text{Eq. 4.173}$$

or backwards, in which case equation 4.168 is replaced by:

$$D_{mijk} = -(\max(0, -d_{mijk}))^n \quad \text{Eq. 4.174}$$

For model types 1–3, if there is a solid saturation, the fluid saturation in the equilibration deviation equation is modified in the same way as in [Reactions involving solid components](#).

The component numbers i and j can be set to 0:

- If the component number i is set to zero, the oil mole fraction x_i is replaced by a value of 1 in equations 4.169 and 4.170, and the water mole fraction a_i is replaced by a value of 1 in equations 4.171 and 4.172.
- If the component number j is set to zero, the gas mole fraction y_i is replaced by a value of 1 in equations 4.169 and 4.170. j should be set to zero if equation 4.171 or equation 4.172 is used.

The [REACTION](#) keyword should be used to specify the total number of equilibrium deviation terms allowed in the simulation and also the number of equilibrium values that are defined by the [EQLDKVCR](#) keyword.

The equilibrium deviation terms D_{mijk} are used in addition to the component concentration terms c_{ri} , they do not replace them. However, if the component orders n_{ri} are set to their default values of zero ([REACCORD](#)), then the component concentration terms disappear and equation 4.167 becomes

$$R_r = V_b \cdot A_r \cdot e^{(-E_r/(RT))} \cdot \Pi D_{mijk} \quad \text{Eq. 4.175}$$

This reaction type is primarily intended for use with “inert” components, such as non-volatile or incondensable components in thermal simulations, or solid or water components. If it is used with live components, then there is a possibility that the reaction will compete with the flash.

If WATERV is selected for item 5 of [REACOPS](#), equation 4.175 is modified similarly to previous cases to be

$$R_r = V_b \theta S_w A_r e^{(-E_r/(RT))} \prod_{o,g} c_{ri}^{n_{ri}} \prod_w (b_w a_i)^{n_{ri}} \prod D_{mijk} \quad \text{Eq. 4.176}$$

The third equilibrium deviation rate, which involves water components, becomes $d_{3ijk} = b_w(a_i - 1/K_k)$

Tracer dependent reaction rates

In runs with tracers the reaction rate can contain extra terms that depend on the tracer concentrations. The tracers act as a catalyst. Equation 4.159 is replaced with:

$$R_r = V_b \cdot A_r \cdot e^{(-E_r/(RT))} \cdot \prod c_{ri}^{n_{ri}} \cdot \prod T_{rj} \quad \text{Eq. 4.177}$$

The tracer multiplier terms T_{rj} are calculated explicitly (at the previous timestep) as

$$T_{rj} = (\theta \cdot t_j \cdot c_c)^{n_{tj}} \quad \text{Eq. 4.178}$$

where

t_j is the tracer concentration

c_c is the concentration of the carrier component c in the tracer phase p_j

n_{tj} is the order of the tracer term.

If the tracer phase is oil then:

$$c_c = b_o \cdot S_o \cdot x_c \quad \text{Eq. 4.179}$$

If the tracer phase is gas:

$$c_c = b_g \cdot S_g \cdot y_c \quad \text{Eq. 4.180}$$

If the carrier is water and the tracer phase is water:

$$c_c = b_w \cdot S_w \quad \text{Eq. 4.181}$$

If the tracer is in all phases:

$$c_c = m_c \quad \text{Eq. 4.182}$$

The reaction number r tracer t_j , tracer phase p_j and tracer order n_{tj} for each tracer dependent term is defined by the [REACTRAC](#) keyword. The [REACTION](#) keyword should be used to specify the total number of tracer dependent terms. The tracer names and carrier components must be defined by the [TRACER](#) keyword. Values of the tracer multipliers can be output using the mnemonic REAC in keywords [RPTRST](#) or [RPTSCHED](#).

Component conservation equations

The residual equations are modified by adding source and sink terms that are proportional to the reaction rates R_r .

The equation for conservation of a fluid component m_{fl} (hydrocarbon or water) in each grid block at each timestep becomes:

$$R_{fl} = \frac{d}{dt} (V_p m_{fl}) - \sum_r [(S_{Pri} - S_{Rri}) \cdot R_r] = 0 \quad \text{Eq. 4.183}$$

where the flow terms between cells and to and from wells have been ignored for simplicity, see "Formulation of the equations".

Energy conservation equation

Thermal option only

The equation for conservation of energy e is modified in a similar manner to the component conservation equations. In each grid block at each timestep:

$$R_e = \frac{d}{dt} (V_b e) - \sum_r [H_r \cdot R_r] = 0 \quad \text{Eq. 4.184}$$

where:

V_b is the bulk volume,

H_r is the reaction molar enthalpy, defined by REACENTH,

and the convective, conduction and well terms have been ignored.

Non-uniqueness

The stoichiometric coefficients S_{Rri} and S_{Pri} are not unique - the same reaction can be defined by multiplying all the coefficients by a constant N . However, the reaction rate R_r should then be divided by N , since N reactions take N times longer. This ensures the component balance equation, equation 4.183 remains unchanged. It is conventional, though not essential, to choose the stoichiometry coefficients so that the coefficient of the first reactant is 1. In this case the rate of reaction R_r represents the rate at which one mole of the first reaction component is reacted.

Thermal option only

In Thermal simulations the reaction molar enthalpy H_r should also be multiplied by N since N reactions produce N times as much energy. This ensures the energy balance equation, equation 4.184 remains the same. If the stoichiometric coefficient of the first reactant is 1 then H_r represents the amount of energy released when one mole of the first reaction component is reacted.

Reaction rate units

The units of the rate constant A_r are not fixed but are chosen to ensure that the units of the reaction rate R_r defined by 4.159 are correct when used in equation 4.183. Therefore they depend on the values entered for the orders of the component terms n_{ri} and may be different for different reactions.

The units of the reaction rate R_r are:

- kg-M/m³/day (METRIC)
- lb-M/bbl/day (FIELD)
- gm-M/cm³/hr (LAB)
- kg-M/m³/day (PVT-M)

Therefore, if the sum of the concentration terms (excluding the gas partial pressure terms) is n , and the sum of the gas partial pressure terms is m then the units of A_r are:

- $(kg - M/m^3)^{1-n}(bar)^{-m}(day)^{-1}$ (METRIC)
- $(lb - M/bbl)^{1-n}(psi)^{-m}(day)^{-1}$ (FIELD)
- $(g - M/cm^3)^{1-n}(atm)^{-m}(hr)^{-1}$ (LAB)
- $(kg - M/m^3)^{1-n}(atm)^{-m}(day)^{-1}$ (PVT-M)

Factors to convert between unit systems are given by:

$$A_{r_{METRIC}} = (0.453593/0.158987)^{1-n} (0.0689476)^{-m} A_{r_{FIELD}} \quad \text{Eq. 4.185}$$

$$A_{r_{PVT-M}} = 1.01325^m A_{r_{METRIC}} \quad \text{Eq. 4.186}$$

$$A_{r_{LAB}} = (10^3)^{n-1} (24)^{-1} A_{r_{PVT-M}} \quad \text{Eq. 4.187}$$

Stability of thermal runs

Thermal option only

To improve the stability when simulating rapid reactions, such as combustion, it may be necessary to interpolate values between the current timestep and the previous one. Interpolation parameters can be specified using the [REACCRIT](#) keyword for the temperature used in the Arrhenius term in equation 4.159:

$$T = \theta_T \cdot T_n + (1 - \theta_T) \cdot T_{n-1}, 0 < \theta_T \leq 1$$

and the oil saturation used in equation 4.160:

$$S_o = \theta_S \cdot S_{o,n} + (1 - \theta_S) \cdot S_{o,n-1}, 0 < \theta_S \leq 1$$

Output

The reaction rate equation 4.159 can be output to the summary file or the RESTART file for graphical display or to the print file.

To output the reaction rate to the summary file the mnemonics BREAC, FREAC and FREAT can be used in the SUMMARY section.

BREAC outputs the reaction rate for specified cells and specified reactions.

FREAC outputs the reaction rate for the whole field for specified reactions.

FREAT outputs the reaction total since the beginning of the simulation for the whole field for specified reactions.

If the reaction number is defaulted in keywords FREAC or FREAT, then data are output for all reactions.

For example:

```
BREAC
-- I J K R -- Rate for reaction 1 in cell (3,4,5)
  3 4 5 1 / -- Rate for reaction 2 in cell (3,4,5)
  3 4 5 2 / -- Rate for reaction 2 in cell (6,7,8)
  6 7 8 2 /
/

FREAC
1 2/ -- Reaction rates summed over all cells for reactions 1 and 2
```



```
FREAT
/  -- Reaction totals summed over all cells for all reactions
```

To output the reaction rate to the RESTART file the REAC mnemonic can be used with the [RPTRST](#) keyword. The reaction rate is printed for each cell and each reaction. For example:

```
RPTRST
REAC/
```

To output the reaction rate to the print file the REAC mnemonic can be used with the [RPTSCHED](#) keyword. The reaction rate is printed for each cell and each reaction. For example:

```
RPTSCHED
REAC/
```

Chemical reaction keywords

RUNSPEC

- [REACTION](#) Number of reactions

PROPS

Restriction	Keyword	Description
	EQLDKVC	K-values for equilibrium deviation reactions
	EQLDREAC	Equilibrium deviation reaction rates
	EQLDTAB	Equilibrium values for water-solid deviation reactions
	REACTACT	Reaction activation molar energy
Thermal option.	REACENTH	Reaction molar enthalpy
	REACCORD	Reaction rate component order
	REACLIMS	Limits on pressures and temperatures used to calculate rates
	REACOPS	Options and convergence criteria
	REACPHA	Reactant phase
	REACPORD	Reaction rate porosity term order
	REACRATE	Reaction rate constant
Solid option	REACSMAX	Reaction rate solid saturation limit.
Solid option.	REACSORD	Reaction rate solid order.
	REACTRAC	Tracer dependent reaction rate terms
	STOPROD	Stoichiometric coefficients of products
	STOREAC	Stoichiometric coefficients of reactants

SUMMARY

Field	Block	Description
FREAC	BREAC	Reaction rate

Field	Block	Description
FREAT		Reaction total

SCHEDULE

Restriction	Keyword	Description
Solid option or Thermal option.	REACCRIT	Reaction criteria

The RPTRST and RPTSCHED arguments are:

Argument	Description
REAC	Reaction rate for each cell
TREACM	Tracer dependent reaction rate multiplier for each cell

Examples

Example 1: Combustion

Coats (1980) gives a thermal example with 5 components plus water:

Component	Name
C1	C ₃₂ +C ₄₇
C2	C ₂₀
C3	C ₆ +C ₈ +C ₁₂
C4	O ₂
C5	N ₂ -CO _x
W	H ₂ O

Table 4.8: Example names for components

Reaction	Stoichiometry
1	C1 + 45.17C4 -> 37.46 C5 + 29.71 W
2	C2+25.08 C4 -> 20.8 C5 + 16.5 W
3	C3 + 11.66 C4 -> 9.65 C5 + 7.67 W

Table 4.9: Example stoichiometry

Reaction	Rate constant	Activation Molar Energy (Btu/lb-M)	Reaction Molar Enthalpy (Btu/lb-M)
1	3.4054×10^{10}	32785	16000
2	0.28164×10^{10}	32785	16000
3	0.4035×10^{10}	32785	16000

Table 4.10: Example chemical reaction parameters

The rate of each reaction is proportional to the partial pressure of oxygen and the square of the oil component concentration.

The following keyword needs to be added to the RUNSPEC section:

```
REACTION
3 /
```

The following keywords need to be added to the PROPS section:

```
STOREAC
-- C1      C2      C3      C4      C5      W
   1        0        0  45.17    0        0 / reaction 1
   0        1        0  25.08    0        0 / reaction 2
   0        0        1  11.66    0        0 / reaction 3

STOPROD
-- C1      C2      C3      C4      C5      W
   0        0        0    0  37.46  29.71 / reaction 1
   0        0        0    0  20.80  16.51 / reaction 2
   0        0        0    0   9.67   7.67 / reaction 3

REACRATE
  3.4054E10  0.28164E10  0.4035E10 / reaction 1,2,3

REACTACT
   32785      32785      32785 / reaction 1,2,3

REACENTH
   16000      16000      16000 / reaction 1,2,3

REACPHA
-- C1      C2      C3      C4      C5      W
  OIL    1*      1*      GPP    1*      1* / reaction 1
  1*      OIL    1*      GPP    1*      1* / reaction 2
  1*      1*      OIL    GPP    1*      1* / reaction 3

REACCORD
-- C1      C2      C3      C4      C5      W
   2        0        0    1    0        0 / reaction 1
   0        2        0    1    0        0 / reaction 2
   0        0        2    1    0        0 / reaction 3
```

Example 2: A dynamic approach to equilibrium

In this example two components are used to model the oil phase and gas phase of a single hydrocarbon. The flash is replaced by an equilibrium-deviation reaction, representing evaporation and condensation.

The following keyword is needed in the RUNSPEC section:

```
REACTION
1 1 1 / one reaction, one deviation term, one EQLD k-value
```

In the PROPS section:

```
CNAMES
C_OIL C_GAS /

CVTYPE
DEAD GAS / Switch off the flash for these components

STOREAC
-- C_OIL  C_GAS  W
   1      0      0 / reaction 1: C_OIL <-> C_GAS

STOPROD
-- C_OIL  C_GAS  W
```

```
      0      1      0 / reaction 1: C_OIL <-> C_GAS
EQLDREAC
-- # N I J K
   1 1 1 2 1 / reaction 1: (X1 - Y2/K1)
EQLDKVCR
1.23E6
833.4E6
0
1.6E4
0          / K1 = (A + B/P + C.P) . EXP (D / (T-E))
REACRATE
  3.4054E10 /
REACTACT
  32785 /
REACENTH
  -1600 / LATENT HEAT
```

Solid phase

	Eclipse 100
x	Eclipse 300

Solid components can be used with the Black oil, Equation of State (EoS), K-value, or Thermal options. Within the reservoir, these components remain in the solid phase throughout the simulation. However, solid components can be converted to fluid components through a chemical reaction. A solid component could be used, for example, to simulate production and take up of coke during combustion.

Solids can be either suspended in the oil phase, or deposited on to the formation. Adsorption isotherms can be defined which determine the fraction of solids in suspension.

Suspended solids are transported with the flow.

Solids that are deposited on the formation reduce the pore volume available for flow. Permeability multipliers can be used to model this effect.

Summary of keywords

RUNSPEC section

- [SOLID](#) Requests solids option

GRID section

- [ROCKDEN](#) Rock density

PROPS section

Restriction	Keyword	Description
	CVTYPE	Define solid phase components
	CVTYPES	Define solid phase components at surface conditions
Reactions only	REACSMAX	Reaction rate solid saturation limit
Reactions only	REACSORD	Reaction rate solid order
	SCREF	Solid compressibility
	SDREF	Solid density at reference conditions
	SOLIDADS	Solid adsorption table
	SOLIDMMC	Solid mobility multiplier (as a function of adsorbed concentration)
	SOLIDMMS	Solid mobility multiplier (as a function of adsorbed saturation)
	SOLWTAB	Solid wettability interpolation parameter (as a function of adsorbed concentration)
	SOLWW	Weight factor for wetting-solid concentration
Thermal only	SPECHS	1st coefficient of solid specific heat
Thermal only	SPECHT	2nd coefficient of solid specific heat
	SPREF	Reference pressure for solid density calculation
	STREF	Reference temperature for solid density calculation
	STHERMX1	Solid thermal expansion coefficient

REGIONS section

- [SOLWNUM](#) Solid wettability region numbers

SOLUTION section

Keyword	Description
SMF	Initial solid molar fraction
SSOLID	Initial solid saturation

In addition, solid specific mnemonics can be used with the [RPTSOL](#) keyword. These are described in the [SCHEDULE keywords section](#).

SUMMARY section

Solid properties and production data can be output to the summary file.

Refer to [Grid block solid quantities](#), and [Solid production totals](#) in the *Eclipse Reference Manual*.

Other Reporting

The following arguments can be used with the [RPTSOL](#), [RPTSCHED](#), and [RPTRST](#) keywords to output grid data specific to the Solid Model:

Mnemonic	Description
BSOL	Solid reservoir molar densities
DENS	Solid reservoir density
HSOL	Solid enthalpies (Thermal option)
SSOLID	Solid saturations
SSFRAC	Suspended solids fraction
SOLADS	Adsorbed solid concentration (mass of solid per mass of rock), requires ROCKDEN
SOLADW	Adsorbed wetting-solid concentration (mass of solid per mass of rock), requires ROCKDEN
SMMULT	Solid mobility multiplier
SOLWET	Solid wettability interpolation parameter

Solid properties

Solid phase

The [SOLID](#) keyword must be used if solid components are present.

The number of components (including solid components) must be specified using the [COMPS](#) keyword, or the [COMPSOL](#) keyword if it is a [BLACKOIL](#) simulation.

Solid-only components

The [CVTYPE](#) keyword can be used to specify solid phase components. The [CVTYPES](#) keyword can be used to specify solid phase components at surface conditions.

These solid components are ignored by the flash - they cannot enter the liquid or gas phase. The only way they can change is through a chemical reaction, when they can become a different component.

Solid phase properties

Properties for solid-only components are defined using special keywords which are described in this section.

Solid phase density

The specific volume of solid phase components V_{sol} is calculated as

$$V_{sol} = m_S / b_S$$

where

m_S is the total number of moles in the solid phase per unit pore volume.

b_S is the solid molar density given by

$$1/b_S = \sum_{c=1}^{N_{comps}} x_S^c \cdot V_{sol}^c$$

where

x_S^c is the mole fraction of component c in the solid phase.

The component molar solid phase volume V_{sol}^c is given by:

$$V_{sol}^c = \frac{MW^c}{\rho^c}$$

where MW^c is the molecular weight of component c given by the [MW](#) keyword, and the component solid phase density ρ^c is given by:

$$\rho^c = \frac{\rho_{ref}^c}{(1 + C_{T1}(T - T_{ref}))(1 - C_P(P - P_{ref}))}$$

The reference density ρ_{ref}^c is defined by the [SDREF](#) keyword

the standard temperature and pressure, T_{ref} and P_{ref} are defined by keywords [STREF](#) and [SPREF](#) (or by [STCOND](#))

the thermal expansion coefficient C_{T1} is defined by the [STHERMX1](#) keyword, and

the component isothermal compressibility C_P is defined by the [SCREF](#) keyword

Trapped gas

When modeling foamy oil it may be necessary to model a component that has a gas-like compressibility in the solid phase (see "Foamy oil"). This can be done using keywords [CCTYPE](#). In this case the component molar solid volume is given by the gas law

Thermal only

$$V_{sol}^c = \frac{Z^c RT}{P} \quad \text{Eq. 4.188}$$

where R is the gas constant and the compressibility terms Z^c can be set for each component using the `ZFACTOR` keyword.

Keyword `CCTYPES` can be used to set the compressibility at surface conditions.

Specific heats

Thermal only

The specific heat C_{sol}^c of a component c in the solid phase is given by

$$C_{sol}^c = C_{sol1}^c + C_{sol2}^c \cdot (T - T_{st})$$

where

the coefficients are defined by keywords `SPECHS` and `SPECHT`

T_{st} is the standard temperature defined by `STCOND`

the specific heat must always be positive.

Solid saturations

When solids are present the pore volume V_p is taken up by both fluids and solids

$$V_p = V_f + V_s = V_p \cdot (V_{flu} + V_{sol})$$

where the specific fluid volume is given by

$$V_{flu} = V_{oil} + V_{gas} + V_{wat}$$

Normalized fluid saturations are defined as fractions of the fluid volume

$$S_o = \frac{V_{oil}}{V_{flu}}, S_g = \frac{V_{gas}}{V_{flu}}, S_w = \frac{V_{wat}}{V_{flu}}$$

The normalized saturations satisfy the equation:

$$S_o + S_g + S_w = 1$$

Saturations can also be defined as fractions of the pore volume:

$$\hat{S}_s = V_{sol}$$

$$\hat{S}_o = V_{oil} = V_{flu} \cdot S_o = (1 - \hat{S}_s) \cdot S_o$$

$$\hat{S}_g = V_{gas} = V_{flu} \cdot S_g = (1 - \hat{S}_s) \cdot S_g$$

$$\hat{S}_w = V_{wat} = V_{flu} \cdot S_w = (1 - \hat{S}_s) \cdot S_w$$

These modified saturations satisfy the equation:

$$\hat{S}_o + \hat{S}_g + \hat{S}_w + \hat{S}_s = 1$$

The solid saturation $\hat{S}$ and the multiplier $1 - \hat{S}$ are used to calculate permeability reduction and initial conditions.

Adsorption model of deposition and erosion

Solid components can be modeled either as suspended in the oil phase, or as adsorbed by the formation rocks.

The suspended solid concentration C_s is defined as the weight of solids per unit pore volume

$$C_s = f_s \cdot W_s \quad \text{Eq. 4.189}$$

where

f_s is the fraction of solids in suspension

W_s is the total weight of solids per unit pore volume.

The adsorbed solid concentration C_a is defined as the weight of solids per weight of rock:

$$C_a = (1 - f_s) \cdot W_s / W_R \quad \text{Eq. 4.190}$$

where

W_R is the weight of rock per unit pore volume:

$$W_R = \rho_R \cdot (1 - \theta) / \theta \quad \text{Eq. 4.191}$$

θ is the porosity and

ρ_R is the rock density given by the [ROCKDEN](#) keyword.

Keyword [SOLIDADS](#) is used to define an adsorption function:

$$C_a = C_a(C_s) \quad \text{Eq. 4.192}$$

which allows equations [4.189](#), [4.190](#) and [4.191](#) to be solved for the suspended solids fraction f_s .

There are two special cases

- If the [SOLIDADS](#) keyword is not used, it is assumed that all solids are adsorbed, $f_s = 0$.
- If the [SOLIDADS](#) keyword is used, but the rock density is zero, then it is assumed that all solids are in suspension, $f_s = 1$.

Output of adsorbed solid concentration and suspended solids fraction

The adsorbed solid concentration C_a and suspended solids fraction f_s can be output to the print file or restart file by using the mnemonics [SOLADS](#) and [SSFRAC](#) with the [RPTRST](#), [RPTSCHED](#) and [RPTSOL](#) keywords. Values in individual grid blocks can be output to the summary file by using the keywords [BSOLADS](#) and [BSSFRAC](#).

Adsorption example: the Langmuir isotherm

An example of an adsorption function is given by the Langmuir isotherm, which is used to describe the adsorption of gas at a fixed temperature. The same function can also be used to describe solid adsorption. The formula for the isotherm is derived by assuming the rate at which solids are adsorbed by the rocks is equal to the rate at which solids are desorbed.

If the rate at which solids are adsorbed per unit pore volume R_a is given by:

$$R_a = K_a \cdot C_s \cdot (C_{amax} - C_a),$$

where

C_{amax} is the maximum adsorbed concentration and

R_a is the adsorption rate

and if the rate at which solids are desorbed per unit pore volume is given by:

$$R_d = K_d \cdot C_a,$$

where K_d is the deposition rate;

then setting the adsorption rate equal to the desorption rate leads to an adsorption function given by the Langmuir isotherm:

$$C_a = C_{amax} \cdot \frac{C_s}{K + C_s}$$

where K is the ratio of the desorption rate to the adsorption rate $K = K_d/K_a$

```
--Langmuir isotherm, Camax = 0.01kg/kg; K = .05 kg/m3
SOLIDADS
--Cs      Ca
--kg/m3   kg/kg
0.00      0.00000
0.01      0.00167
0.0125    0.00200
0.0333    0.00400
0.05      0.00500
0.075     0.00600
0.15      0.00750
0.20      0.00800
0.35      0.00875
0.45      0.00900
0.95      0.00950
/
```

Adsorption of dissolved water components

In the previous section, the physical processes of deposition and erosion are modeled by adsorption functions. Adsorption can also be used to model the chemical process where a dissolved solid comes out of solution and is adsorbed to the rock. In this case a water component is used to represent the dissolved solid and a solid component is used to represent the component in the adsorbed state.

A reversible chemical reaction is used to model the dissolution process. See ["Equilibrium deviation reaction rates"](#).

The equilibrium value can be set in two different ways:

- Using the [EQLDKVCR](#) keyword to specify an equilibrium k-value for the water component.
- Using the [EQLDTAB](#) keyword to specify the equilibrium value for the adsorbed solid as a function of the water mole fraction of the dissolved component.

The reaction can be one way, to model just adsorption, or just desorption, or reversible, to model both processes simultaneously.

If the reaction rate is set to a high value, the adsorption / desorption process will be close to instantaneous, but this may be harder to converge. In this case a smaller reaction rate may improve the run, although it will take longer to reach equilibrium.

If the water component and solid component occupy different volumes, the adsorption reaction will change the fluid + solid volume which will lead to a pressure change to compensate. This can be avoided by setting similar values for the water and solid reference densities and compressibilities (see for example [DREFW](#) and [SDREF](#)).

In Eclipse 100, keywords such as [ALKADS](#), [FOAMADS](#), [PLYADS](#) and [SURFADS](#) define an adsorbed concentration as a function of a dissolved concentration $C_a = C_a(C_d)$. This data can be converted for use with [EQLDTAB](#). The water mole fraction of the dissolved components is given by:

$$a_d = \frac{MW_w}{\rho_{ws}} \cdot \frac{C_d}{MW_d}$$

where

MW_w and MW_d are the molecular weight of the water mixture and the dissolved component (which are the same if the dissolved component is treated as a tracer), and

ρ_{ws} is the surface density of water.

Permeability reduction caused by plugging

When a solid phase is present the mobility M_p^c for component c in phase p is given by

$$M_p^c = x_p^c \cdot k_s \cdot k_{rp}(S_p) \cdot \frac{b_p}{\mu_p}$$

where

x_p^c is the mole fraction of component c in phase p ,

k_s is the mobility multiplier due to the solids,

k_{rp} is the relative permeability of phase p

S_p is the normalized saturation of phase p ,

b_p is the molar density of phase p , and

μ_p is the viscosity of phase p .

The term k_s represents the reduction in mobility caused by adsorbed solids plugging the pores. There are two ways of entering k_s data. The [SOLIDMMC](#) keyword allows the mobility multiplier to be entered as a function of the adsorbed concentration:

$$k_s = k_{sc}(C_a).$$

Alternatively the [SOLIDMMS](#) keyword can be used to enter the mobility multiplier as a function of adsorbed saturation S_a :

$$k_s = k_{ss}(S_a), S_a = (1 - f_s) \cdot \hat{S}_s.$$

Output of the mobility multiplier

The mobility multiplier k_s can be output to the print file or restart file by using the mnemonic **SMMULT** with the **RPTRST**, **RPTSCHED** and **RPTSOL** keywords. Values of the mobility multiplier in individual grid blocks can be output to the summary file by using the keyword **BSMMULT**.

Solid wettability effects

Adsorbed solids can change the wettability of the rock. Two sets of saturation functions can be specified representing different wettability conditions. An interpolation parameter can be defined as a function of the adsorbed wetting-solid concentration using the **SOLWTAB** keyword. The concentration of wetting-solid is calculated by

$$C_{wa} = (1 - f_s) \left[\sum w^c m^c M W^c \right] / W_R$$

Here w^c is a wettability weight factor given by the **SOLWW** keyword. A value of $w^c = 0$ means the solid component has no effect on wettability, while a value of $w^c = 1$ means the full weight of the component is included in the wettability calculation.

The saturation functions corresponding to low adsorbed solids are identified using the **SATNUM** keyword, and the saturation functions corresponding to high adsorbed solids are identified using the **SOLWNUM** keyword. In this model the adsorbed solids only affect oil and water saturation functions (relative permeability and capillary pressure). The gas saturation functions are determined by the **SATNUM** functions alone.

This model cannot be used with other saturation function options, such as hysteresis, **ODD3P** and **IKU3P**. However it can be used with the oil-water miscibility model described in the "**Surfactant Model**".

Output of the solid wettability data

The wetting-solid concentration, C_{wa} , and the solid wettability interpolation parameter can be output to the print file or restart file by using the mnemonics **SOLADW** and **SOLWET** with the **RPTRST**, **RPTSCHED** and **RPTSOL** keywords. Values of the concentration and interpolation parameter in individual grid blocks can be output to the summary file by using the keywords **BSOLADW** and **BSOLWET**.

Transport of suspended solids

Solids adsorbed to the rock are immobile, but the suspended fraction will be transported with the oil phase. It is assumed that the concentration of suspended solids is small, so that they can be treated as a tracer - that is the oil flow is not modified by the presence of suspended solids, and the mobility of a solid component c is given by

$$M_o^c = x_{so}^c \cdot k_s \cdot k_{ro}(S_o) \cdot \frac{b_o}{\mu_o}$$

where x_{so}^c is the number of moles of component c in the solid phase that are suspended in the oil phase, per mole of oil

$$x_{so}^c = \frac{f_s \cdot m^c}{S_o \cdot b_o}$$

Initial solids

Initial values for fluid components can be determined in three ways:

- explicitly user-defined
- calculated by the simulator from equilibrium conditions
- read from restart files

In restart runs, initial values of solid components are read from the restart file.

For solid components, explicit initialization is used in cases 1 and 2. This is because the solid components do not affect the flash and therefore do not affect the equilibration. The molar concentration of component c is given by

$$m^c = (1 - \hat{S}_s) \cdot m_{fl}^c + \hat{S}_s \cdot b_s \cdot x_s^c \quad \text{Eq. 4.193}$$

where

$\hat{S}_s$ is the initial solid saturation, defined by keyword `SSOLID`

x_s^c is the initial solid molar fraction, defined by keyword `SMF`

m_{fl}^c is the concentration of the fluid components, set either explicitly or by equilibration.

For solid components the initial fluid concentration (which can be specified by `ZI`, `ZMFVD`, `COMPVD`, `XMF`, `YMF`, `ZMF` or `NEI`) should be zero. For fluid components the initial solid concentration should be zero.

If the initial solid saturation is not defined then a default value of zero is used and equation 4.193 becomes:

$$m^c = m_{fl}^c$$

Initial porosity

The terms $(1 - \hat{S}_s)$ and $\hat{S}_s$ in equation 4.193 represent the fractions of the pore-volume initially containing fluid and solid. The initial pore volume, which is usually calculated from the porosity θ should therefore include both a fluid volume and a solid volume:

$$\theta = \frac{V_f + V_s}{V_f + V_s + V_r}$$

where V_r is the rock volume.

In some circumstances the measured porosity θ' may only take into account the initial fluid volume

$$\theta' = \frac{V_f}{V_f + V_s + V_r} = \theta \cdot (1 - \hat{S}_s)$$

If these values are used to define the porosity (`PORO`) then the initial pore volumes should in theory be multiplied by a factor $1/(1 - \hat{S}_s)$ using `MULTPV`. However if the initial solid saturations are small or zero this term can be neglected.

Initial energy

Thermal only

In thermal runs the initial energy density e is calculated from:

$$V_b \cdot e = V_p \cdot ((1 - \hat{S}_s) \cdot e_{fl} + \hat{S}_s \cdot b_s \cdot e_s) + V_r \cdot e_r \quad \text{Eq. 4.194}$$

where

V_b , is the bulk volume $V_b = V_p + V_r$

e_{fl} is the fluid energy density

e_s is the solid energy density,

e_r is the rock energy density.

Chemical reactions

Solid components can be used in chemical reactions, see ["Reactions involving solid components"](#). This provides a mechanism for modeling the appearance and disappearance of solid-only components.

When the reservoir contains solids, the reaction rates will depend on the solid saturation. This dependence can be modified by using the [REACSORD](#) and [REACSMAX](#) keywords.

Solid production

If suspended solids are allowed, then solids will be produced by producer wells. Solid production data can be output to the summary file using the keywords specified in [Grid block solid quantities](#) and [Solid production totals](#) in the *Eclipse Reference Manual*.

Gas plants and NGLs

	Eclipse 100
x	Eclipse 300

A gas plant table may be used instead of a flash calculation for a separator stage (see item 9 of keyword [SEPCOND](#)). In almost all cases, a single stage only will be required. Gas plant tables are defined with keyword [GPTABLE](#), [GPTABLEN](#) or [GPTABLE3](#). The table defined with [GPTABLE](#) specifies the oil recovery fraction (the number of moles of liquid output per mole of feed) for each component as a function of the 'heavy' mole fraction of the feed. The components that comprise the heavy fraction are also defined in the table.

The [GPTABLEN](#) keyword should be used instead of [GPTABLE](#) if NGL (natural gas liquid) output is also required. The table defined with [GPTABLEN](#) specifies both the oil recovery fractions and the NGL recovery fractions for each component as a function of the heavy mole fraction of the feed. The [DNGL](#) keyword should also be used to set the partial densities in the NGL stream.

The [GPTABLE3](#) keyword defines a table with recovery fractions for oil, NGL and gas for each component as a function of the heavy mole fraction of the feed. Use this keyword instead of [GPTABLE](#) or [GPTABLEN](#) if it is required that, for one or more components, the sum of the oil, NGL and gas recovery fractions is not unity. You can use this keyword to reduce or exclude one or more components in the post separator oil, NGL or gas streams. The [DNGL](#) keyword should also be used to set the partial densities in the NGL stream.

The way in which the gas plant table splits the input streams is described below:

Suppose the input composition to the gas plant is z , and the oil, NGL and gas recovery factors are R_o , R_{NGL} and R_g . Then the moles of each component in the oil, NGL and gas streams are:

$$m_{oi} = R_{oi} z_i \quad \text{Eq. 4.195}$$

$$m_{NGLi} = R_{NGLi} z_i \quad \text{Eq. 4.196}$$

$$m_{gi} = R_{gi} z_i \quad \text{Eq. 4.197}$$

Where R_g is either taken directly from [GPTABLE3](#), or calculated from the requirement that the factors sum to unity when using [GPTABLE](#) or [GPTABLEN](#).

To obtain the oil, NGL and gas compositions, the streams are normalized:

$$x_i = \frac{m_{oi}}{\sum m_{oi}} \quad \text{Eq. 4.198}$$

$$n_i = \frac{m_{NGLi}}{\sum m_{NGLi}} \quad \text{Eq. 4.199}$$

$$y_i = \frac{m_{gi}}{\sum m_{gi}} \quad \text{Eq. 4.200}$$

The molar output rates of oil, NGLs and gas per mole of feed are given by:

$$L = \sum_i m_{oi} \quad \text{Eq. 4.201}$$

$$N = \sum_i m_{NGLi} \quad \text{Eq. 4.202}$$

$$V = \sum_i m_{gi} \quad \text{Eq. 4.203}$$

In order to define a volume rate of NGLs, the molar density must be obtained. This uses Amagat's law which states that the mixture molar volume is the mole-fraction weighted sum of the component molar volumes:

$$V_{mix} = \sum_i n_i V_i \quad \text{Eq. 4.204}$$

The partial molar volumes are obtained from the density values entered with keyword DNGL, using

$$V_i = \frac{Mw_i}{\rho_i} \quad \text{Eq. 4.205}$$

Production control

Production wells may be made subject to NGL production limits or set NGL production rate targets using the [WCONPROD](#), [WCONHIST](#) and [WELTARG](#) keywords.

The recovery plant option

The recovery plant option takes all the gas exported from the field and processes this into NGLs and remaining gas. All other field quantities, such as GOR and gas-injection are reported prior to this processing. The aim of this facility is to model processing of the gas produced by the reservoir through facilities such as coolers, which can extract additional liquids from the stock tank gas.

Recovery plant calculations are turned on if the [RECOVERY](#) keyword is encountered. The recovery plant table may be specified in the [SOLUTION](#) or [SCHEDULE](#) sections, and may be re-specified during the [SCHEDULE](#) section.

The [RECOVERY](#) keyword has two arguments, followed by a table of one or more rows of data.

The first argument specifies the lower component index used to define the 'heavy' fraction. This defaults to the last component.

The second argument specifies the upper component index used to define the 'heavy' fraction. This defaults to the last component.

The table consists of $N_{comps} + 1$ columns, where N_{comps} is the number of components in the run, listing the NGL recovery fractions of each component as a function of the 'heavy' mole fraction. The recovery fraction is the number of moles of NGL per mole of feed for each component.

The defaults for items 1 and 2 imply that the recovery plant table is normally composed of recovery fractions against the mole fraction of the last component in the feed.

An example would be:

```
RECOVERY
6 9
--Z9 CO2 N2 C1 C2 C3 C4-6 CZ7 CZ8 CZ9
0.0 0.0 0.002 0.001 0.45000 0.65 0.98 1.0 1.0 1.0
0.2 0.0 0.002 0.001 0.48000 0.68 0.99 1.0 1.0 1.0
/
```

The feed used by the recovery table is the export gas from the field, which is the total gas production minus gas used for fuel and for reinjection.

The volume of NGL produced is obtained using Amagat's law, adding the partial molar volumes. The molar density used is that entered with DNGL.

The rate of production of recovered NGLs and of the remaining gas is reported in the printed output, and may be obtained from the [SUMMARY](#) section using the following keywords:

FNRR NGL recovery volume rate

FNRT NGL recovery volume total

FNRM NGL recovery molar rate

FGRR gas remaining volume rate

FGRT gas remaining volume total

FGRM gas remaining molar rate

FXNR NGL recovery composition

FYGR gas remaining composition

The rates and totals for the recovery process are also output to the PRINT file under the REPORT header, in the table headed 'NGL recovery'. The compositions are reported in the same section in the table headed 'Recovered NGLs and remaining gas'.

Multiphase flash

	Eclipse 100
x	Eclipse 300

Multiphase components can be used in EoS runs for asphaltene prediction. These components can appear in the solid, liquid or gas phases.

Using multiphase

The **SOLID** keyword must be used if multiphase components are present.

Multiphase components

The simulator can be used to predict when asphaltene precipitation first occurs. At the end of each timestep a multiphase flash is performed to determine whether a stable asphaltene phase exists. The multiphase flash first determines whether one or more liquid phases are stable. A liquid phase is then identified as an asphaltene phase if:

- the heaviest hydrocarbon is an aromatic (see the **HYDRO** keyword), and
- the heaviest hydrocarbon represents more than a third of the phase (by weight).

The saturation, density and molar density of the asphaltene phase are stored in the simulator as solid properties. At present the asphaltene phase is ignored in the flow calculations, so this model should only be used as an indicator of whether asphaltene precipitation occurs in the reservoir.

Asphaltene prediction is carried out in compositional runs using an Equation of State, if the **SOLID** keyword is used, and if the heaviest hydrocarbon is an aromatic (see the **HYDRO** keyword).

Multiphase keywords

RUNSPEC

Keyword	Description
MULTIPHA	Requests the multiphase flash which can be used to predict asphaltene precipitation.
SOLID	Requests Solids option.

PROPS

- HYDRO** Define multiphase components.

SUMMARY section keywords

Refer to [Grid block solid quantities](#) in the *Eclipse Reference Manual*.

SCHEDULE section keywords

Solid properties that can be output with **RPTRST**, **RPTSCHED** and **RPTSOL**

Keyword	Description
BSOL	Solid reservoir molar densities.

Keyword	Description
DENS	Solid reservoir density.
SSOLID	Solid saturations.

Temperature option

x	Eclipse 100
x	Eclipse 300

The temperature option enables Eclipse to model temperature effects, for example the injection of cold water into a hot reservoir.

The major effect of temperature changes in the vicinity of the injection wells is to modify the fluid viscosities. In addition, changes in the reservoir temperature will induce additional stresses within the reservoir which may modify the rock properties. Eclipse can output the temperature distribution to the RESTART file; this may be used by a separate program to model the effect of the temperature change on the reservoir rock.

The temperature option is initiated by the keyword `TEMP` in the RUNSPEC section.

Particularly for dual porosity calculations, you should refer to the documentation on the thermal option in "[Dual porosity](#)" for information on how the rock volume is handled in the presence of more than one porosity.

Eclipse 100 and 300 differences

Differences between the temperature option in Eclipse 100 and Eclipse 300

The temperature option is handled differently in the two simulators. In Eclipse 100 temperature is treated like a tracer, while in Eclipse 300 the `TEMP` keyword invokes the thermal option, see "[Thermal option](#)". This leads to a number of important differences:

- In Eclipse 100 an energy conservation equation is solved at the end of each converged timestep, and the grid block temperatures are updated. The new temperatures are then used to calculate the oil and water viscosities for the subsequent timestep. In Eclipse 300 the energy conservation equation is solved simultaneously with the flow equations. In both cases, the rock and all fluids in a grid block are assumed to be at the same temperature.
- In Eclipse 300 water is allowed in the vapor phase.
- In Eclipse 300 the equilibrium state depends on the temperature.
- In Eclipse 300 the reservoir temperature changes if the reservoir pressure changes. Similarly the injection temperature changes as the well BHP changes.
- The Eclipse 300 thermal option is designed for a number of specialized processes, including steam injection, hot fluid or gas injection, well bore heaters and combustion. Note that when modeling steam injection, it should not be used for pressures or temperatures above the critical point of water (221.2 bar / 3208 psi, 374.15°C / 705.5°F)

These differences mean that there will be a difference in performance between the two simulators, and there may be some difference in the results.

Differences between the temperature option in Eclipse 300 and the thermal option

The Thermal option can be used to run Temperature option data sets by replacing the `TEMP` keyword with the `THERMAL` and `BLACKOIL` keywords. This uses exactly the same code as the Eclipse 300 temperature option, but allows `THERMAL` keywords to be used as well. For example, heat loss, steam injection and heater keywords can all be added to the data set.

Thermal properties

The minimum thermal properties required by temperature option are the specific heat capacity of the rock and the fluids present in the reservoir. The rock specific heat is entered as a volume specific heat, tabulated against temperature, using the [SPECROCK](#) keyword. The fluid specific heats are mass specific heats, again tabulated against temperature, entered using the [SPECHEAT](#) keyword.

Viscosity modification

Both the oil and water viscosities can optionally be specified as functions of temperature using the [OILVISCT](#) and [WATVISCT](#) keywords. The viscosities are supplied at the reference conditions specified using the [VISCREF](#) keyword. The oil viscosity at the prevailing temperature, pressure and gas/oil ratio is calculated as follows:

$$\mu_o(T, P, R_s) = \mu_T(T) \cdot \frac{\mu_p(P, R_s)}{\mu_p(P_{ref}, R_{sref})} \quad \text{Eq. 4.206}$$

where

μ_T is the viscosity from the [OILVISCT](#) keyword

μ_p is the viscosity from the [PVTO](#) (or [PVDO](#)) keyword

P_{ref} is the reference pressure defined by the [VISCREF](#) keyword

R_{sref} is the reference gas/oil ratio defined by the [VISCREF](#) keyword.

Similarly, the water viscosity at the prevailing temperature, pressure and optionally salt concentration is calculated as follows:

$$\mu_w(T, P) = \mu_T(T) \cdot \frac{\mu_p(P)}{\mu_p(P_{ref})} \quad \text{Eq. 4.207}$$

or, if the brine option is active, as

$$\mu_w(T, P, c_s) = \mu_T(T) \cdot \frac{\mu_p(P, c_s)}{\mu_p(P_{ref}, c_{sref})} \quad \text{Eq. 4.208}$$

where

μ_T is the viscosity from the [WATVISCT](#) keyword

μ_p is the viscosity from the [PVTW](#) or [PVTWSALT](#) keyword

P_{ref} is the reference pressure defined by the [VISCREF](#) keyword

c_{sref} is the reference salt concentration defined by item 2, record 1 of the [PVTWSALT](#) keyword.

Thermal conduction

The thermal conduction of heat in the rock is often a small effect compared with the convection of heat with the injected water. Thermal conduction is optional and is initiated by using either the [THCONR](#)

keyword, or for Eclipse 300, keywords [THCROCK](#), [THCOIL](#), [THCWATER](#), [THCGAS](#) and [THCSOLID](#), in the grid section, to specify the cell rock and fluid thermal conductivity.

Eclipse 100

The thermal conductivity and thermal transmissibility can be printed out using the [THCONR](#) and [TRTHER](#) arguments in the [RPTGRID](#) keyword. The thermal transmissibilities calculated by Eclipse can be modified using inter-region thermal transmissibility multipliers specified using keywords [MULTREGH](#) and [MULTNUM](#).

Eclipse 300

The average cell thermal conductivity can be printed out using the argument [THCAVE](#) in the [RPTGRID](#) keyword. The thermal transmissibility can be printed out using the [HEATTX](#), [HEATTY](#), and [HEATTZ](#) arguments in the [RPTGRID](#) keyword. The thermal transmissibilities calculated by Eclipse can be modified using keywords [HEATTX](#), [HEATTY](#) and [HEATTZ](#).

Heat losses/gains

No account is taken of heat loss to and from areas outside the reservoir model. If these heat losses are likely to be significant, the reservoir model must be extended to include large blocks to act as heat sinks. These additional blocks have to be active cells (pore volume > zero), but the permeability may be set to zero.

Equilibrium state

Eclipse 300

In Eclipse 300 a thermal flash is performed to determine the state. K-values are calculated from R_s and R_v values as functions of pressure and temperature:

$$K_{oil} = \frac{R_s + F}{1 + F \cdot R_v} \cdot R_v \cdot e^{D_{oil} \cdot (1/T_0 - 1/T)}$$

$$K_{gas} = \frac{R_s + F}{1 + F \cdot R_v} \cdot \frac{1}{R_s} \cdot e^{D_{gas} \cdot (1/T_0 - 1/T)}$$

The temperature T_0 is taken to be the initial reservoir temperature [RTEMPA](#) and the coefficients for the temperature dependence D_{oil} and D_{gas} are taken from equation 4.250 with default values of the acentric factors and critical temperatures given in table 4.11.

The constant F is the ratio of the molar volumes of gas to oil at surface conditions:

$$F = \frac{V_{gas,s}}{V_{oil,s}} = \frac{MW^{gas}}{MW^{oil}} \cdot \frac{\rho_{oil,s}}{\rho_{gas,s}}$$

Values of $\rho_{oil,s}$ and $\rho_{gas,s}$ are taken from the [DENSITY](#) keyword, while default values for the molecular weights are given in the following table.

Property	Symbol	Units	Oil	Gas
Acentric factor	Ac		0.348	0.056
Critical temperature	Tc	K	603	200
Molecular weight	MW	kg/kg-M	120	25

Table 4.11: Default values used in calculating K-values

Setting the initial conditions

The initial temperature is either assumed to be constant throughout the reservoir or can be specified as function of depth. If the initial temperature distribution is specified as a function of depth then the reservoir will not be in thermal equilibrium as there are no heat sources/sinks at the top and bottom of the model. The initial temperature is set using either the [RTEMPA](#) or [RTEMPVD](#) keywords in the `SOLUTION` section.

Wells

Eclipse 100

In Eclipse 100 the temperature of the injected fluid can be specified for each well in the `SCHEDULE` section, using the [WTEMP](#) keyword. If a well does not have an injection temperature specified, the fluid is taken to be at zero degrees.

Eclipse 300

In Eclipse 300 the enthalpy of the injected water must be specified. This can be done by specifying the temperature and pressure of the injected water for each well in the `SCHEDULE` section, using the [WINJTEMP](#) keyword.

Solution output

Eclipse 100

In Eclipse 100 the temperature can be output using the `TEMP` argument in [RPTSCHED](#), and the field and region energy balance reports can be output by using the `FIPTEMP` argument.

Eclipse 300

In Eclipse 300 temperature data can be output using Thermal option keywords, see "[Thermal keywords](#)".

Summary output

Eclipse 100

Refer to [Temperature option quantities](#) in the *Eclipse Reference Manual*.

Note that the heat in place is quoted relative to the initial conditions.

Eclipse 300

In Eclipse 300 temperature data can be output using Thermal option keywords, see "[Thermal keywords](#)".

Restrictions

In Eclipse 300 this option is only valid for black oil simulation using the [BLACKOIL](#) keyword.

If a cell is defined with zero pore volume in the `GRID` section, then making it active by changing the pore volume in the `EDIT` section using [PORV](#), is not allowed with the temperature option.

Convergence issues

Eclipse 100

The effects of unconverged timesteps forced to be accepted by Eclipse 100 might be significant although they tend to dissipate rapidly. These anomalies can be corrected if desired by choosing smaller timesteps that ensure the convergence of the solution.

Note: The [MESSOPTS](#) keyword can be used to control the severity type for messages in the PRT file that report timesteps that are forced to be accepted. Note also that the total number of timesteps forced to be accepted is output in the Error Summary.

Example data set

The example data sets HEAT100.DAT and HEAT300.DAT which are in the directory e300\data\thermal can be used with Eclipse 100 and Eclipse 300 respectively.

Thermal option

	Eclipse 100
x	Eclipse 300

Thermal recovery methods are typically used in heavy oil reservoirs, where the oil viscosity is high at reservoir temperatures, but reduces as the temperature increases. A number of thermal recovery processes can be simulated in Eclipse. These include:

- steam injection, such as cyclic steam injection (huff and puff), steam flood, or steam assisted gravity drainage (SAGD)
- hot fluid or gas injection
- well bore heaters (see ["Heaters"](#))
- combustion (see ["Combustion"](#)).

Four phases are allowed in the simulation:

- an oil phase that contains only hydrocarbon components,
- a gas phase that can contain both hydrocarbon and water components, and
- a water phase that contains only a water component,
- a solid phase (for use with chemical reactions, ["Reactions involving solid components"](#))

‘Foamy oil’ can be modeled using the thermal option. Although foamy oils occur at reservoir temperatures, the phenomenon is associated with heavy oils, so is included with the thermal option, see ["Foamy oil"](#).

The thermal option can be used for dual porosity grids, but the combination of the thermal option and dual porosity makes extra assumptions about the distribution of the rock volume between the porosities: see ["Dual porosity"](#).

The thermal option can be run in one of three modes:

- live oil, using K-values to define equilibrium,
- dead oil, when the hydrocarbon components are non-volatile,
- black oil, using Temperature option keywords, see ["Temperature option"](#).

The next section contains a list of keywords that can only be used with the Thermal live oil and dead oil options, and those Eclipse keywords that have special meaning when used with the Thermal option. This is followed by sections describing the thermal properties and features that can be modeled. Example data sets are also provided.

Formulation

The equations used to describe thermal processes are similar to those used in the compositional simulation, ["Formulation of the equations"](#), but with three important differences: the addition of an energy variable and an energy equation; the presence of a water component in the gas phase as well as the water phase; and temperature dependence of properties.

The thermal simulator does not use an equation of state to determine the thermodynamic properties. Instead you must supply K-values to determine equilibrium and densities, viscosities and enthalpies for each component in each phase.

Primary variables

The primary solution variables, used for cells in all phase states, are

$$X = (P, m_1, \dots, m_N, m_w, e)$$

where

P is the pressure,

$m_1, \dots, m_N$ are molar densities of each hydrocarbon component,

m_w is the molar density of the water component, and

e is the bulk internal energy density.

The molar densities of the fluid components are measured as moles per unit reservoir volume, while the energy density is measured as energy per unit bulk volume. The bulk volume includes both the reservoir volume and the rock volume.

Conservation equations

The primary variables are found by solving conservation equations for each component, for energy and for volume. These equations are calculated using a fully implicit scheme.

The non-linear residual, R_{fl} , is derived from the component conservation equation for each fluid component (hydrocarbon and water) in each grid block at each timestep:

$$R_{fl} = \frac{d}{dt} (V_p m_{fl}) + F_{fl} + Q_{fl} = 0 \quad \text{Eq. 4.209}$$

where

V_p is the pore volume,

F_{fl} is the net flow rate into neighboring grid blocks and

Q_{fl} is the net flow rate into wells during the timestep.

The non-linear residual, R_e , is derived from the energy conservation equation in each grid block at each timestep:

$$R_e = \frac{d}{dt} (V_b e) + F_e + C_e + Q_{HL} + Q_e = 0 \quad \text{Eq. 4.210}$$

where

V_b is the bulk volume,

F_e is the convective enthalpy flow rate into neighboring grid blocks,

C_e is the conductive energy flow rate into neighboring grid blocks,

Q_{HL} is the conductive energy flow rate to the surrounding rocks (heat loss), and

Q_e is the net enthalpy flow rate into wells during the timestep.

The non-linear residual, R_v , is derived from the conservation of volume in each grid block at each timestep:

$$R_v = V_p - V_f = 0 \quad \text{Eq. 4.211}$$

where V_f is the fluid volume.

Thermodynamic equilibrium condition

The phase properties used in the residual and Jacobian equations (that is the fluid volume and the flow terms) can be calculated from the component properties once the mole fractions of each component in each phase have been determined.

The number of moles of each component (including water) can be written as

$$m_c = m_t \cdot L \cdot x_o^c + m_t \cdot V \cdot x_g^c + m_t \cdot W \cdot x_w^c \quad \text{Eq. 4.212}$$

where

$m_t \cdot L \cdot x_o^c$, $m_t \cdot V \cdot x_g^c$ and $m_t \cdot W \cdot x_w^c$ are the number of moles in the oil, vapor and water phases.

m_t is the total fluid molar density given by

$$m_t = \sum_c m_c + m_w \quad \text{Eq. 4.213}$$

L , V and W are the oil, vapor and water mole fractions and

x_p^c are the component phase fractions.

Two simplifying conditions are imposed, that there is no water in the oil phase, and no hydrocarbon in the water phase:

$$x_w^c = 0, c = 1, \dots, N$$

$$x_w^w = 1,$$

if $W > 0$

It is also assumed the three phases are in thermodynamic equilibrium, determined by user-defined K-values:

$$x_g^c = K^c(P, T) \cdot x_o^c \quad \text{Eq. 4.214}$$

$$x_g^w = K^w(P, T) \cdot x_w^w \quad \text{Eq. 4.215}$$

The K-values are functions of pressure P and temperature T only.

Using these equations the mole fractions of each component in each phase can be expressed in terms of the primary variables and the four flash variables, L , V , W , and T :

$$x_o^c = \frac{m_c}{m_t} \cdot \frac{1}{L + K^c \cdot V} \quad c = 1, \dots, N \quad L > 0 \quad \text{Eq. 4.216}$$

$$x_g^c = \frac{m_c}{m_t} \cdot \frac{K^c}{L + K^c \cdot V} \quad c = 1, \dots, N \quad V > 0 \quad \text{Eq. 4.217}$$

$$x_g^w = K^w \quad V > 0 \quad W > 0 \quad \text{Eq. 4.218}$$

Total mole fractions

The total mole fractions, which are used in equations 4.216 and 4.217 are defined as

$$z_c = \frac{m_c}{m_T} \quad z_w = \frac{m_w}{m_T}$$

Equation 4.213 can be written in terms of the mole fractions

$$\sum_c z_c + z_w = 1 \quad \text{Eq. 4.219}$$

The total mole fractions differ from the hydrocarbon mole fractions:

$$\zeta_c = \frac{m_c}{m_T - m_w} = \frac{z_c}{1 - z_w} \quad \text{Eq. 4.220}$$

Substituting this into equation 4.219 gives

$$\sum_c \zeta_c = 1 \quad \text{Eq. 4.221}$$

Thermal flash

The four flash variables can be found from the primary variables by solving the four flash equations in each cell:

$$\sum_c x_o^c = 1 \quad \text{if } L > 0 \quad \text{Eq. 4.222}$$

$$\sum_c x_g^c + x_g^w = 1 \quad \text{if } V > 0 \quad \text{Eq. 4.223}$$

$$L + V + W = 1 \quad \text{Eq. 4.224}$$

and

$$V_b \cdot e = V_p \cdot m_t \cdot (L \cdot e_o + V \cdot e_g + W \cdot e_w) + (V_b - V_p) \cdot e_r \quad \text{Eq. 4.225}$$

where

e_o , e_g and e_w are the user-defined molar energies of the oil, gas and water phases, and

e_r is the user-defined energy per unit volume of rock.

Fluid volume

The specific volume of each phase can be determined from user-defined molar densities for each phase (b_o , b_g and b_w):

$$V_o = \frac{m_t L}{b_o}, \quad V_g = \frac{m_t V}{b_g}, \quad V_w = \frac{m_t W}{b_w}, \quad \text{Eq. 4.226}$$

The volume balance residual equation 4.211 can then be written as:

$$R_v = V_p - V_p \cdot (V_o + V_g + V_w) = 0 \quad \text{Eq. 4.227}$$

The phase saturations S_o , S_g and S_w are defined as

$$S_o = \frac{V_o}{V_o + V_g + V_w}, S_g = \frac{V_g}{V_o + V_g + V_w}, S_w = \frac{V_w}{V_o + V_g + V_w}, \quad \text{Eq. 4.228}$$

When equation 4.227 holds, the phase saturations are equal to the phase specific volumes. Therefore

$$S_o = \frac{m_o L}{b_o} \quad S_g = \frac{m_g V}{b_g} \quad S_w = \frac{m_w W}{b_w} \quad \text{Eq. 4.229}$$

The number of moles of each component equation 4.212 and the energy density, equation 4.225 can now be written in terms of the saturations:

$$m_c = S_o \cdot b_o \cdot x_o^c + S_g \cdot b_g \cdot x_g^c + S_w \cdot b_w \cdot x_w^c \quad \text{Eq. 4.230}$$

$$V_b \cdot e = V_p \cdot (S_o \cdot b_o \cdot e_o + S_g \cdot b_g \cdot e_g + S_w \cdot b_w \cdot e_w) + (V_b - V_p) \cdot e_r \quad \text{Eq. 4.231}$$

These two equations can be used for initializing the simulation - when the primary solution variables need to be calculated from user-defined saturations.

Inter-block flows

The net flow rate of a hydrocarbon or water component c from cell i into neighboring cells is obtained by summing the component flow over all phases p and over the neighboring cells n ,

$$F_{ci} = \sum_n \gamma_{ni} \sum_p (M_p^c dP_{pni}) \quad \text{Eq. 4.232}$$

The net flow rate of energy from cell i into neighboring cells is obtained in a similar manner:

$$F_{ei} = \sum_n \gamma_{ni} \sum_p (M_p^e dP_{pni}) \quad \text{Eq. 4.233}$$

In these formulae

γ_{ni} is the transmissibility between cells n and i ,

M_p^c is the generalized mobility of component c in phase p ,

M_p^e is the generalized mobility of enthalpy in phase p and

dP_{pni} is the potential difference of phase p between cells n and i .

The generalized mobilities are given by

$$M_p^c = x_p^c \cdot k_{rp}(S_p) \cdot \frac{b_p}{\mu_p} \quad \text{Eq. 4.234}$$

and

$$M_p^e = H_p \cdot k_{rp}(S_p) \cdot \frac{b_p}{\mu_p} \quad \text{Eq. 4.235}$$

where

x_p^c is the mole fraction of component c in phase p ,

H_p is the molar enthalpy in phase p .

k_{rp} is the relative permeability of phase p ,

S_p is the saturation of phase p ,

b_p is the molar density of phase p , and

μ_p is the viscosity of phase p

The fluid mobilities are evaluated in the upstream cell for each phase p separately (cell n if dP_{pni} is positive, cell i if dP_{pni} is negative). The upstream calculation applies separately for each phase (oil, water, gas) so that, for example, oil may flow from cell i to cell n while water flows from cell n to cell i .

The potential difference is made up of a pressure term, a capillary pressure term and a hydrostatic term,

$$dP_{pni} = P_n - P_i + P_{cpn} - P_{cpi} - \rho_{pni} g (D_n - D_i) \quad \text{Eq. 4.236}$$

where

P_{cp} is the capillary pressure for phase p ,

ρ_p is the mass density of phase p ,

g is the acceleration due to gravity, and

D is the cell center depth.

Well flows

The rate of flow of component c into a production well w from cell i is obtained by summing the component flow over all phases p :

$$Q_{cwi} = -\mathcal{I}_{wi} \cdot \sum_p (P_{pi} - H_{iw} - P_{bh}) \cdot M_{pi}^c \quad \text{Eq. 4.237}$$

Similarly the rate of flow of energy into a production well w from cell i is

$$Q_{ewi} = -\mathcal{I}_{wi} \cdot \sum_p (P_{pi} - H_{iw} - P_{bh}) \cdot M_{pi}^e \quad \text{Eq. 4.238}$$

Here

$\mathcal{I}_{wi}$ is the well connection transmissibility factor,

H is the hydrostatic head correction,

P_{bh} is the bottom hole pressure,

M_{pi}^c is the generalized mobility of component c in phase p and

M_{pi}^e is the enthalpy mobility in phase p .

See ["Well inflow performance"](#) for further information.

Heat conduction

The heat conduction term for each cell i is given by summing conduction between all neighboring cells n :

$$C_{ei} = - \sum_n \Psi_{ni} \Lambda (T_i - T_n) \quad \text{Eq. 4.239}$$

where

Ψ_{ni} is the conduction transmissibility between cells n and i

Λ is the upstream saturation dependent conductivity multiplier.

Solid phase

Solid components can be used in thermal simulations. A solid component cannot also appear in the liquid or gas phase, but it can be created by chemical reactions, see ["Chemical reactions"](#). If solid components are modeled, then a number of modifications are required to the formulation described above.

The volume balance residual equation [4.211](#) is modified to include the solid volume V_s :

$$R_v = V_p - V_f - V_s = 0$$

The total fluid molar density m_t is still calculated by equation [4.213](#), but the sum is taken over the fluid components only.

The energy density equation, [4.225](#), is modified to include the solid energy e_s :

$$V_b \cdot e = V_p \cdot m_t \cdot (L \cdot e_o + V \cdot e_g + W \cdot e_w) + V_p \cdot e_s + (V_b - V_p) \cdot e_r$$

Equation [4.227](#), which is derived from the volume balance equation, is replaced by

$$R_v = V_p - V_p \cdot (V_o + V_g + V_w + V_{sol}) = 0$$

where V_{sol} is the specific solid volume.

The equations for the saturations, equation [4.228](#) are modified to include the solid volume:

$$\hat{S}_o = \frac{V_o}{V_o + V_g + V_w + V_{sol}} = \frac{V_f}{V_f + V_{sol}} \cdot S_o$$

$$\hat{S}_g = \frac{V_g}{V_o + V_g + V_w + V_{sol}} = \frac{V_f}{V_f + V_{sol}} \cdot S_g$$

$$\hat{S}_w = \frac{V_w}{V_o + V_g + V_w + V_{sol}} = \frac{V_f}{V_f + V_{sol}} \cdot S_w$$

$$\hat{S}_{sol} = \frac{V_{sol}}{V_o + V_g + V_w + V_{sol}}$$

The modified saturations satisfy

$$\hat{S}_o + \hat{S}_g + \hat{S}_w + \hat{S}_{sol} = 1$$

Equation 4.228 is still used to define normalized saturations S_o , S_g and S_w , representing the fraction of the fluid volume that is in each fluid phase. The normalized saturations satisfy

$$S_o + S_g + S_w = 1$$

The normalized saturations are used in calculating the relative permeabilities in equations 4.234 and 4.235.

Solution method

For each cell there are $N + 3$ variables (pressure, N hydrocarbon components, one water component, and energy) and $N + 3$ equations ($N + 1$ component conservation equations, including water, equation 4.209, energy conservation, equation 4.210 and volume balance, equation 4.211). The equations can be solved using a fully implicit method (**FULLIMP** / default), an implicit pressure, explicit properties method (**IMPES**) or a combination of both, (**AIM**).

The component conservation equations are written as

$$R_{fl} = \frac{V_p(P_t + \Delta t) \cdot m_{fl}(t + \Delta t) - V(P_t) \cdot m_{fl}(t)}{\Delta t} + F_{fl} + Q_{fl} = 0$$

where, in an implicit cell the flow terms and well terms are evaluated at the new time:

$$F_{fl} = F_{fl}(P_t + \Delta t, T_t + \Delta t, S_t + \Delta t)$$

$$Q_{fl} = Q_{fl}(P_t + \Delta t, T_t + \Delta t, S_t + \Delta t)$$

and in the IMPES method the flow terms and well terms are evaluated using the new pressure but with properties at the previous timestep:

$$F_{fl} = F_{fl}(P_t + \Delta t, T_t, S_t)$$

$$Q_{fl} = Q_{fl}(P_t + \Delta t, T_t, S_t)$$

The energy conservation equation is dealt with in a similar manner, but the volume balance equation is always evaluated at the new timestep.

The non-linear residual equations

$$R(X) = 0$$

are solved iteratively,

$$X_n = X_{n-1} + \Delta X_{n-1} \quad \text{Eq. 4.240}$$

The solution increment ΔX_{n-1} is found by solving the linear matrix equation

$$0 = R(X_{n-1}) + J \cdot \Delta X_{n-1}$$

where J is the Jacobian matrix:

$$J = \left(\frac{\partial R}{\partial X} \right) \Big|_{X = X_{n-1}}$$

The diagonal Jacobian element corresponding to an IMPES cell has a simple form which can be solved relatively easily:

$$J = \begin{matrix} & (P & m_1 & m_2 & \dots & e) \\ \text{Comp1} & \begin{bmatrix} x & x & 0 & \dots & 0 \end{bmatrix} \\ \text{Comp2} & \begin{bmatrix} x & 0 & x & \dots & 0 \end{bmatrix} \\ \dots & \begin{bmatrix} \dots & \dots & \dots & \dots & \dots \end{bmatrix} \\ \text{Energy} & \begin{bmatrix} x & 0 & 0 & \dots & x \end{bmatrix} \\ \text{Volume bal} & \begin{bmatrix} x & x & x & \dots & x \end{bmatrix} \end{matrix}$$

Linear solvers

To compute the solution increment ΔX the linear system involving the Jacobian matrix must be inverted. Whilst the default WARP linear solver may be adequate for relatively simple simulations better performance is usually achieved with the JALS solver, which is used by default for thermal simulations. The JALS linear solver employs the constrained pressure residual (CPR) technique in which an approximate pressure system is solved for each outer iteration. The solution of the pressure system is referred to as the inner iteration. The GMRES algorithm is used for both the outer and inner iteration. The GMRES algorithm is similar to the ORTHOMIN algorithm and requires storing a set of search directions. The JALS solver options can be controlled with the [JALS](#) keyword, which can be inserted in the RUNSPEC section.

Thermal properties

Rocks

Normally in Eclipse, a cell is inactive if its porosity is zero. However, in the thermal option, rock filled cells are considered as active cells since they conduct heat. Therefore, a cell is inactive if both its pore volume and rock volume are zero, or, if the [ACTNUM](#) keyword has been used to make it inactive.

Rock heat capacity

The enthalpy per unit volume of rock is given by

$$H_r = C_{r0}(T - T_{\text{ref}}) + \frac{C_{r1}(T - T_{\text{ref}})^2}{2} \quad \text{Eq. 4.241}$$

where

the volumetric heat capacity C_{r0} is defined by the [HEATCR](#) keyword,

the temperature coefficient C_{r1} is defined by the [HEATCRT](#) keyword

the reference temperature T_{ref} is specified with the [STCOND](#) keyword.

Thermal conductivity

There are two ways to define the thermal conductivity, triggered using either the [THCONR](#) or [THCROCK](#) keyword in the model:

- Using keyword [THCONR](#) to define the thermal conductivity of a rock and liquid filled cell, and optionally [THCONSF](#)
- Using keyword [THCROCK](#), and depending on the phases present in the model, the phase thermal conductivities, [THCOIL](#), [THCWATER](#), [THCGAS](#) and [THCSOLID](#).

For both options, we define the cell average thermal conductivity, $THCAVE$, which is used to determine the thermal transmissibilities. Depending on the option activated, thermal transmissibilities can be multiplied by a saturation-dependent factor. More details are supplied below.

The keywords from these two separate options must not be mixed together in the data set.

Option 1: using $THCONR$

The $THCONR$ keyword defines the thermal conductivity of a rock and liquid filled cell.

It can be calculated from the rock, oil, gas and water conductivities (K_r , K_o , K_g , K_w) by first determining an average liquid conductivity

$$K_l = (K_o + K_w)/2$$

The conductivity for a liquid filled cell can then be calculated as a porosity weighted average of the liquid and rock conductivities:

$$THCAVE = THCONR = \theta \cdot K_l + (1-\theta) \cdot K_r$$

$THCAVE$ is used to calculate the thermal transmissibilities in the reservoir.

Option 2: using $THCROCK$

Depending on the phases in the model, the average cell thermal conductivity ($THCAVE$) is defined as a porosity weighted average of the phase and rock conductivities using the expression below:

$$THCAVE = \varphi \frac{(THCOIL + THCGAS + THCWATER + THCSOLID)}{NPHAS} + (1-\varphi)THCROCK$$

In this expression, φ is the grid porosity; $NPHAS$ denotes the number of phases in the model; $THCOIL$, $THCGAS$, $THCWATER$ and $THCSOLID$ specify the phase thermal conductivities; and $THCROCK$ specifies the rock thermal conductivity.

If some phases are not present in the model, then the relevant phase thermal conductivity will be ignored.

$THCAVE$ is used to calculate the thermal transmissibilities in the reservoir.

Thermal transmissibility

Thermal transmissibilities are calculated in a similar manner to the flow transmissibilities, with the thermal conductivity being used instead of the permeabilities. For example, using corner point geometry, the thermal transmissibility between cells i and j is given by $H^{i \rightarrow j}$ where

$$1/H^{i \rightarrow j} = 1/H_i + 1/H_j \quad \text{Eq. 4.242}$$

$$H_i = THCAVE_i \cdot \frac{A \cdot D_i}{D_i \cdot D_i} \quad \text{Eq. 4.243}$$

$$A \cdot D_i = A_X \cdot D_{iX} + A_Y \cdot D_{iY} + A_Z \cdot D_{iZ} \quad \text{Eq. 4.244}$$

and

$$D_i \cdot D_i = D_{iX}^2 + D_{iY}^2 + D_{iZ}^2 \quad \text{Eq. 4.245}$$

Here

A_X , A_Y and A_Z are the X-, Y- and Z- projections of the mutual interface area of cell i and cell j ,

and D_{iX} , D_{iY} and D_{iZ} are the X-, Y- and Z- components of the distance between the center of cell i and the center of the relevant face of cell i .

See "Transmissibility Calculations" for further details.

Thermal transmissibilities can also be entered directly, using the keywords [HEATTX](#), [HEATTY](#), [HEATTZ](#) (or [HEATTR](#), [HEATTTHT](#), [HEATTZ](#) in radial coordinates).

Definition of multiplier for Heat Conduction

Option 1: using THCONR

Thermal transmissibilities can be multiplied by a saturation dependent factor Λ . Following [Coats \(1980\)](#), it is assumed that the thermal transmissibility is a function of the gas saturation, S_g , only. This method does not distinguish between the thermal transmissibility of a rock filled with oil and the thermal transmissibility of a rock filled with water.

The multiplier Λ is defined to be:

$$\Lambda = (1 - \alpha S_g) \quad \text{Eq. 4.246}$$

where α is specified by the [THCONSF](#) keyword.

From the definition of THCONR above, a value for α can be determined as:

$$\alpha = \theta \cdot (K_l - K_g) / \text{THCONR}$$

To check these values, we can calculate the conductivities for a liquid filled cell $S_g = 0$ and a gas filled cell $S_g = 1$:

$$\text{THCONR} \cdot \Lambda(0) = \theta \cdot K_l + (1 - \theta) \cdot K_r$$

$$\text{THCONR} \cdot \Lambda(1) = \theta \cdot K_g + (1 - \theta) \cdot K_r$$

which are just the porosity weighted averages of the rock and fluid conductivities

Option 2: using THCROCK

Thermal transmissibilities are multiplied by a saturation dependent factor Λ . For this option, Λ takes the form:

$$\Lambda = (\varphi(\sum_p S_p K_p) + (1 - \varphi)K_{\text{rock}}) / \text{THCAVE}$$

In the above expression, φ is the porosity; [THCOIL](#) is denoted by K_{oil} , [THCGAS](#) is denoted by K_{gas} , [THCWATER](#) is denoted by K_{water} , [THCSOLID](#) is denoted by K_{solid} and [THCROCK](#), specifying the rock thermal conductivity, is denoted by K_{rock} . The phase saturations are represented by S_p .

For a perfectly homogeneous reservoir, with $\text{THCAVE}_i = \text{THCAVE}_j$, the THCAVE term in the Λ formula will cancel the THCAVE term in the thermal transmissibilities, and the heat flow will be proportional to $(\varphi(\sum_p S_p K_p) + (1 - \varphi)K_{\text{rock}})$. However, for heterogeneous reservoirs, the THCAVE terms will not cancel out, since the transmissibility uses a harmonic average of neighbor cell THCAVE values, while the Λ formula uses the upstream cell THCAVE value.

Dual porosity or dual permeability

A matrix-fracture thermal conductivity can be set using the [THCONMF](#) keyword. The [SIGMATH](#) keyword can be used to set the multiplier used in calculating the matrix-fracture thermal transmissibilities.

Hysteretic rock compaction

During high pressure cyclic steam injection (huff and puff), the rock in the reservoir may be stressed beyond its elastic limits. A simple hysteretic model of rock compaction, the `BOBERG` option of keyword `ROCKCOMP`, designed specifically for these circumstances, can be used: see "[Rock Compaction](#)".

Relative permeability curves

Keywords `ENPTVT`, `ENKRVT`, and `ENPCVT` specify the temperature-dependence of relative permeability curves. The form of these keywords is similar to the standard Eclipse 100/300 end-point-scaling keywords `ENPTVD`, `ENKRVD`, and `ENPCVD`. In the thermal option, end-point-scaling keywords cannot be used in the same data set as temperature-dependent relative permeability keywords.

Live oil

For a live oil run you must specify temperature dependent properties for each component. The properties required for live components are, K-values to determine equilibrium between the oil and gas phase, and the phase enthalpies, densities and viscosities. These properties are output to the print file in tabular form. The program `PVT i` can be used to output K-values, enthalpy and density data.

Volatility type

In a live oil run, not all the components need be live. The `CVTYPE` keyword can be used to define a component as non-volatile, non-condensable or solid.

Non-volatile and non-condensable components

A component can be made non-volatile or non-condensable using the `CVTYPE` keyword.

For versions of E300 prior to 2023.1, this keyword also reset the component critical temperature, to 10^4 for non-volatile components and to 10^{-6} for non-condensable components. From 2023.1, this is no longer the case and the user-supplied critical temperature will be used in any calculations where it is needed, for instance Lorentz-Bray-Clark viscosity.

For back-compatibility with earlier versions of Eclipse, components with critical temperatures of 10^4 and above are also assumed to be non-volatile, while those with temperatures of 10^{-6} are assumed to be non-condensable.

For surface calculations, the volatility types are set by default to the reservoir values. If different volatility types are required at the surface, the `CVTYPES` keyword can be used. In this case the `TCRITS` keyword should also be used to ensure the correct critical temperatures are used in the surface calculations.

Solid phase components

A component can be made a solid using the `CVTYPE` keyword. Solid components have density and enthalpy. A solid component cannot appear in the oil phase or the liquid phase, so its K-values can be defaulted.

K-values

Distribution of volatile components between the oil and gas phases is determined by K-values. The mole fraction y^c of volatile component c in the gas phase is given by $y^c = K^c(P, T) \cdot x^c$, where x^c is the mole fraction of component c in the oil phase and $K^c(P, T)$ is the equilibrium K-value, which depends on the pressure P and the temperature T . Four methods are available for specifying oil component K-values:

- Keyword **KVCR** uses the K-value correlation given in *Coats (1980)* and *Crookston et al (1979)*, where the K-value for each oil component is given by

$$K(P, T) = (A + B/P + C \cdot P) \cdot e^{-D/(T-E)} \quad \text{Eq. 4.247}$$

Some care needs to be taken when using this formula to ensure that K is always positive and increases as T increases. A simplified version that avoids these problems can be used:

$$K(P, T) = (B/P) \cdot e^{-D/T} \quad \text{Eq. 4.248}$$

Values for B and D can be calculated by setting

$$K = P_{sat}/P$$

and finding the best straight line

$$\ln(P_{sat}) = \ln(B) - D/T$$

through a set of points $(1/T, P_{sat})$. Suggested values for some components are given in the following tables.

		B (Psi)	D (°R)
C1	Methane	168900	1904
C2	Ethane	170400	3062
C3	Propane	528400	4571
C6	Hexane	686800	6598
C10	Decane	1214000	9966

Table 4.12: K-value coefficients (Field units)

		B (Bars)	D (K)
C1	Methane	11650	1058
C2	Ethane	11750	1701
C3	Propane	36440	2539
C6	Hexane	47360	3666
C10	Decane	83740	5537

Table 4.13: K-value coefficients (Metric units)

- Keyword **KVWI** uses the Wilson correlation (*Wilson, 1968*), where the K-value for each oil component is given by

$$K(P, T) = e^{5.372697 \cdot (1 + A_c) \cdot (1 - T_{crit}/T)} \cdot P_{crit}/P \quad \text{Eq. 4.249}$$

This is the same as equation 4.248 with coefficients

$$B = e^{5.372697 \cdot (1 + A_c)} \cdot P_{crit}$$

$$D = 5.372697 \cdot (1 + A_c) \cdot T_{crit} \quad \text{Eq. 4.250}$$

determined from the acentric factor A_c , critical temperature T_{crit} and critical pressure P_{crit} , which are defined by keywords **ACF**, **TCRIT** and **PCRIT** respectively.

- Keywords **KVTABTn** and **KVTEMP** can be used to enter the K-values as tables.

- Keywords [KVG](#)EN and [KVTEMP](#) can be used to generate the K-values tables, using an equation of state, if one of the keywords [CGDTYPE](#), [CODTYPE](#), [CGVTYPE](#) or [COVTYPE](#) are used (activating input using an equation of state (EoS)).

Surface K-values

The default K-values for surface calculations are the reservoir K-values. Different values can be used if required:

- If keyword [KVCR](#) is used to set the reservoir K-values, keyword [KVCRS](#) can be used to set the surface K-values
- If keyword [KVWI](#) is used to set the reservoir K-values, keywords [ACFS](#), [TCRITS](#) and [PCRITS](#) can be used to set the coefficients of the Wilson correlation, equation 4.249 for surface conditions.

Density

Oil phase density

The molar density b_o of the oil phase is given by:

$$b_o = 1/V_{oil} \quad \text{Eq. 4.251}$$

where the molar specific volume V_{oil} of the oil phase is calculated using Amagat's law of partial volumes:

$$V_{oil} = \sum_{c=1}^{N_{comps}} x^c V_{oil}^c \quad \text{Eq. 4.252}$$

and x^c is the mole fraction of component c in the oil phase. The component oil phase volume V_{oil}^c is given by:

$$V_{oil}^c = \frac{MW^c}{\rho^c} \quad \text{Eq. 4.253}$$

where

MW^c is the molecular weight of component c given by the [MW](#) keyword, and

the component oil phase density ρ^c is given by:

$$\rho^c = \frac{\rho_{ref}^c}{(1 + C_{T1}(T - T_{ref})) (1 - C_p(P - P_{ref}))} \quad \text{Eq. 4.254}$$

The reference density ρ_{ref}^c is defined by the [DREF](#) keyword; the standard temperature and pressure, T_{ref} and P_{ref} are defined by keywords [TREF](#) and [PREF](#) (or by [STCOND](#)); the thermal expansion coefficient C_{T1} is defined by the [THERMEX1](#) keyword; and the component isothermal compressibility C_p is defined by the [CREF](#) keyword.

Oil phase density using an EoS

An equation of state (EoS) can be used to compute the specific molar volume for the oil phase. The keyword [CODTYPE](#) specifies the components that form the mixture for the EoS calculation.

Let $x_e^c = \begin{cases} x^c & \text{for } c \text{ in EOS} \\ 0 & \text{otherwise} \end{cases}$ as defined by the keyword `CODTYPE`

and $x_n^c = \begin{cases} 0 & \text{for } c \text{ in EOS} \\ x^c & \text{otherwise} \end{cases}$ the components not included by the EoS calculation

then $x_{esum} = \sum_{c=1}^{N_{comps}} x_e^c$ and $x_{emix}^c = \frac{x_e^c}{x_{esum}}$.

The oil specific molar volume is computed as

$$V_{oil} = \sum_{c=1}^{N_{comps}} x_n^c V_{oil}^c + x_{esum} V_{eos}(P, T, x_{emix}^1, \dots, x_{emix}^{N_{comps}})$$

There are two options when computing the EoS specific molar volume. Using the value `EMIX` for a component in `CODTYPE`, the lowest root (Liquid root) of the cubic equation of state is chosen. For a mixture that cannot exist as an oil under the given pressure and composition, this volume could be unphysical. Another choice is to let the volume used correspond to the stable phase if the composition $x_{emix}^1, \dots, x_{emix}^{N_{comps}}$ was one-phase, by choosing the root with the lowest Gibbs free energy. If the value `EGFE` is selected for a component in `CODTYPE`, this choice is applied. Typically this could lead to a gas-phase-like specific volume being used together with the overall oil specific volume.

Note: When only a subset of the components are selected for the EoS, usually all heavy components should be selected.

Oil phase density at the surface

For surface calculations, the optional keywords `MWS`, `DREFS`, `TREFS` and `PREFS` can be used to supply the values used in equations 4.253 and 4.254. The default, if these are not supplied, is to use the reservoir values given by `MW`, `DREF`, `TREF` and `PREF` (or `STCOND`).

Gas phase density

The molar density of the gas phase b_g is given by:

$$b_g = 1/V_{gas} \quad \text{Eq. 4.255}$$

where the molar specific volume V_{gas} of the gas phase is

$$V_{gas} = \sum_{c=1}^{N_{comps}} y^c V_{gas}^c + y^w V_{gas}^w \quad \text{Eq. 4.256}$$

and y^c is the mole fraction of component c in the gas phase. Each hydrocarbon component gas phase molar volume V_{gas}^c is obtained from the gas law,

$$PV_{gas}^c = Z^c RT \quad \text{Eq. 4.257}$$

The component Z -factor Z^c is given by an empirical formula

$$Z^c = Z_0^c - Z_1^c \cdot \frac{P}{RT} \quad \text{Eq. 4.258}$$

where the coefficients Z_0^c and Z_1^c are user-specified with the `ZFACTOR` and `ZFACT1` keywords.

The molar specific water vapor volume V_{gas}^w used in equation 4.256 is calculated from the steam density.

Determining Z-factor coefficients from the Joule-Thomson coefficient

For an ideal gas the coefficients Z_0^c and Z_1^c are 1 and 0 respectively. The default values are in fact 0.96 and 0, which represents a non-ideal gas, but with constant Z-factors at reservoir conditions. This simple approach can be extended by calculating average reservoir values of Z_0^c in PVT *i* and exporting the data using the `ZFACTOR` keyword. However, in some circumstances it may be necessary to use a temperature dependent Z-factor by specifying values of the coefficient Z_1^c . It can be seen by substituting equation 4.258 into 4.257 that the coefficient Z_1^c has dimensions volume / mole:

$$V_{gas}^c = Z_0^c \cdot \frac{RT}{P} - Z_1^c \quad \text{Eq. 4.259}$$

Z_1^c can be determined from the Joule-Thomson coefficient μ . For a steady state (constant enthalpy) gas flow the change of temperature as a result of a pressure change is given by *Abbott and van Ness (1989)*:

$$\mu = \left(\frac{\partial T}{\partial P} \right)_H = \frac{1}{(MW \cdot C_p)} \cdot \left(T \cdot \left(\frac{\partial V}{\partial T} \right)_P - V \right) \quad \text{Eq. 4.260}$$

where MW and C_p are the fluid molecular weight and specific heat. Substituting equations 4.256 and 4.260 gives

$$\mu = \sum_{c=1}^{N_{comps}} y^c \cdot \mu^c + y^w \cdot \mu^w \quad \text{Eq. 4.261}$$

where

$$\mu^c = \frac{Z_1^c}{(MW_{gas} \cdot C_{gas})} \quad \text{Eq. 4.262}$$

Hence the coefficient Z_1^c can be determined using equation 4.262 from the component Joule-Thomson coefficient μ^c and the average molecular weight and heat capacity of the gas phase. The coefficient Z_1^c is also used to calculate the component enthalpy, equations 4.268 and 4.281.

Gas phase density using an EoS

An equation of state (EoS) can be used to calculate the specific molar volume for the gas phase. The keyword `CGDTYPE` specifies the components that form the mixture for the EoS calculation.

Let $y_e^c = \begin{cases} y^c & \text{for } c \text{ in EOS} \\ 0 & \text{otherwise} \end{cases}$ as define by the keyword `CGDTYPE`

and $y_n^c = \begin{cases} 0 & \text{for } c \text{ in EOS} \\ y^c & \text{otherwise} \end{cases}$ the components not included by the EoS calculation

then $y_{esum} = \sum_{c=1}^{N_{comps}} y_e^c$ and $y_{emix}^c = \frac{y_e^c}{y_{esum}}$.

The gas specific molar volume is computed as

$$V_{gas} = \sum_{c=1}^{N_{comps}} y_n^c V_{gas}^c + y_{esum} V_{eos}(P, T, y_{emix}^1, \dots, y_{emix}^{N_{comps}}) + y^w V_{gas}^w$$

There are two options when computing the EoS specific molar volume. By using the value `EMIX` for a component in `CGDTYPE`, the highest root (Vapor root) of the cubic equation of state is chosen. For a mixture that cannot exist as a gas under the given pressure and composition this volume could be unphysical. Another choice is to let the volume used correspond to the stable phase if the composition $y_{emix}^1, \dots, y_{emix}^{N_{comps}}$ was one-phase, by choosing the root having the lowest Gibbs free energy. If the value `EGFE` is selected for a component in `CGDTYPE`, this choice is applied. Typically this could lead to an oil-phase-like specific volume being used together with the overall gas specific volume.

Note: When only a subset of the components are selected for the EoS, usually all light components should be selected.

Foamy oil

When modeling foamy oil it may be necessary to model a component that has a gas-like compressibility in the oil phase (see "Foamy oil"). This can be done using keywords `CCTYPE`. In this case equation 4.253 is replaced by:

$$V_{oil}^c = \frac{Z^c RT}{P} \quad \text{Eq. 4.263}$$

Keyword `CCTYPES` can be used to set (or reset) the compressibility at surface conditions.

Solid phase density

If a solid phase is modeled, the molar specific volume of solid phase components V_{sol} is calculated in a similar manner to the oil volume:

$$V_{sol} = \sum_{c=1}^{N_{comps}} s^c V_{sol}^c \quad \text{Eq. 4.264}$$

where s^c is the mole fraction of component c in the solid phase. The component solid phase volume V_{sol}^c is given by:

$$V_{sol}^c = \frac{MW^c}{\rho^c} \quad \text{Eq. 4.265}$$

where MW^c is the molecular weight of component c given by the `MW` keyword, and the component solid phase density ρ^c is given by:

$$\rho^c = \frac{\rho_{ref}^c}{(1 + C_{T1}(T - T_{ref})) (1 - C_P(P - P_{ref}))} \quad \text{Eq. 4.266}$$

The reference density ρ_{ref}^c is defined by the `SDREF` keyword; the standard temperature and pressure, T_{ref} and P_{ref} are defined by keywords `STREF` and `SPREF` (or by `STCOND`); the thermal expansion coefficient C_{T1} is defined by the `STHERMX1` keyword; and the component isothermal compressibility C_P is defined by the `SCREF` keyword.

Enthalpy

The enthalpy of the oil phase is calculated using a mole fraction weighted average of the component enthalpies:

$$H_{oil} = \sum_{c=1}^{N_{comps}} x^c \cdot MW^c \cdot H_{oil}^c(T) \quad \text{Eq. 4.267}$$

where x^c is the mole fraction of component c in the oil phase and MW^c is the component molecular weight given by the `MW` keyword.

The enthalpy of the hydrocarbon components in the gas phase is calculated using a mole fraction weighted average of the component enthalpies:

$$H_{gas} = \sum_{c=1}^{N_{comps}} y^c \cdot (MW^c \cdot H_{gas}^c(T) + H_{JT}^c(P)) \quad \text{Eq. 4.268}$$

where y^c is the mole fraction of component c in the gas phase. The gas phase component enthalpies include both a temperature dependent term H_{gas}^c and a pressure dependent (Joule-Thomson) term H_{JT}^c .

The enthalpy of the hydrocarbon components in the solid phase is calculated using a mole fraction weighted average of the component enthalpies:

$$H_{sol} = \sum_{c=1}^{N_{comps}} s^c \cdot MW^c \cdot H_{sol}^c(T) \quad \text{Eq. 4.269}$$

where s^c is the mole fraction of component c in the solid phase.

The component enthalpies in the oil and gas phases are calculated from component specific heats and heats of vaporization. Specific heat values must be given for each hydrocarbon component in at least one fluid phase. If they are only specified in one phase, then the heat of vaporization should be given. The following options are therefore available for the fluid phase enthalpies:

- Set the specific heat of components in the oil and gas phases.
- Set the specific heat of components in the oil phase and the heat of vaporization.
- Set the specific heat of components in the gas phase and the heat of vaporization.

Solid phase enthalpies are only required in simulations where a solid phase is present. The component enthalpies in the solid phase are calculated from component specific heats.

Specific heats

The specific heats C_{oil}^c , C_{gas}^c and C_{sol}^c of a component c in the oil, gas and solid phases are given by

$$C_{oil}^c = C_{oil1}^c + C_{oil2}^c \cdot (T - T_{st}) \quad \text{Eq. 4.270}$$

$$C_{gas}^c = C_{gas1}^c + C_{gas2}^c \cdot (T - T_{st}) \quad \text{Eq. 4.271}$$

$$C_{sol}^c = C_{sol1}^c + C_{sol2}^c \cdot (T - T_{st}) \quad \text{Eq. 4.272}$$

where the oil phase coefficients are defined by keywords `SPECHA` and `SPECHB`, the gas phase coefficients are defined by keywords `SPECHG` and `SPECHH` and the solid phase coefficients are defined by keywords

[SPECHS](#) and [SPECHT](#). T_{st} is the standard temperature defined by [STCOND](#). The specific heats must always be positive.

If oil phase specific heats are defined, the molar enthalpy of component c in the oil phase is:

$$H_{oil}^c = \int_{T_{st}} C_{oil}^c dT = C_{oil1}^c \cdot (T - T_{st}) + \frac{1}{2} \cdot C_{oil2}^c \cdot (T - T_{st})^2 \quad \text{Eq. 4.273}$$

For temperatures above the critical temperature, $T > T_{crit}^c$, the oil phase molar enthalpy is set equal to the gas phase molar enthalpy.

If gas phase specific heats are defined, the molar enthalpy of component c in the gas phase is:

$$\begin{aligned} H_{gas}^c &= H_{vaps}^c + \int_{T_{st}} C_{gas}^c dT \\ &= H_{vaps}^c + C_{gas1}^c \cdot (T - T_{st}) + \frac{1}{2} \cdot C_{gas2}^c \cdot (T - T_{st})^2 \end{aligned} \quad \text{Eq. 4.274}$$

where H_{vaps}^c is the heat of vaporization at the standard temperature, defined by [HEATVAPS](#). The oil and gas phase component enthalpies are illustrated in the following figure.

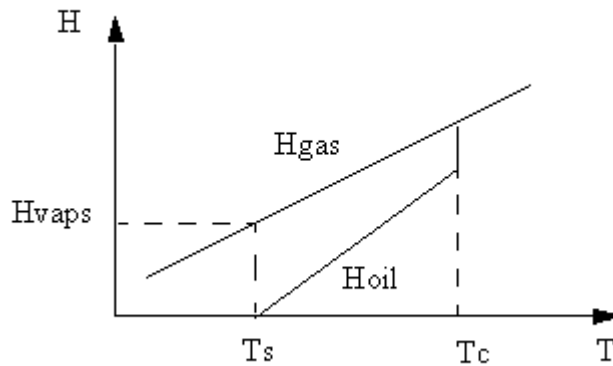


Figure 4.1. Component enthalpy defined by specific heats

If a solid phase is simulated, the molar enthalpy of component c in the solid phase is

$$H_{sol}^c = \int_{T_{st}} C_{sol}^c dT = C_{sol1}^c \cdot (T - T_{st}) + \frac{1}{2} \cdot C_{sol2}^c \cdot (T - T_{st})^2 \quad \text{Eq. 4.275}$$

Heat of vaporization

A temperature dependent heat of vaporization $\Delta H^c(T)$ can be specified instead of either the oil or gas specific heat.

If the specific heat of a component c in the oil phase is not specified, then the oil phase enthalpy is calculated from the gas component enthalpy by:

$$H_{oil}^c = H_{gas}^c - \Delta H^c(T) \quad \text{Eq. 4.276}$$

This is illustrated in figure 4.2 where a linear function has been used for the heat of vaporization $\Delta H^c(T)$.

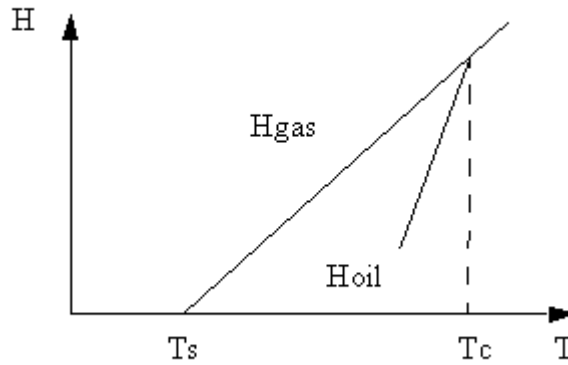


Figure 4.2. Component enthalpy defined by gas specific heat and heat of vaporization

Alternatively, if the specific heat of a component c in the gas phase is not specified, then the gas phase enthalpy is calculated from the oil component enthalpy

$$H_{gas}^c = H_{oil}^c + \Delta H^c(T) \quad \text{Eq. 4.277}$$

This is illustrated in figure 4.3 where a linear function has been used for the heat of vaporization $\Delta H^c(T)$.

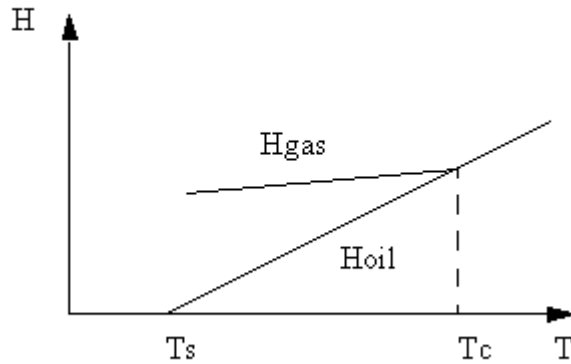


Figure 4.3. Component enthalpy defined by oil specific heat and heat of vaporization

The heat of vaporization is given by [Edminster and Lee \(1984\)](#):

$$\Delta H^c(T) = A \cdot (1 - T/T_{crit}^c)^B \quad T < T_{crit}^c \quad \text{Eq. 4.278}$$

$$\Delta H^c(T) = 0 \quad T \geq T_{crit}^c \quad \text{Eq. 4.279}$$

where the constant A is defined by the [HEATVAP](#) keyword, the exponent B is defined by the [HEATVAPE](#) keyword, and T_{crit}^c is the component critical temperature defined by the [TCRIT](#) keyword.

Heats of vaporization are usually obtained at the normal boiling point T_{nb} . The constant A , defined by the HEATVAP keyword can be found by inverting equation 4.278:

$$A = \frac{\Delta H^c(T_{nb})}{(1 - T_{nb}/T_{crit})^B} \quad \text{Eq. 4.280}$$

where $\Delta H^c(T_{nb})$ is the heat of vaporization at the normal boiling point. The exponent B , entered with keyword HEATVAPE, is defaulted to 0.38, which is the value found in the literature ([Edminster and Lee, 1984](#)). However, this can lead to convergence problems, as discussed in the next section.

Pressure dependent gas enthalpy

The gas enthalpy equation 4.268 includes a pressure dependent term H_{JT}^c . This term is only used when the temperature dependent terms in the oil and gas enthalpies are set using specific heats (equations 4.273 and 4.274). It is not used when the heat of vaporization HEATVAP is used (equation 4.276 or 4.277). The pressure dependent term is given by

$$H_{JT}^c = -Z_1^c \cdot (P - P_{st}) \quad \text{Eq. 4.281}$$

where Z_1^c is specified with the ZFACT1 keyword and P_{st} is the standard pressure defined by STCOND.

Restrictions on specifying enthalpy data

The component enthalpies must obey the following relationships

$$H_{gas}^c \geq H_{oil}^c \text{ for } T < T_{crit}^c$$

$$\frac{dH_{oil}^c}{dT} > 0$$

$$\frac{dH_{gas}^c}{dT} > 0$$

Eclipse will issue an error message if any of these relationships does not hold. Errors can occur when specific heats are incorrectly specified, see Figure 4.4; or when using a non-linear heat of vaporization, see figure 4.5. These problems can be avoided by specifying specific heats for both phases, with the oil specific heats equal to the gas specific heats figure 4.6.

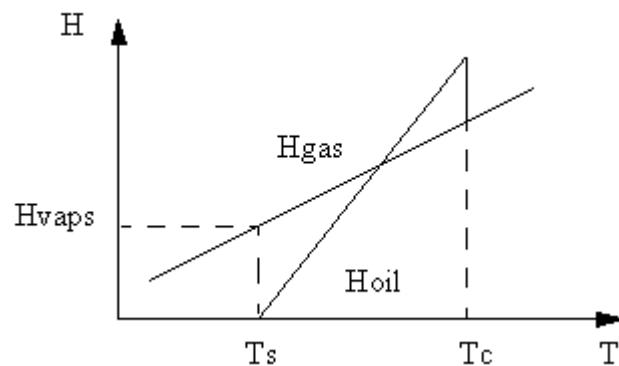


Figure 4.4. Specific heat data leading to $H_{oil} > H_{gas}$

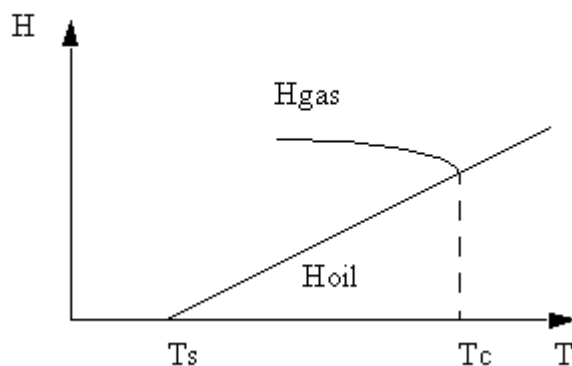


Figure 4.5. Heat of vaporization data leading to decreasing gas enthalpy

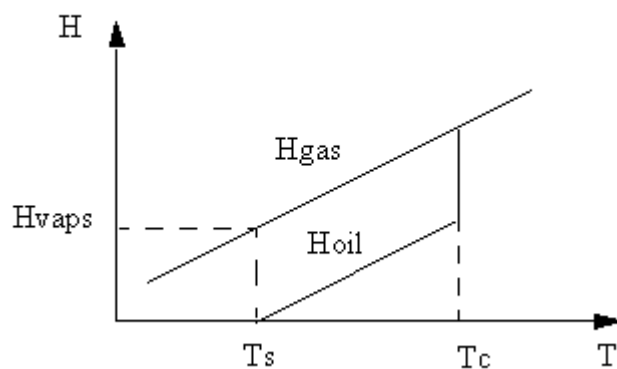


Figure 4.6. Component enthalpy defined by equal specific heats

Sources of enthalpy data

Molar specific heats are often quoted as a third order polynomial

$$C_p = A + B \cdot T + C \cdot T^2 + D \cdot T^3$$

These can be converted to Eclipse input values C_{p1} and C_{p2} used in equations 4.270 - 4.272 as follows:

$$C_{p1} = C_p(T_{st})/(MW) = (A + B \cdot T_{st} + C \cdot T_{st}^2 + D \cdot T_{st}^3)/(MW) \quad \text{Eq. 4.282}$$

$$C_{p2} = \left. \frac{dC_p}{dT} \right|_{T_{st}} / (MW) = (B + 2 \cdot C \cdot T_{st} + 3 \cdot D \cdot T_{st}^2)/(MW) \quad \text{Eq. 4.283}$$

where MW is the component molecular weight.

Note: The Eclipse specific heat input data is a function of the standard temperature T_{st} set by keyword `STCOND`

Specific heats calculated using equations 4.282 and 4.283 from data in *Reid et al (1987)* are shown in tables 4.14 and 4.15.

The following table contains calculated specific heats in metric units at $T_s = 20^\circ\text{C}$ for selected components. These are calculated from data in *Reid et al (1987)*.

Component	C_{p1} kJ/kg/K	C_{p2} kJ/kg/K ²
Methane	2.199	3.505e-3
Ethane	1.725	4.645e-3
Propane	1.650	5.026e-3
n-Butane	1.669	4.570e-3
Isobutane	1.656	4.885e-3
n-Pentane	1.641	4.847e-3
Isopentane	1.621	5.008e-3
n-Hexane	1.637	4.826e-3
n-Heptane	1.633	4.809e-3
n-Octane	1.630	4.798e-3
n-Nonane	1.627	4.808e-3
n-Decane	1.625	4.779e-3
n-Unadecane	1.624	4.772e-3
Dodecane	1.623	4.766e-3
n-Tridecane	1.621	4.763e-3
n-Tetradecane	1.620	4.758e-3
n-Pentadecane	1.619	4.754e-3
Hexadecane	1.618	4.752e-3
n-Heptadecane	1.618	4.751e-3

Component	C_{p1} kJ/kg/K	C_{p2} kJ/kg/K ²
Octadecane	1.618	4.747e-3
n-Nonadecane	1.616	4.743e-3
n-Eicosane	1.615	4.775e-3

Table 4.14: Calculated specific heats in metric units at $T_s = 20$ degrees C

The following table contains calculated specific heats in field units at $T_s = 20$ °C (68 °F) for selected components. These are calculated from data in [Reid et al \(1987\)](#).

Component	C_{p1} Btu/lb/°R	C_{p2} Btu/lb/°R ²
Methane	5.226e-1	4.659e-4
Ethane	4.080e-1	6.201e-4
Propane	3.895e-1	6.718e-4
n-Butane	3.946e-1	6.099e-4
Isobutane	3.911e-1	6.528e-4
n-Pentane	3.876e-1	6.480e-4
Isopentane	3.827e-1	6.696e-4
n-Hexane	3.866e-1	6.452e-4
n-Heptane	3.857e-1	6.430e-4
n-Octane	3.849e-1	6.415e-4
n-Nonane	3.843e-1	6.430e-4
n-Decane	3.838e-1	6.391e-4
n-Unadecane	3.836e-1	6.381e-4
Dodecane	3.833e-1	6.374e-4
n-Tridecane	3.829e-1	6.370e-4
n-Tetradecane	3.828e-1	6.363e-4
n-Pentadecane	3.825e-1	6.358e-4
Hexadecane	3.823e-1	6.355e-4
n-Heptadecane	3.821e-1	6.353e-4
Octadecane	3.821e-1	6.349e-4
n-Nonadecane	3.818e-1	6.344e-4
n-Eicosane	3.815e-1	6.387e-4

Table 4.15: Calculated specific heats in field units at $T_s = 20$ degrees C

Viscosity

Oil viscosity temperature dependence

Oil component viscosities can be input in functional form ([OILVISCC](#) or [OILVISCF](#)) or tabular form ([OILVISCT](#)). It is also possible to compute Lorentz-Bray-Clark viscosities for the oil phase by using the

keyword `COVTYPE`. The interpolation method for the tabular form can be specified using the `OILVTIM` keyword.

If the `OILVISC` keyword is used the viscosity μ_o^c for oil component c can be specified using one of four standard formulae:

The ASTM correlation:

$$\log_{10}(\mu_o^c + A) = B \cdot T^{-C} \quad \text{Eq. 4.284}$$

The Andrade formula (*Perry and Green (1984)* and *Reid et al (1987)*):

$$\log_{10}(\mu_o^c) = A + B/T \quad \text{Eq. 4.285}$$

The Vogel formula (*Perry and Green (1984)* and *Reid et al (1987)*):

$$\log_{10}(\mu_o^c) = A + B/(T + C) \quad \text{Eq. 4.286}$$

A logarithmic formula:

$$\log_{10}(\mu_o^c) = A + B \cdot \log_{10}(T) \quad \text{Eq. 4.287}$$

In these equations:

T is the temperature.

The coefficients A , B and C can either be specified directly, or values of viscosity and temperature can be specified, in which case the simulator calculates the correlation coefficients.

The ASTM correlation can be expressed in a form similar to *Prats (1982)*:

$$\mu_o^c = \left(\mu_{o1}^c + A \right)^{10^{-x}} - A \quad \text{Eq. 4.288}$$

where

$$x = C \cdot \log_{10}(T/T_1),$$

$(1 - A)$ is the limiting viscosity at infinite temperature,

$$B = \log_{10}(\mu_{o1}^c + A) \cdot T_1^{-C} \text{ and}$$

μ_{o1}^c is the viscosity at temperature T_1 .

Note that the formula is given in terms of the viscosity μ and not the kinematic viscosity $\nu = \mu/\rho$.

The keyword `OILVISC` can be used to fit each oil component viscosity according to the equation

$$\mu_o^c = D \cdot T^B \quad \text{Eq. 4.289}$$

that is the same as the logarithmic formula specified by the `OILVISC` keyword when $A = \log_{10}D$.

When all component viscosities have been calculated, the oil phase viscosity is calculated from

$$\log(\mu_o) = \sum_{c=1}^{N_{\text{comps}}} f_c(x^c) \cdot \log\left(\mu_o^c\right) \quad \text{Eq. 4.290}$$

where the functions f_c are given by keyword `OILVINDX`, or are defaulted to

$$f_c(x^c) = x^c.$$

Oil viscosity pressure dependence

The oil phase viscosity can be modified by a factor $\mu_p(P)/\mu_p(P_{vref})$ where μ_p and P_{vref} are defined by the keywords `PVCO` and `VISCREF`.

Lorentz-Bray-Clark oil viscosities

The oil phase viscosities can be computed by the Lorentz-Bray-Clark (LBC) method. Here the components that should be used to form the mixture for the LBC calculation are selected using the keyword `COVTYPE`. The LBC viscosity is computed based on the fluid density, pressure and temperature. If `LMIX` is selected in this keyword, the components with this entry compute their densities and corresponding viscosities using the LBC method. If the `LEMIX` entry is present, the densities for these components are computed with an EoS and then applied for the LBC viscosity calculation. The entry `NO` means that the regular component viscosity calculation is applied for this component. Care must be taken when this method is applied for heavy oils.

Gas viscosity temperature dependence

Gas component viscosities can be input in functional form (`GASVISCF`) or tabular form (`GASVISCT`).

If the functional form is used, each gas component viscosity is fitted according to the equation

$$\mu_g^c = A \cdot T^B \quad \text{Eq. 4.291}$$

where T is the temperature and c is the component number. The gas phase viscosity is calculated to be

$$\mu_g = \sum_{c=1}^{N_{\text{comps}}+1} y^c \cdot \mu_g^c \quad \text{Eq. 4.292}$$

Lorentz-Bray-Clark gas viscosities

The gas phase viscosities can be computed by the Lorentz-Bray-Clark (LBC) method. Here the components that should be used to form the mixture for the LBC calculation are selected using the keyword `CGVTYPE`. The LBC viscosity is computed based on the fluid density, pressure and temperature. If `LMIX` is selected with this keyword, the components with this entry compute their densities and corresponding viscosities using the LBC method. If the `LEMIX` entry is present, the densities for these components are computed with an EoS and then applied for the LBC viscosity calculation. The entry `NO` means that the regular component viscosity calculation is applied for this component.

Example

For a two hydrocarbon component study, the `PROPS` section may look something like this

```
PROPS =====
-- Temperature variation with depth - deg F
TEMPVD
624 105 /
1500 105 /

-- Standard Conditions Temp deg F Press psia
STCOND
```

```

60 14.7 /

-- Water Saturation Rel Perm Function
SWFN
.2000 .0000 0.0
.4000 .00035 0.0
.5000 .00101 0.0
.6000 .00213 0.0
.7000 .0038 0.0
/

-- Gas Saturation Rel Perm Function
SGCN
.0000 .0000 0.0
.0200 .0000 0.0
.1000 .07059 0.0
.2000 .15882 0.0
.3000 .24706 0.0
.4000 .33529 0.0
.5000 .42353 0.0
.6000 .51176 0.0
.7000 .6000 0.0
/

-- Two Phase Oil Saturation Rel Perm Functions
SOF3
.2000 .0000 .0000
.3000 .0000 .1666
.4000 .02468 .33333
.5000 .12155 .500000
.6000 .30885 .66667
.7000 .59856 .83333
.8000 1.0000 1.0000
/

-- Stone 3-phase oil rel perm
STONE

-- Relative permeability endpoints vs temperature (deg F)
ENKRVT
-- temp Krwmax Krgmax Kromax Krwro Krgro Krorg Krorw
100 1* 1* 1* 0.0038 1* 1* 1*
450 1* 1* 1* 0.1 1* 1* 1* /

-- Saturation endpoints vs temperature (deg F)
ENPTVT
-- temp Swc Swir Swmax Sgc Sgr Sgmax Sorw Sorg
100 0.20 0.20 1* 1* 1* 1* 0.3 0.2
450 0.45 0.45 1* 1* 1* 1* 0.1 0.05 /

-- Component gas viscosities cp
GASVISCT
100 .0112 1
200 .0128 1
300 .0145 1
400 .016 1
/

-- Component oil phase viscosities cp
OILVISCT
75 7500 7500
100 2000 2000
125 532 532
150 210 210
200 50 50
250 15 15
300 8.1 8.1
400 3.0 3.0
500 1.5 1.5
/

-- Crookston K-value coefficients
KVCR
1.89 1*
3226.8 1*
0 1*

```

```

362.7 1*
0      1*
/

-- Heats of vaporization Btu/lb
HEATVAP
350 /
-- Critical temperatures deg R
TCRIT
358 10000 /
-- Critical pressures psia
PCRIT
666.37 1* /

-- Component reference densities in oil phase lb/ft**3
DREF
60.975 60.975 /
-- Component compressibilities in oil phase 1/psi
CREF
0.00002 0.0000126 /
-- Component specific heats Btu/lb/deg R
SPECHA
0.52 0.432 /
CNAMES
SGAS HEAVY /
MW
17.3 500 /

-- Thermal expansion coefficients 1/deg R
THERMEX1
.00035 .000315 /

-- Oil component gas phase compressibilities 1/psi
ZFACTOR
0.95 /

-- Water properties
PVTW
-- Pref      Bw      Cw      Vw      Cvw
-- PSIA      RB/STB  1/PSI    CPOISE  1/PSI
-- 75.000    1.0     3.E-08  .3      7.E-09
/
-- Ref press rock compressibility
-- psia      1/psi
ROCK
100 5.0E-05 /

-- global mole fraction vs depth
ZMFVD
624 0.0001 0.9999 /

```

Dead oil

If the oil phase is represented by a single non-volatile component, then the [DEADOIL](#) option can be used and a simple set of keywords can be used to specify the effective properties of the hydrocarbon component.

Molecular weight

The [OILMW](#) keyword can be used to specify the molecular weight.

Specific heat

A constant specific heat is used with DEADOIL. It is defined using the [OILSPECH](#) keyword.

Density

The oil density is defined as

$$\rho_o = \frac{\rho_{ref,o}}{\left[1 - c_{Po}(P - P_{ref})\right] \left[1 + c_{T1}(T - T_{ref}) + c_{T2}(T - T_{ref})^2\right]}$$
Eq. 4.293

where

the oil compressibility c_{Po} and the thermal expansion coefficients c_{T1} and c_{T2}

are defined by the [OILCOMPR](#) keyword,

$\rho_{ref,o}$

is defined by the [DENSITY](#) keyword and

T_{ref} and P_{ref}

are defined by the [STCOND](#) keyword.

Viscosity

Oil viscosity can be specified in tabular form ([OILVISCT](#)) or functional form ([THANALV](#)). The interpolation method for the tabular form can be specified using the [OILVTIM](#) method. The functional form follows [Wingard \(1988\)](#):

$$\log_{10}(\mu_o) = A_o + B_o/T_K$$
Eq. 4.294

where

$A_o = -0.8772$, $B_o = 356.681$ and

T_K is the temperature in K.

This is the same as the live oil viscosity given by the Andrade formula, equation [4.285](#).

Oil viscosity pressure dependence

The oil viscosity can be modified by a factor $\mu(P)/\mu(P_{vref})$ where $\mu(P)$ and P_{vref} are defined by the keywords [PVCO](#) and [VISCREF](#).

Note: Oil viscosity in the [DEADOIL](#) option cannot be lower than 0.1 cP. It will be reset to 0.1 cP

Example

A typical set of dead oil keywords would be:

```
OILCOMPR
--Compressibility      therm.expansion coeff.
  5.00E-06              3.80E-04 0.0 /
OILSPECH
  specific heat
    0.5 /
OILMW
--effective molecular weight
  600.0 /
DENSITY
--for thermal, only oil density is needed.
  60.68 2* /
```

If multiple PVT regions are being used, a set of data should be entered for each PVT region. The number of PVT regions is entered as the second parameter of [TABDIMS](#).

Water and steam

Molecular weight

The molecular weight of water is defined as 18.015 gm/gm-M.

K-values

The K-value of water, K_{ww} , is defined as the saturation pressure of water $P_b(T)$ divided by the pressure:

$$K_{ww} = \frac{P_b}{P} \quad \text{Eq. 4.295}$$

By default the saturation pressure and temperature are obtained from the formula of [Farouk Ali \(1970\)](#):

$$P_b = \left(\frac{T}{a}\right)^{1/b} \quad \text{Eq. 4.296}$$

$$T_b = aP^b \quad \text{Eq. 4.297}$$

[Ejiougu and Fiori \(1987\)](#) give the following values for the constants a and b when the temperature is measured in °C, and in MPa:

$$a = 180.89, b = 0.2350$$

If the [THSTT97](#) keyword is used, the saturation pressure and temperature are calculated from an analytical formulation, as detailed by *The International Association for the Properties of Water and Steam* (IAPWS-IF97).

Alternatively, the water K-value can be obtained from a correlation of the form

$$K_{ww} = \frac{P_b}{P} = (A + B/P + C \cdot P) \cdot e^{-D/(T-E)}$$

where the coefficients A , B , C , D and E are specified using the [KVCRWAT](#) keyword.

Water density

The water density can be calculated in one of four ways; using an internal analytic function; using an analytic function with user-defined coefficients; using internal steam tables; or using an industrial analytical formulation.

Analytic formula with internally defined coefficients

Trangenstein's modification of Kell's 1975 correlations ([Trangenstein, 1992](#)) is used for the water density (default / [THANALB](#)):

$$\rho_w = \frac{A_0 + A_1 T_C + A_2 T_C^2 + A_3 T_C^3 + A_4 T_C^4 + A_5 T_C^5}{1 + A_6 T_C} e^{c_{Pw}(P_P - A_7)} \quad \text{Eq. 4.298}$$

where

ρ_w is in kg/m³,

the temperature T_C is in °C,

the pressure P_P is in MPa,

$$\begin{aligned}
A_0 &= 999.83952 & A_1 &= 16.955176 \\
A_2 &= -7.987 \times 10^{-3} & A_3 &= -46.170461 \times 10^{-6} \\
A_4 &= 105.56302 \times 10^{-9} & A_5 &= -280.54353 \times 10^{-12} \\
A_6 &= 16.87985 \times 10^{-3} & A_7 &= 10.2
\end{aligned}$$

The water compressibility c_{pw} (in MPa^{-1}) is defined by the [PVTW](#) keyword or calculated from the [WATERTAB](#) data.

Analytic formula with user-defined coefficients

If the [WATDENT](#) keyword is used then the water density can be specified in a similar manner to the oil component densities, equation, [4.254](#):

$$\rho_w = \frac{\rho_{ref,w}}{\left[1 - c_{pw}(P - P_{ref})\right] \left[1 + c_{T1}(T - T_{ref}) + c_{T2}(T - T_{ref})^2\right]} \quad \text{Eq. 4.299}$$

where

the reference density is given by $\rho_{ref,w} = \frac{\rho_{s,w}}{B_w(P_{ref})}$ and

$\rho_{s,w}$ is the water density at surface conditions given by the [DENSITY](#) keyword

P_{ref} is the reference pressure given by the [PVTW](#) keyword

$B_w(P_{ref})$ is the formation volume factor given by the [PVTW](#) keyword

c_{pw} is the water compressibility defined by the [PVTW](#) keyword or calculated from the [WATERTAB](#) data.

T_{ref} is the reference temperature given by the [WATDENT](#) keyword

c_{T1} is the first coefficient of thermal expansion, given by the [WATDENT](#) keyword

c_{T2} is the second coefficient of thermal expansion, given by the [WATDENT](#) keyword

Steam tables

If the [THTABB](#) keyword is used, the water density is calculated from steam tables ([Trangenstein, 1992](#)).

Industrial formulation

If the [THSTT97](#) keyword is used, the water density is based on an analytical formulation, as detailed by *The International Association for the Properties of Water and Steam* (IAPWS-IF97).

Steam density

The steam density can be calculated in one of three ways; using an internal analytic function; using internal steam tables; or using an industrial analytical formulation.

Analytic formula with internally defined coefficients

The analytic formula is used by default or if the `THANALB` keyword is used. The steam density is calculated from the real gas law:

$$\rho_g = \rho_{gb} \frac{T_b}{T} \quad \text{Eq. 4.300}$$

where

T_b is the boiling point temperature and

ρ_{gb} is the density at the boiling point.

The saturated water vapor density is taken from *Tortike and Farouq Ali (1989)*:

$$\rho_{gb} = \exp \left[C_0 + C_1 T_{bK} + C_2 T_{bK}^2 + C_3 T_{bK}^3 + C_4 T_{bK}^4 + C_5 T_{bK}^5 \right] \quad \text{Eq. 4.301}$$

where,

ρ_{gb} is in kg/m^3 ,

T_{bK} is the boiling point temperature in K and

$$C_0 = -93.7072 \quad C_1 = 0.833941 \quad C_2 = -0.003208$$

$$C_3 = 6.57652 \times 10^{-6} \quad C_4 = -6.93747 \times 10^{-9} \quad C_5 = 2.97203 \times 10^{-12}$$

Steam tables

If the `THATABB` keyword is used, the steam density is calculated from steam tables (*Perry and Green, 1984*).

Industrial formulation

If the `THSTT97` keyword is used, the steam density is based on an analytical formulation, as detailed by *The International Association for the Properties of Water and Steam (IAPWS-IF97)*.

Energies and enthalpies

Water and steam energies and enthalpies are read from steam tables. The default is to use tables from *Perry and Green (1984)*. Tables from *Haywood (1990)* can be selected using the `OPTIONS3` keyword. Analytic formulae, based on Perry's tables, can also be used, using keyword `THANALH`. If the `THSTT97` keyword is used, the water and steam energies and enthalpies are based on an analytical formulation, as detailed by *The International Association for the Properties of Water and Steam (IAPWS-IF97)*.

The enthalpy of pure water or steam increases with temperature, as shown below. It also varies with pressure, decreasing as pressure increases for steam (PT values below the saturation curve), but increasing for water (PT values above the saturation curve).

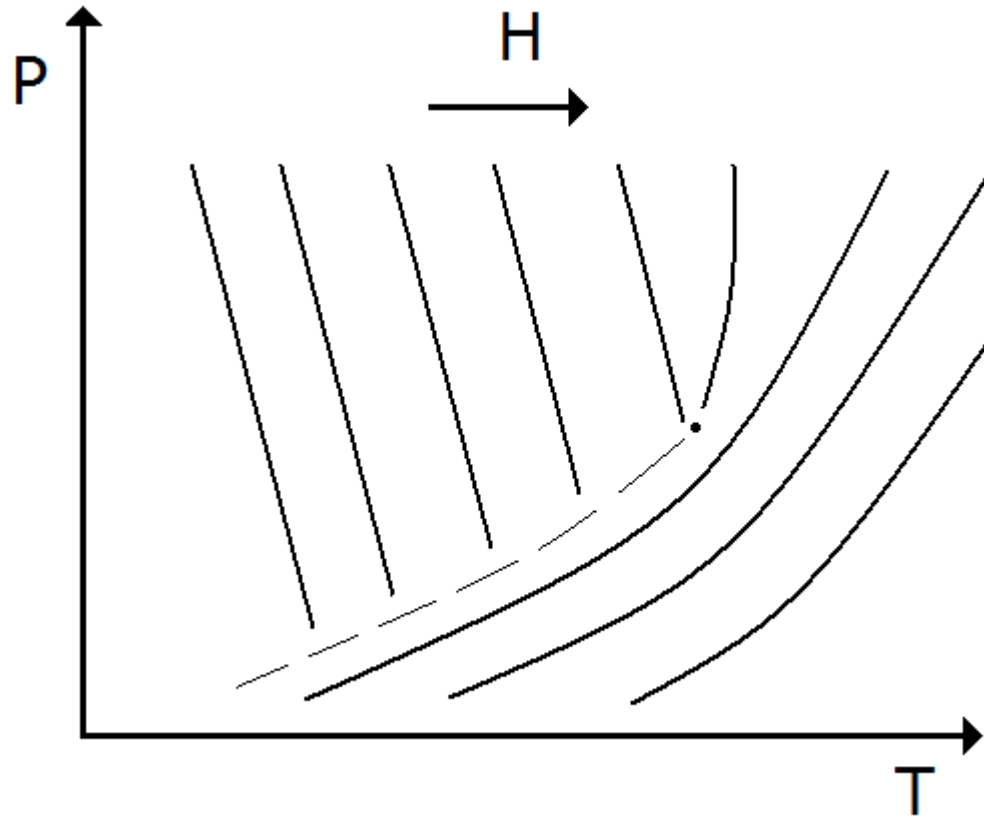


Figure 4.7. Constant enthalpy curves for pure water

For isenthalpic flows of pure water, the change in temperature is fairly constant over a range of pressures and temperatures, increasing by approximately 1 °C for a 50 bar pressure drop (approximately 1 °F for a 400 psi pressure drop). For isenthalpic flows of pure steam the temperature change is more variable, decreasing by the order of 1 °C for a pressure drop of 1 bar (approximately 1 °F for a 10 psi pressure drop).

Water viscosity

The water viscosity can be defined in tabular form ([WATVISCT](#)) or in functional form (default). The functional form is taken from [Grabowski and Rubin \(1981\)](#).

$$\mu_w = \frac{A_w}{-1 + B_w T_F + C_w T_F^2} \quad \text{Eq. 4.302}$$

where

the temperature T_F is in °F and

the default coefficients are $A_w = 2.1850$ $B_w = 0.04012$ $C_w = 5.1547 \times 10^{-6}$

The coefficients maybe modified using the [THWVC](#) keyword.

Water viscosity pressure dependence

The water viscosity can be multiplied by a factor $\mu_p(P)/\mu_p(P_{vref})$ where μ_p and P_{vref} are defined by the keywords [PVTW](#) and [VISCREF](#).

Steam viscosity

The steam viscosity is taken from [Ejiogu and Fiori \(1987\)](#).

$$\mu_g = A_g + B_g T_C + C_g P_P^{D_g} \quad \text{Eq. 4.303}$$

where

the temperature T_C is in °C,

the pressure P_P is in MPa

and the default coefficients are:

$$A_g = 4.9402 \times 10^{-3}, B_g = 5.0956 \times 10^{-5}, C_g = 2.9223 \times 10^{-6}, D_g = 2.5077$$

The coefficients maybe modified using the [THSVC](#) keyword.

Combustion

The chemical reaction keywords, "[Chemical Reactions](#)" can be used to model combustion. An example is given in "[Example 1: Combustion](#)".

Foamy oil

‘Foamy oil’ occurs in some heavy oil reservoirs during cold primary production. It is believed that the highly viscous heavy oil traps small gas bubbles as they form, hindering the development of the gas phase. The result is lower than expected gas production, a prolonged solution gas drive and higher than expected oil production.

A number of methods of modeling foamy oil have been proposed, using one, two or three components to represent the gas:

- In the simplest method, a single component is used to represent the gas and the foamy oil effect is modeled by increasing the critical gas saturation. This method has been successfully used to simulate laboratory studies. However, it does not simulate the dynamic processes of formation and dispersion of the trapped gas, which are important in reservoir studies.
- In the two gas component model, one component is used to represent the gas in the oil phase and the (gaseous) foam phase, and a second component is used to represent the free gas. Formation of trapped gas is modeled by the usual flash equations. The dispersion of the trapped gas into free gas is modeled by a chemical reaction, see "[Chemical reactions](#)". The reduced mobility of the trapped gas can be modeled by increasing the gas phase viscosity of the first gas component. However, since the gas mobility is a weighted average of all the gas component mobilities, this will result in the trapped gas moving too fast, and the free gas moving too slowly.
- In the three gas component model, different components are used to represent the gas in the oil phase, foamy oil phase and gas phase. In the foamy oil phase, the trapped gas is modeled as an oil-phase component, or even a solid component. This ensures that the trapped gas moves with the oil phase (or is immobile if it is modeled as a solid). A chemical reaction is needed to represent the formation of trapped gas from dissolved gas, replacing the flash. This has the added advantage of allowing non-equilibrium effects to be modeled.

In this model the reported oil saturation (or solid saturation) includes a contribution from the trapped gas. This contribution can be stored as a property `SFOIL` (or `SFSOL` for ‘foamy solid’) which can be output using keywords `RPTRST`, `RPTSCHED` and `RPTSOL` or with the SUMMARY keywords `BSFOIL` (or `BSFSOL`).

Example

In the following example four components are used, one to represent the heavy oil and three to represent the gas in the oil, foam and gas phases. Keyword `CVTYPE` is used to set the volatility types of the three gas components. Keyword `CCTYPE` is used to set the compressibility of the trapped gas component.

Two reactions are used to represent the liberation of gas:

1. dissolved gas to trapped gas
2. trapped gas to free gas.

Reaction 1 uses equilibrium deviation reaction rates, defined by the `EQLDREAC` keyword.

To get the correct quantities of oil and gas at the surface, the keywords `CVTYPES`, `TCRITS` and `KVCRS` are used to reset values for the surface flash, and keyword `CCTYPES` is used to reset the oil phase density.

The RUNSPEC section will contain the following keywords:

```
COMPS
  4 / 1 heavy component and 3 components representing gas

REACTIONS
  2 1 1 / 2 reactions, 1 equilibrium deviation reaction,
  -- 1 equilibrium k-value
```

The PROPS section will contain the following keywords:

```
CNAMES
      HEAVY      DISS_GAS TRAP_GAS FREE_GAS /

-- Volatility

CVTYPE
      DEAD      DEAD      DEAD      GAS      / gas trapped in oil, or
      DEAD      DEAD      SOLID     GAS      / trapped gas is immobile

KVCR
      0          1*        1*        1*
      1.214E6    1*        1*        1*
      0          1*        1*        1*
      9966       1*        1*        1*      /

-- Compressibility

CCTYPE
      1*        1*        GAS      1*        /

-- Reactions
-- 1: DISS_GAS <-> TRAP_GAS
-- 2: TRAP_GAS -> FREE_GAS

STOREAC
      0          1          0          0          0 / reaction 1
      0          0          1          0          0 / reaction 2

STOPROD
      0          0          1          0          0 / reaction 1
      0          0          0          1          0 / reaction 2

REACCORD
      0          0          0          0          0 / reaction 1
      0          0          1          0          0 / reaction 2
```

```

REACRATE
  0.1  0.5  / reactions 1 - 2

EQLDREAC
-- #   TYPE   XC   YC   KV
--   1     1     2     0   1 / reaction 1 proportional to X2-1/K1

EQLDKVCRC
0 1.704E5 0 3062 / K1 (K-value for ethane in field units)

-- For surface conditions use 'normal' component values

CVTYPES
1*          1*          1*          1*          /
CCTYPES
1*          1*          1*          1*          /
KVCRC
0           0           0           0
1.214E6    1.704E5    1.704E5    1.704E5
0           0           0           0
9966       3062       3062       3062       /
TCRITS
1409.7     549.7     549.7     549.7     /

```

The [REACENTH](#) keyword could be used here to define a latent heat, for example a negative value could be used for reaction 1.

Thermal features

Heat loss

Heat loss to the overburden, underburden and sides of the reservoir can be modeled. The [ROCKDIMS](#), keyword should be used to define the number of rock types outside the reservoir. Each rock type can have different properties, defined using the [ROCKPROP](#) keyword. The [ROCKCON](#) keyword is used to define which cells on the edge of the reservoir connect to each rock type, and the [ROCKCONL](#) keyword is used to specify the local grid refinement connection data for these cap and base rocks.

Two methods can be used to calculate the heat loss, a numerical method and an analytic method. In the numerical method the heat loss rate per unit area is given by

$$-k \cdot \frac{\partial T}{\partial n} \Big|_{n=0}$$

where

k is the conductivity, and

the temperature gradient outside the reservoir $\frac{\partial T}{\partial n}$ is found by solving the temperature equation

$$c \cdot \frac{\partial T}{\partial t} = k \cdot \frac{\partial^2 T}{\partial n^2}$$

where c is the rock heat capacity.

In the analytic method, Vinsome and Westerveld's approximate solution of the temperature equation is used ([Vinsome and Westerveld, 1980](#)). The heat loss per unit area in time Δt is given by

$$\int_0^\infty c \cdot T(t + \Delta t) dn - \int_0^\infty c \cdot T(t) dn$$

Summary keywords `FHLR` and `FHLT` can be used to obtain information on the field heat loss rate and field heat loss total.

Initialization

Initial values of all the primary variables, P , m_c and e need to be determined at the beginning of the simulation. Four methods are available:

- The solution can be read from a save file or a `RESTART` file generated by an earlier run.
- The solution can be determined from properties set directly in each grid block.
- The solution can be calculated by the equilibrium facility.
- The pressure can be calculated by equilibrium and the other variables determined from properties set directly in each grid block.

Restarting from a previous run

There are two different ways of restarting from a previous run:

- With a fast restart the `LOAD` keyword can be used to read data that had been previously saved using the `SAVE` keyword. This will only work if the same version of the simulator is used for both runs.
- With a flexible restart the `RESTART` keyword can be used to read data that had been previously output using the `RPTRST` keyword.

Explicit initialization

In an explicit initialization the component molar densities, equation 4.230 and the energy density, equation 4.231 can be determined from the component phase mole fractions and the saturations, equation 4.229:

$$m_c = S_o \cdot b_o \cdot x_o^c + S_g \cdot b_g \cdot x_g^c + S_w \cdot b_w \cdot x_w^c \quad \text{Eq. 4.304}$$

$$e = \frac{V_p}{V_b} \cdot (S_o \cdot b_o \cdot e_o + S_g \cdot b_g \cdot e_g + S_w \cdot b_w \cdot e_w +) \left(1 - \frac{V_p}{V_b}\right) \cdot e_r \quad \text{Eq. 4.305}$$

The phase molar densities b_p^m and energies e_p are functions of the pressure, temperature, component phase mole fractions and the component and rock properties. You must therefore specify

- pressure `PRESSURE`,
- temperature `TEMPI` or `TEMPVD`,
- two of the three saturations `SOIL`, `SGAS` and `SWAT`.

Data can also be read from `RESTART` files using the `GETSOL` keyword.

You must also specify the component liquid and vapor mole fractions x_o^c and x_g^c . There are a three ways this can be done:

- directly using the `XMF` and `YMF` keywords,
- directly, as functions of pressure using the `XMFP` and `YMFVP` keywords,
- using the `NEI` keyword to specify sample liquid mole fractions.

When specifying the initial component data directly, it is up to you to ensure the fluid composition is in thermodynamic equilibrium, see "[Thermodynamic equilibrium condition](#)". Otherwise, the solution and

properties change the first time a flash calculation is performed. This difficulty is avoided if the `NEI` keyword is used.

Initialization with NEI

When the `NEI` keyword is used, the vapor mole fractions are set so that the composition is in thermodynamic equilibrium, equation 4.215. The simulator tests to see if this composition gives the correct phase saturations - if not it adjusts the composition of the component with the highest gas phase mole fraction, so that the phase saturations are exactly matched.

Equilibration

In the equilibrium calculation a steady state solution of the flow equations is sought. This can be achieved if the pressure gradient in equation 4.236 is zero. For thermal simulations the heat conduction terms, equation 4.239 should also be zero, although this will in fact be determined by the initial temperatures that you specify. The equilibrium calculation is discussed in detail in the Equilibration section of [Initializing the study](#).

The `EQUIL` keyword is used to define the depths of the gas-oil contact and water oil-contact. If the reservoir does not contain a gas-oil contact the `EQUIL` keyword can be used to define the pressure at a datum depth.

The `TEMPVD` keyword is used to define initial temperatures.

Either the `COMPVD`, `ZMFVD` or the `ZI` keyword is used to define the hydrocarbon mole fractions ζ_c , see equation 4.220.

Mixed initialization

It is possible to engineer a mixed initialization where the pressures are calculated from a reference pressure at a given depth and the other properties are set explicitly. To do this an initial equilibration run is performed using the `EQUIL` keyword. The pressures from this run are output to a restart file:

```
SOLUTION
EQUIL
... /      set pressure at reference depth
RPTRST
PRESSURE /  output pressure
...
SCHEDULE
END
```

The equilibration run can be stopped at the start of the `SCHEDULE` section. In the main run, explicit initialization is used, but the pressures are read from the equilibration run:

```
SOLUTION
GETSOL
PRESSURE EQUILRUN 0 /  get pressure from previous run
/
SOIL
... /      set saturations
...
```

Although the equilibration run does not give the correct saturations, the pressures are reasonably accurate, since the oil and water densities are similar. These pressures are combined with the desired saturations in the main run.

Aquifers

The temperature of the aquifer water or gas is specified in the aquifer definition keyword (see the [AQUCT](#), [AQUFETP](#), [AQUCHWAT](#), [AQUCHGAS](#) or [AQUFLUX](#) keywords) for analytic aquifers. If the temperature of the analytic aquifer is defaulted and the [TEMPVD](#) keyword is specified, the aquifer temperature will be interpolated from the [TEMPVD](#) table using the aquifer depth. Otherwise if the temperature of the analytic aquifer is defaulted and the [TEMPVD](#) keyword is not specified, the [RTEMP](#) temperature will be used.

The temperature of the numerical aquifer water is constant. If the [TEMPVD](#) keyword is used, the aquifer temperature will be the temperature at the aquifer depth. Otherwise the [RTEMP](#) temperature is used.

Heaters

Heaters can be placed in any grid cell using the [HEATER](#) keyword, or in an LGR cell using the [HEATERL](#) keyword. These keywords specify the maximum rate of heating H_{max} and a maximum temperature at which the heater should operate T_{max} . The number of heaters allowed is controlled by the [HEATDIMS](#) keyword.

Depending on the arguments in the [HEATER](#) keyword the heater rate is given either by

$$H = \min[(T_{max} - T_c) \times R, H_{max}]$$

where

T_c is the cell temperature and

R the temperature-dependent heat injection rate,

or

$$H = \min\left(H_{max}, V \cdot \frac{\Delta E}{\Delta t}\right) \quad \text{Eq. 4.306}$$

where

V is the cell volume, Δt is the timestep, and

the estimated change in energy density ΔE is given by

$$\Delta E = e_t(T_{max}) - e_{t-\Delta t}$$

where

$e_t(T_{max})$ is the cell energy density when the cell temperature is T_{max} and

$e_{t-\Delta t}$ is the cell energy density at the previous timestep.

In conditions where there is no heat flow from a cell, a heater governed by equation 4.306 will heat the connection cell at a constant rate H_{max} until the cell reaches a temperature T_{max} . However, if there is energy flow into or out of the cell, then equation 4.306 is only approximate. In this case the heater rate may be less than H_{max} for temperatures near T_{max} , or the cell temperature may exceed T_{max} slightly; the discrepancies depend on the size of the heat flow and the size of the timestep.

Output of heater performance data to the print file can be controlled by using item 14 of the [RPTPRINT](#) keyword.

SUMMARY keywords WHTR and WHTT contain the energy rate and energy total for individual heaters and require a list of heater names. Vectors FHTR and FHTT contain the energy rate and energy total for all heaters.

Well inflow

Keywords [COMPMOBI](#) and [COMPMBIL](#) can be used to set the mobility used in calculating the completion flow for injection wells. Alternatively keywords [COMPKRI](#) and [COMPKRIL](#) can be used to set the relative permeability used in calculating the completion flow for injection wells.

Use of these keywords may improve the speed and stability of thermal runs - although there may be a slight loss in accuracy of the calculated injector BHP. This is not usually a matter of concern if the well is on a rate control. Furthermore, for steam injection, the injection pressure may not depend strongly on the mobility.

Keywords [COMPAGH](#) and [COMPAGHL](#) can be used to set average gravity heads for completions.

See "[Well Inflow Performance](#)" for further information on well inflow calculations.

Water, steam and gas injection

An injection fluid made up of water, steam, gas, oil or a mixture of these, can be specified with the [WCONINJE](#) keyword.

If the injection fluid contains hydrocarbon gas, the [WINJGAS](#) keyword can be used to specify the gas composition, and if the injection fluid contains hydrocarbon oil, the [WINJOIL](#) keyword can be used to specify the oil composition.

The enthalpy of the injected fluid is calculated from data entered using the [WINJTEMP](#) keyword.

If the injection fluid contains just steam or water there are four ways of entering the data:

1. Specify the steam quality Q_i and temperature T_i , pressure derived from the temperature.
2. Specify the steam quality Q_i and pressure P_i , temperature derived from the pressure.
3. Specify the pressure P_i and temperature T_i .
4. Specify the enthalpy H_i and pressure P_i .

The first two methods can be used for injection of two-phase saturated water and steam when the steam quality is between 0 and 1. The enthalpy of each phase is set to the saturated value at the specified temperature or pressure, and the injection fluid enthalpy is then calculated by interpolation:

$$H_i = Q_i \cdot H_s^{sat}(P_i, T_i) + (1 - Q_i) \cdot H_w^{sat}(P_i, T_i)$$

where

$$P_i = P_{sat}(T_i) \text{ and } T_i = T_{sat}(P_i)$$

The third method can be used for injection of single phase water or steam. The enthalpy is determined from steam tables

$$H_i = H_w(P_i, T_i) \text{ if } P_i > P_{sat}(T_i),$$

or

$$H_i = H_s(P_i, T_i) \text{ if } P_i < P_{sat}(T_i).$$

This method should not be used for two-phase injection, since the enthalpy of a saturated fluid is not uniquely defined by the pressure and temperature.

The fourth method can be used in both one-phase and two-phase cases.

If steam and gas are injected together then the same four ways can be used to specify the enthalpy. In methods 1,2 and 3 the enthalpy of the gas is calculated from the pressure and temperature and added to the steam enthalpy. If method 4 is used, the specified enthalpy is taken to be the enthalpy of the fluid.

If hydrocarbon gas is injected without steam then only methods 3 and 4 should be used to specify the fluid enthalpy.

Temperature at the bottom hole pressure

At the bottom hole pressure P_w , the injection enthalpy is equal to the enthalpy specified by the `WINJTEMP` keyword. Therefore the temperature of the well T_w can be found by solving

$$H(P_w, T_w) = H_i(P_i, T_i)$$

The well temperature will only be equal to the temperature specified using the `WINJTEMP` keyword, T_i , if the pressure P_i is equal to the bottom hole pressure P_w .

Steam trap

In `DEADOIL` simulations, steam production can be prevented by using either the `WELLPROD` or `WCONPROD` keyword to set the well BHP above the saturated water pressure in the well cells. The difference between the BHP and the saturated water pressure can be controlled either by specifying a pressure offset or a temperature offset.

The same keywords can be used to control the BHP in `LIVEOIL` simulations. However, if the reservoir contains volatile hydrocarbons, some steam may be produced at pressures above the water saturation pressure.

Steam production and injection rates

The steam production rate can be controlled by using either the `WELLPROD` or `WCONPROD` keyword, or modified using the `WELTARG` keyword. The steam rate should be specified as a cold water equivalent (C.W.E.) volume/time. The rate is imposed at reservoir conditions, that is the well steam rate is calculated by summing the number of moles of water in the gas phase entering each completion. The simulator converts the molar rate to a C.W.E. rate by multiplying by the surface molar volume of water.

The same value of surface molar volume of water is used for steam rate control and water rate control. In dead oil runs it is calculated from the internal steam tables at the standard conditions (`STCOND`). In live oil runs it is calculated from the `DENSITY` keyword.

Well voidage rates are calculated using the average hydrocarbon weighted pressure for the field (or fluid in place region, if item 13 of keyword `WELSPECS` is specified). Due to large pressure gradients that may exist in high viscosity heavy oil reservoirs, the average pressure can differ significantly from the local well pressures. Because of these pressure differences, combined with high steam compressibility and localised phase changes, the well voidage rates may not reflect the actual pore volume occupied by the fluids in the reservoir.

Dual porosity

The use of dual porosity grids in a thermal calculation makes additional assumptions about the distribution of rock volume between the porosities. In single porosity calculations, the rock volume is assumed to be the bulk volume minus the pore volume. This approach cannot be used with dual porosity, as it leads to the rock volume being represented multiple times, one in each porosity. Instead, default assumptions are made about the rock volume distribution, which can be overridden by you.

The default behavior assumes that the rock is associated with the matrix porosity, as matrix pore volume is contained in pores in the rock, as compared with fracture pore volume, which is associated with gaps in the matrix rock. However, to represent the local thermal effect of rock bounding the fracture, which may cool or heat the fracture fluids before a thermal transient advances far into the matrix rock, an amount of rock equal to the fracture pore volume is associated with the fracture. To prevent negative rock volumes being calculated if the porosities are large, this fracture rock volume will be proportionately reduced, so that the matrix/fracture rock volume ratio equals that of the matrix/fracture pore volume ratio in the case that negative volumes would otherwise be produced.

This approach may not be appropriate in all cases, and you have the option of overriding this behavior by providing a weighting factor for each cell in each porosity, using the `ROCKSPLV` keyword. In this case, the total rock volume, defined as the average bulk volume for all porosities for a particular cell, less the total pore volume over all porosities for that cell, is multiplied by the cell weighting factor specified by `ROCKSPLV`, to give the rock volume for that porosity. It is not required that the weighting factors sum to unity, and so this keyword can also be used as a multiplier.

The default behavior may also be appropriate for triple porosity, where the non-matrix porosities are larger voids. For general multiple porosity, where the sub-matrix porosities can have a variety of user-defined interpretations, the default behavior may or may not be appropriate, depending on whether the sub-matrix cells are used to model transients in the matrix, or for other purposes.

The use of the `ACTNUM` keyword with dual porosity and the thermal option creates additional options for combining the various data inputs, and you should check carefully that the final result is what is expected, as the `ACTNUM` and rock proportioning behaviors are applied separately from each other.

Multisegment wells

Multisegment wells can be used with the Thermal option. Heat storage in the tubing and casing can be specified using the `WELSEGS` keyword. This keyword can also be used to specify longitudinal conduction in the tubing and casing between successive segments.

Heat conduction to the reservoir, and heat loss outside the reservoir can be specified using keyword `WSEGHEAT`. Heat conduction to the reservoir occurs between segments and completion grid cells. Therefore it may be necessary to define completions in each grid cell the well passes through. These completions can be shut if no flow occurs between the grid cell and the segment. The length of well in each completion cell can be defined using the `COMPSEGS` keyword.

`WSEGHEAT` can also be used to model radial heat conduction between non-successive segments, for example between tubing segments and casing segments.

For multisegment well source terms, the keyword `WSEGEXSS` can be used to specify the temperature of imported gas.

Temperature dependent well cutback

Well production rates can be reduced if the well temperature exceeds given limits. Keywords `TCBDIMS` and `WELLTCBT` can be used to set up tables of temperatures and cut back factors. The keyword `WELLTCB`

can be used to associate a well with a table and the keyword `WELLTCBC` can be used to control the switching from one rate to another. Note, these keywords have no effect if the well is controlled by BHP.

The `WECON` keyword can also be used to close a well if the temperature exceeds a given limit.

Temperature range checking

The keyword `TRANGE` can be used to check that the reservoir temperature stays within expected limits.

Thermal output

Temperature, energy, enthalpy and steam data can all be output to the SUMMARY file, PRINT file or for input to graphical post-processors. See the keywords `SUMMARY`, `RPTSCHEM` and `RPTRST`.

Convergence criteria

By default, the convergence criteria are chosen to be reliable across a wide range of thermal models. If necessary, the default convergence criteria can be altered with the `CVCRT` keyword.

Limiting non-linear solution updates

Limits P_{lim} , S_{lim} and T_{lim} can be set on the pressure change, specific volume change and temperature change during a non-linear iteration to help control the convergence process. The solution update ΔX_{n-1} in equation 4.240 is multiplied by a reduction factor λ , $0 < \lambda \leq 1$ chosen so that for every cell:

$$\lambda \cdot |\Delta P| \leq P_{lim}$$

$$\lambda \cdot \left| \Delta X' \cdot \frac{\partial V}{\partial X'} \right| \leq S_{lim},$$

$$\lambda \cdot \left| \Delta e \cdot \frac{\partial T}{\partial e} \right| \leq T_{lim}$$

where X' is any solution variable except the pressure

$$X' = (m_1, \dots, m_N, m_w, e)$$

and V is the total specific volume, equal to the sum of the phase specific volumes defined in equation 4.226. The temperature limit is only used in thermal simulations, and only in fully implicit cells. Limits S_{lim} , T_{lim} and P_{lim} are set using items 13, 14 and 15 of the `CVCRT` keyword.

Appleyard chop

After the solution is updated new values of the saturations and temperatures are calculated by performing a flash. If the saturations in an implicit cell change significantly near a critical value - for example if a phase appears or disappears or becomes mobile / immobile - then the solution change in that cell is chopped, so that the new saturations are close to the critical point.

The Appleyard chop is also performed if a property change in an implicit cell is large. Chops are performed if the saturation change is greater than 0.2, or if the temperature change is greater than T_{lim} .

The saturation based chop can be switched off by using item 18 of the `OPTIONS3` keyword. The temperature based chop can be switched off by using item 27 of `OPTIONS3`. Both chops can be switched off by using item 25 of `OPTIONS3`.

Non-linear convergence

The non-linear iterations are deemed to have converged when the reduction factor λ is greater than 0.9 and the pressure change, specific volume change and temperature change during a non-linear iteration are less than limits P_{con} , S_{con} and T_{con} :

$$\lambda \cdot |\Delta P| \leq P_{con}$$

$$\lambda \cdot \left| \Delta X' \cdot \frac{\partial V}{\partial X'} \right| \leq S_{con}$$

$$\lambda \cdot \left| \Delta e \cdot \frac{\partial T}{\partial e} \right| \leq T_{con}$$

The temperature condition is only used in thermal simulations, and only in fully implicit cells. Limits P_{con} , S_{con} and T_{con} are set using items 1, 7 and 8 of the [CVCRIIT](#) keyword.

Timestepping criteria

By default the timestepping strategy employed in thermal simulations is based upon controlling the number of non-linear iterations that a timestep is expected to take. This is a heuristic timestep selection method in which the overall run time of a simulation can be minimized. In general this is a compromise between taking many cheap short timesteps and taking fewer longer, but computationally more expensive steps. The algorithm is based primarily on selecting a new timestep length depending on the previous timestep length and the number of non-linear iterations required to converge it.

The default timestep selection strategy is generally found to be robust across a wide range of models. However, in some instances, it may be possible to improve upon the default timestep selection by using the [TSCRIT](#) keyword to specify solution change targets and solution change limits.

Rejecting a timestep

A timestep can be rejected even though the non-linear iterations have converged, if the solution change during the timestep is too big. In these cases a smaller timestep will be used. A timestep will only be accepted if the change in specific volume, the throughput, TPT and TTE are less than limits S_{max} , TPT_{max} and TTE_{max} :

$$\sum_{X'} |X'_t - X'_{t-\Delta t}| \cdot \left| \overline{\frac{\partial T}{\partial X'}} \right| \leq S_{max} \quad \text{Eq. 4.307}$$

$$TPT \leq TPT_{max} \quad \text{Eq. 4.308}$$

$$TTE \leq TTE_{max} \quad \text{Eq. 4.309}$$

where

X' is any solution variable except the pressure $X' = (m_1, ..., m_N, m_w, e)$,

$\left| \overline{\frac{\partial V}{\partial X'}} \right|$ is the average value of $\left| \frac{\partial V}{\partial X'} \right|$ over all cells

The throughput is the volume flow divided by the cell volume and is discussed further in "[Throughput target \(TPT\)](#)". The throughput limit is only applied to [IMPES](#) cells.

The time truncation error is a measure of the “acceleration” of the specific volume and is discussed further in "[Time truncation error \(TTE\)](#)".

In thermal simulations three additional convergence criteria can be applied. The changes in saturation SAT_p of each phase p , temperature and energy in every cell must be less than limits SAT_{max} , T_{max} and e_{max} :

$$|SAT_{p,t} - SAT_{p,t-\Delta t}| \leq SAT_{max} \quad \text{Eq. 4.310}$$

$$|T_t - T_{t-\Delta t}| \leq T_{max} \quad \text{Eq. 4.311}$$

$$|e_t - e_{t-\Delta t}| \leq e_{max} = c_p \cdot T_{max} \quad \text{Eq. 4.312}$$

The maximum energy change is defined by the maximum temperature change, using a typical value of specific heat $c_p = 2347 \text{ kJ/m}^3/\text{K} = 35 \text{ Btu/ft}^3/\text{R}$. The saturation limit can be used instead of the solution change limit.

Limits TTE_{max} , TPT_{max} , S_{max} , SAT_{max} and T_{max} are set using items 7, 9, 13, 16 and 18 of the [TSCRIT](#) keyword.

Determining a new timestep

New timesteps Δt are chosen so that the predicted solution change is less than a specified target. Targets TTE_{tar} , TPT_{tar} , S_{tar} , P_{tar} , SAT_{tar} , T_{tar} are set for the time truncation error, throughput and changes in specific volume, pressure, saturation and temperature. An energy target e_{tar} is calculated from the temperature target.

$$e_{tar} = T_{tar} \cdot c_p$$

Changes in parameters for the new timestep are estimated by multiplying the rate of change in the previous timestep by the new timestep length.

Values for targets TTE_{tar} , TPT_{tar} , S_{tar} , P_{tar} , SAT_{tar} and T_{tar} are set using items 6, 8, 10, 12, 15 and 17 of the [TSCRIT](#) keyword.

Determining implicit cells in AIM runs

The throughput and changes in specific volume and temperature for a timestep are compared with targets:

$$R_P = (TPT_t)/(TPT_{tar})$$

$$R_S = (\Delta S_t)/(S_{tar})$$

$$R_T = (\Delta T_t)/(T_{tar})$$

These ratios can be output to the restart file by using arguments [RATP](#), [RATS](#) or [RATT](#) with the [RPTRST](#) keyword. The ratios are extrapolated to the new timestep $t + \Delta t$ and cells are made implicit if these extrapolated ratios exceed values set in [AIMCON](#).

Instability in AIM runs due to thermal conduction

In [AIM](#) runs the implicit cells are determined using throughput criteria. However in thermal runs, conduction can cause instability in explicitly solved cells. To prevent this instability:

- Rock filled cells (or near rock filled cells) should be solved fully implicitly. Therefore, in AIM runs item 3 of the [AIMCON](#) keyword should be set to a positive value.
- Problems where conduction is significant should not be solved using AIM.

Restriction	Parameter	Non linear iteration ΔX lim	Non linear iteration ΔX con	Timestep change max	Timestep change tar	Timestep change argument
	Pressure	CVCRIT item 15	CVCRIT item		TSCRIT item 12	PCT
	Specific volume	CVCRIT item 13	CVCRIT item 7	TSCRIT item 13	TSCRIT item 10	SCT
	Time truncation error			TSCRIT item 7	TSCRIT item 6	TTE
IMPES cells	Throughput			TSCRIT item 9	TSCRIT item 8	TPT
Thermal only	Temperature	CVCRIT item 14	CVCRIT item 8	TSCRIT item 18	TSCRIT item 17	TEC
Thermal only	c/cp			TSCRIT item 18	TSCRIT item 17	EnC
Thermal only	Saturation			TSCRIT item 16	TSCRIT item 15	SAC

Table 4.16: Summary of convergence criteria and timestepping criteria

Note: If items 13 and 14 of keyword **CVCRIT** are not set, the default values for the non-linear step limits S_{lim} and T_{lim} are calculated from items 10 and 17 of keyword **TSCRIT**.

This means that if the target specific volume is relaxed (item 10 of **TSCRIT**) - which may be the case if the target saturation is used (item 15 of **TSCRIT**) - then item 13 of keyword **CVCRIT** should be used to override the default value for the non-linear step limit S_{lim} .

Tuning thermal models for performance

The additional dependence of the reservoir properties on temperature, as modeled with the Thermal option, can make models significantly more computationally expensive to run. In general, the default tuning is found to be the best compromise across a wide range of models and it is recommended that all Thermal models are run without any additional tuning initially to establish a baseline. This is particularly relevant if the model predates the introduction of the JALS linear solver (default in 2009.1) which was found to significantly improve the performance of many models and render previous tuning redundant and sometimes detrimental. However, obtaining accurate and reliable performance timings for simulations, especially on clusters, is not trivial and it is advised that the trend in elapsed time is also compared with the total number of linear iterations (MSUMLINS) since these two quantities are usually closely correlated. Having established a baseline there are a number of non-default performance options which can be trialed for better performance if necessary.

Item 161 of the **OPTIONS3** keyword can be used to activate an alternative linear algebra procedure for simulations with rock-filled cells. In a thermal simulation, for cells with zero pore volume, only the energy conservation equation has to be solved. By employing a slightly different, but mathematically identical, lineup of equations and variables it is possible to eliminate some of the variables from the linear system. With this approach the size of the system passed to the linear solver is reduced by one set of variables and hence can prove to generate a noticeable reduction in simulation time. However, the system passed to the linear solver may also prove to be numerically less favorable so an improvement in performance is not always seen. This option is only worth pursuing if the model contains one or more rock-filled cells.

[TSCRIT](#) item 11 can be used to increase the maximum length of timestep after a well modification, and is active due to the well control modifications or a well opening/closing. Prior to the 2012.1 software release, the default value was set to a low value for thermal simulations (0.5 days) since in some processes the state of the near well zone is very sensitive to the manner in which the well is operating. Short timesteps assist in resolving these features as well as others, including simulation models containing complex geometry and/or heterogeneity (where timesteps less than 0.5 day may be required). However, this constraint can be over cautious especially when using the [JALS](#) solver (the default solver) for simulations where the well modifications are relatively smooth or have less of an impact on the reservoir conditions. The default value for [TSCRIT](#) item 11 has now increased to 5 days.

Current limitations

At present a number of options cannot be used with the Thermal option. These include:

- "Diffusion"
- "Transport coefficients"
- "Non-Darcy flow"
- "Optimization"
- "Surface tension effects"
- [CO2SOL](#), [H2SSOL](#), [GASSOL](#) and [GASWAT](#).

Converting a black-oil data set to thermal

To convert a black-oil data set to run with the thermal option, the black-oil fluid description must be modified to a compositional fluid description and the thermal and temperature dependent properties of the reservoir, fluids and wells must be described. A first step in this process is to use thermal keywords to create a data set that mimics the isothermal black-oil run. At this stage many of the temperature dependent properties can be defaulted. Once it has been confirmed that this gives similar results, the data set can be enhanced, by adding more components to improve the fluid description, and by including temperature effects and tuning the temperature dependent properties. This section gives suggestions for carrying out the first step of this process - running a black-oil simulation with the thermal simulator.

Converting Eclipse 100 to Eclipse 300 black oil

If the black-oil data set is an Eclipse 100 data set, then you should check that it runs in Eclipse 300 too. If necessary remove any Eclipse 100 keywords that are not recognized by Eclipse 300.

Eclipse 100 data sets using the Temperature option ("[Temperature option](#)") can be run with Eclipse 300 if the [SCHEDULE](#) section data is converted to Thermal keywords - in particular [WTEMP](#) data should be changed to [WINJTEMP](#) data.

RUNSPEC section

Add the keyword [THERMAL](#) and remove the keyword [BLACKOIL](#).

If there is no gas in the black-oil simulation, the thermal simulation can be run as a dead oil simulation by adding keyword [DEADOIL](#). Otherwise the thermal simulation can be run as a live oil with two components, one representing the black-oil oil, and the other representing the black-oil gas.

The number of hydrocarbon components must be defined with the **COMPS** keyword. As a first step use two components if both oil and gas are being simulated, and one component if only one hydrocarbon phase is being simulated.

If the data set contains horizontal wells, add the **HWELLS** keyword to the RUNSPEC section.

Reservoir properties

The rock heat capacity (**HEATCR**) and thermal conductivity (**THCONR**) must be specified in the GRID section. For an isothermal run the actual values are not important - a high rock heat capacity can be used to dampen any temperature fluctuations and the thermal conductivity can be set to zero.

In the PROPS section the reservoir temperature can be set using the **RTEMP** or **TEMPVD** keyword. The surface temperature and pressure can be set using the **STCOND** keyword.

Dead-oil fluid properties

Keywords, **OILCOMPR**, **OILVISCT**, **OILSPECH** and **OILMW** have to be added. The black-oil keywords (**PVTO** and **PVDG**) must be removed.

Live oil fluid properties

If black oil data is derived from PVT experiments, it should be possible to go back to the original data and derive thermal properties too. If this is not possible the keywords can be converted directly as described below. It is assumed that there are two components, an “oil” component which is entirely in the oil phase at surface conditions, and a “gas” component that is entirely in the gas phase at surface conditions. The component names should be defined using the **CNAMES** keyword

Molecular weights

Molecular weights MW^c of each component should be specified. Suggested values are 120 for the oil component and 25 for the gas component.

Volatility types

The surface volatility types (keyword **CVTYPES**) should be set to DEAD for the “oil” component and GAS for the “gas” component to ensure the surface oil and gas phases contain only the “oil” and “gas” components.

If the black-oil simulation does not contain vaporized oil (**PVDG**) then the reservoir volatility type (keyword **CVTYPE**) can be set to DEAD for the “oil” component. Alternatively, a small K-value should be used for the “oil” component.

If the black-oil simulation does not contain dissolved gas (**PVDO**) then the reservoir volatility type (keyword **CVTYPE**) can be set to GAS for the “gas” component. Alternatively, a large K-value should be used for the “gas” component.

K-values for live components

K-values can be calculated from R_s and R_v values as functions of pressure:

$$K_{oil} = \frac{R_s + F}{1 + F \cdot R_v} \cdot R_v$$

$$K_{gas} = \frac{R_s + F}{1 + F \cdot R_v} \cdot \frac{1}{R_s}$$

where

F is the ratio of the molar volumes of gas to oil at surface conditions:

$$F = \frac{V_{gas,s}}{V_{oil,s}} = \frac{MW^{gas}}{MW^{oil}} \cdot \frac{\rho_{oil,s}}{\rho_{gas,s}}$$

Values of $\rho_{oil,s}$ and $\rho_{gas,s}$ can be taken from the [DENSITY](#) keyword. The units of F should be the same as those used for R_s and $1/R_v$. In field units, a factor of 0.005615 can be used to convert ft^3/ft^3 to $Mscf/stb$.

A K-value table (keyword [KVTA B T n](#)) at reservoir temperature can be constructed from the R_s and R_v data taken from the [PVTO](#) and [PVTG](#) keywords. At least one more K-value table is required at a different temperature. If no other data is available equation 4.248 can be used:

$$K \propto e^{-D/T}$$

with values of D taken from tables 4.12 or 4.13 to give K-values at a second temperature:

$$K(T_2) = K(T_1) \cdot e^{D \cdot (1/T_1 - 1/T_2)}$$

Oil phase density

The oil formation volume factor B_o is given by

$$B_o = \left(V_{oil,r}^{oil} + \frac{R_s}{F} \cdot V_{oil,r}^{gas} \right) / V_{oil,s} \quad \text{Eq. 4.313}$$

where $V_{oil,r}^c$ is the reservoir oil phase volume per mole of component c .

Values of $V_{oil,r}^c$ are determined from user-defined densities and compressibilities (see equations 4.253 and 4.254), which can be chosen to give a good fit for equation 4.313 at the reservoir temperature. As a first step the component compressibility [CREF](#) and thermal expansion coefficient [THERMEX1](#) can be ignored to get approximate values for the reference density [DREF](#):

$$B_o = \frac{\rho_{oil,s}}{\rho_{ref}^{oil}} + R_s \cdot \frac{\rho_{gas,s}}{\rho_{ref}^{gas}} \quad \text{Eq. 4.314}$$

Gas phase density

The gas formation volume factor B_g is given by

$$B_g = \left(F \cdot R_v \cdot V_{gas,r}^{oil} + V_{gas,r}^{gas} \right) / V_{gas,s} \quad \text{Eq. 4.315}$$

where $V_{gas,r}^c$ is the reservoir gas phase volumes per mole of component c :

$$V_{gas,r}^c = \frac{Z^c \cdot R \cdot T}{P} \quad \text{Eq. 4.316}$$

where

R is the gas constant,

P is the pressure,

T the temperature and the compressibility terms Z^c can be set for each component using the [ZFACTOR](#) keyword to match equation 4.315 at the reservoir temperature.

The temperature dependence of these terms are fixed. A surface compressibility can be set using the [ZFACTORS](#) keyword to ensure the correct surface gas density:

$$V_{gas,s} = \frac{Z^{gas} \cdot R \cdot T}{P} = \frac{MW^{gas}}{\rho_{gas,s}} \quad \text{Eq. 4.317}$$

Oil phase viscosity

The thermal oil phase viscosity is given by equation 4.290 as a function of temperature and mole fractions. It can be rewritten in terms of R_s to give:

$$\log \mu_{oil} = \frac{1}{F + R_s} \cdot (F \cdot \log \mu_{oil}^{oil} + R_s \cdot \log \mu_{oil}^{gas}) \quad \text{Eq. 4.318}$$

If necessary an extra pressure dependent factor can included in this formula. Values of the component oil phase viscosities μ_{oil}^c can be found at the reservoir temperature to give a good fit for equation 4.318. These values can be entered in the [OILVISCT](#) keyword at two temperatures to give a temperature independent oil viscosity.

Gas phase viscosity

The thermal gas phase viscosity is given by equation 4.292 as a function of temperature and mole fractions. It can be rewritten in terms of R_v to give:

$$\mu_{gas} = \frac{1}{F \cdot R_v + 1} \cdot (F \cdot R_v \cdot \mu_{gas}^{oil} + \mu_{gas}^{gas}) \quad \text{Eq. 4.319}$$

Values of the component gas phase viscosities μ_{gas}^c can be found at the reservoir temperature to give a good fit for equation 4.319. These values can be entered in the [GASVISCT](#) keyword at two temperatures to give a temperature independent gas viscosity.

Specific heats

For an isothermal run, the actual values of the specific heats are not important. However, the enthalpy of each component in the gas phase should be larger than the enthalpy of the same component in the oil phase. This can be checked by looking at the property output in the `PRT` file.

Other data

The [TCRIT](#) and the [PCRIT](#) keywords must be included in the data set. There are no component-wise default values. You are recommended not to default these to your internal default values.

Water properties

In thermal simulations the temperature dependence of water properties is usually more significant than the pressure dependence. Internal tables are used to define the water density and viscosity. To mimic a black-oil run, keywords [WATDENT](#) and [WATVISCT](#) should be used to define the temperature dependence of the water density and viscosity while the pressure dependence is taken from the [PVTW](#) keyword.

Initial conditions

Data from a black-oil run can be used to initialize a thermal run. The keyword [GETSOL](#) can be used to read pressure and saturation data from a black-oil restart file. Keywords [XMF](#), [YMF](#) have to be added.

If the [EQUIL](#) keyword is used item 10 should be set.

Initial compositions can be set by noting that at the saturation pressure

$$z_{oil} = \frac{F}{F + R_s}$$

$$z_{gas} = \frac{R_s}{F + R_s}$$
Eq. 4.320

In thermal runs, water is allowed in the gas phase. This can cause differences in the initialization if a gas cap is present. If the initial composition is not in equilibrium then temperatures and saturations will change the first time the composition is flashed. It is therefore recommended that initial pressures, temperatures and saturations are output to the print file ([RPTSOL](#)) or restart file ([RPTSCHED](#)) and checked for consistency with the input data.

Well properties

The fluid enthalpy for any injection wells can be specified with the [WINJTEMP](#) keyword.

Well BHPs (keywords [WCONINJE](#) and [WCONPROD](#)) should be set, since the default values in the Thermal option are different from the default values used with black oil.

Example

An example is provided on the release DVD. CASE3T is a thermal version of the black oil data set CASE3B.

Summary of keywords

The Eclipse 300 Thermal option can be selected by entering the [THERMAL](#) keyword in the RUNSPEC section of the DATA file. Fluid properties can be specified in three different ways:

LIVEOIL	allows the simulation of multiple hydrocarbon components in both oil and gas phases.
DEADOIL	the hydrocarbon component cannot appear in the gas phase. A simpler set of property keywords is used. More than one hydrocarbon component is not allowed.
BLACKOIL	properties are specified using the Temperature option keywords, see the " Temperature option ".

Thermal keywords can be used in the SUMMARY section and the SCHEDULE section. This option is very similar to the Eclipse 300 Temperature option (which is selected by using the [TEMP](#) keyword instead of the [THERMAL](#) keyword). However extra features are allowed with the black-oil thermal option (for example, heat loss, heaters and steam injection)

RUNSPEC

Keyword	Description
BLACKOIL	Request use of Black Oil properties
DEADOIL	Request use of Dead Oil properties

Keyword	Description
JALS	Switch on the more powerful JALS solver. Recommended for thermal simulations
HEATDIMS	Heater keyword dimensions
LIVEOIL	Request use of Live Oil properties (the default)
ROCKDIMS	Base and cap rock dimensions
TCBDIMS	Thermal cut-back table dimensions
THERMAL	Request use of Thermal option

GRID

The following table contains those keywords that can only be used with the Thermal option, as well as general Eclipse 300 keywords (marked*) that have extra arguments or may be used in different ways in thermal runs.

Keyword	Description
ACTNUM *	Active grid block identification
HEATCR	Rock volumetric heat capacity
HEATCRT	Rock volumetric heat capacity T coefficient
MINROCKV	Minimum rock volume
MINRV	Minimum rock volume (alias for MINROCKV)
NNC *	Ninth argument is conductivity across the NNC
ROCKCON	Base and cap rock connections to the reservoir
ROCKPROP	Base and cap rock properties
ROCKSPLV	Non-default proportions of the rock in each porosity (for dual porosity).
SIGMATH	Matrix-fracture thermal conductivity multiplier
THCONMF	Matrix-Fracture thermal conductivity
THCGAS	Thermal conductivity for gas phase
THCOIL	Thermal conductivity for oil phase
THCONR	Thermal conductivity of rock and fluid
THCROCK	Thermal conductivity for rock
THCSOLID	Thermal conductivity for solid phase
THCWATER	Thermal conductivity for water phase
THCONSF	Thermal conductivity saturation-dependence

Edit

Keyword	Description
HEATTR	Heat transmissibility overwrite
HEATTTHT	Heat transmissibility overwrite
HEATTX	Heat transmissibility overwrite

Keyword	Description
HEATTY	Heat transmissibility overwrite
HEATTZ	Heat transmissibility overwrite
ROCKV	Rock volume

PROPS

Property keywords are given in the following lists. Some keywords can be used with [DEADOIL](#), some with [LIVEOIL](#), and some with both.

The rock fluid property keywords can be used with both [LIVEOIL](#) and [DEADOIL](#).

- [ENKRVT](#) Endpoint relative permeabilities versus temperature
- [ENPCVT](#) Endpoint capillary pressures versus temperature
- [ENPTVT](#) Endpoint saturations versus temperature

The following table of hydrocarbon properties contains those keywords that can only be used with the Thermal option, as well as general Eclipse 300 keywords (marked*) that have extra arguments or may be used in different ways in thermal runs. These are reservoir keywords that can only be used with [LIVEOIL](#) unless otherwise stated.

Restriction	Keyword	Description
	ACF *	Component acentric factors (used with KVWI).
	CCTYPE * and CCTYPES * (surface keyword)	Component liquid compressibility type.
LIVEOIL and DEADOIL	CGDTYPE	Component gas density type, EoS density.
	CGVTYPE	Component gas viscosity type, LBC viscosity.
	CODTYPE	Component oil density type, EoS density
	COVTYPE	Component oil viscosity type, LBC viscosity.
	CREF	Component compressibilities.
	CVTYPE and CVTYPES (surface keyword).	Component volatility (dead, live, gaseous).
	DREF * and DREFS * (surface keyword)	Component densities
	DREFT * and DREFTS * (surface keyword)	Component densities (alias for DREF)
	GASVISCF	Gas viscosity function of temperature
	GASVISCT	Gas viscosity table as a function of temperature
	GREF *	Reference gravities for live oil components
	HEATVAP	Component heat of vaporization
	HEATVAPE	Heat of vaporization exponent
	HEATVAPS	Component heat of vaporization at standard temperature

Restriction	Keyword	Description
	KVCR and KVCRS (surface keyword)	K-value correlation parameters
	KVGEN	Generate tables of K-values from an EoS
	KVTABTn	Tables of K-values
	KVTEMP	Temperatures for K-value tables
	KVWI	Wilson K-value correlation parameters
DEADOIL only	OILCOMPR	Oil compressibility, expansion coefficient
DEADOIL only	OILMW	Oil molecular weight
DEADOIL only	OILSPEC	Oil specific heat
	OILVINDX	Oil viscosity index
	OILVISCC	Oil viscosity-temperature correlations
	OILVISCF	Oil viscosity function
LIVEOIL and DEADOIL	OILVISCT	Oil viscosity table as a function of T.
LIVEOIL and DEADOIL	OILVTIM	Oil viscosity temperature interpolation method.
	PCRIT	Component critical pressure (used with KVWI)
	PREF and PREFS (surface keyword).	Component reference pressures
	PREFT * and PREFTS * (surface keyword)	Component reference pressures (alias for PREF)
LIVEOIL and DEADOIL	REACENTH	Reaction molar enthalpy
	SPECHA	Oil specific heat, 1st coefficient
	SPECHB	Oil specific heat, 2nd coefficient
	SPECHG	Gas specific heat, 1st coefficient
	SPECHH	Gas specific heat, 2nd coefficient
	SPECHS	Solid specific heat, 1st coefficient
	SPECHT	Solid specific heat, 2nd coefficient
	TCRIT *	Component critical temperature (used with KVWI)
LIVEOIL and DEADOIL	THANALV	Analytic oil viscosities.
	THERMEX1	Liquid thermal expansion coefficient
	TREF * and TREFS * (surface keyword)	Component reference temperatures
	TREFT * and TREFTS *	Component reference temperature (alias for TREF)
LIVEOIL and DEADOIL	VISCREF	Reference conditions for the viscosity/temperature tables.

Restriction	Keyword	Description
	ZFACT1 and ZFACT1S (surface keyword)	Z-factor first coefficient
	ZFACTOR and ZFACTORS (surface keyword)	Z-factor zeroth coefficient

The following water keywords can be used with both `LIVEOIL` and `DEADOIL`.

Keyword	Description
KVCRWAT	Water K-value correlation
THANALB	Analytic water and steam densities
THANALH	Analytic water and steam enthalpies
THSVC	Steam viscosity coefficients
THTABB	Tabular water and steam densities
THWVC	Water viscosity coefficients
WATDENT	Water density temperature dependence
WATVISCT	Water viscosity table as a function of T

REGIONS

The `REGIONS` keywords are:

- [THERMNUM](#) Thermal region numbers

SOLUTION

The following table contains those keywords that can only be used with the Thermal option, as well as general Eclipse 300 keywords (marked*) that have extra arguments or may be used in different ways in thermal runs. The `SOLUTION` keywords are:

Keyword	Description
RPTRST *	Controls solution output for restart files: see SOLUTION and SCHEDULE section output
RPTSOL *	Controls PRINT file output: see SOLUTION and SCHEDULE section output

SUMMARY

Refer to the following tables in the *Eclipse Reference Manual*.

- [The well and group energy rate options](#)
- [Field, grid block, region quantities \(Eclipse 300 Thermal\)](#)
- The segment keywords for multisegment wells are given in [Multisegment wells](#)
- [Analytic aquifer quantities](#)
- [Simulator performance keywords](#)

SCHEDULE

The following list contains those keywords that can only be used with the Thermal option, as well as general Eclipse 300 keywords (marked*) that have extra arguments or may be used in different ways in thermal runs. The SCHEDULE keywords are:

Keyword	Description
AIMCON *	AIM selection criteria
CVCRIT *	Convergence criteria
HEATER	Position and properties of heater connections
HEATERL	Position and properties of heater connections in LGR
REACCRIT	Reaction interpolation criteria
RPTPRINT *	Item 14 can be used to control heater output
RPTRST *	Controls solution output: see
RPTSCHED *	Controls print file solution output: see
TRANGE	Specify minimum and maximum reservoir temperature
TSCRIT *	Timestepping control
WCONINJE *	Specify well injection control Specify mixed steam/gas injection
WCONPROD *	Specify well production targets Specify steam production rate
WECON*	Specify well economic limits Specify well temperature limit
WELLTCB	Specify thermal cut-back for wells
WELLTCBC	Well Temperature Cut-Back critical temperatures
WELLTCBT	Tables to be used for thermal cut-back
WELSEGS *	Defines thermal properties of the tubing/casing Defines the multisegment well structure
WELTARG *	Change well production targets. Change steam production rate
WINJTEMP	Specify steam quality or temperature or enthalpy of injection fluid
WSEGEXSS	Defines external source / sinks for multisegment wells. Specify the temperature of imported gas
WSEGHEAT	Specify the segment heat transfer to a completion, another segment or to an external source / sink.

SOLUTION and SCHEDULE section output

Grid data, for output to the restart file or the PRINT file, can be output from the SOLUTION section and the SCHEDULE section. A list of the arguments that can be used is given below. All arguments can be used for RPTRST, RPTSOL and RPTSCHED output unless otherwise noted,

Keyword	Description
ENERGY	Total energies/bulk volume (energy/(rock + pore volume))
FLOE	Inter-block energy flow

Keyword	Description
GMEG	Gas phase energy mobility
GMEO	Oil phase energy mobility
GMEW	Water phase energy mobility
HGAS	Gas enthalpy
HOIL	Oil enthalpy
HSOL	Solid enthalpy
HWAT	Water enthalpy
TCMULT	Saturation dependent thermal conductivity transmissibility multiplier
RATT	Ratio of actual temperature change to temperature target (RPTRST only)
SL	Liquid mole fractions
SPENGAS	Gas internal energy
SPENOIL	Oil internal energy
SPENWAT	Water internal energy
SV	Vapor mole fractions
SW	Water mole fractions
T2	Temperature at the two to three phase transition
T3	Bubble point temperature
TEMP	Grid block temperature
TET	Temperatures at last time level
TO	Temperature at which the oil phase leaves the system
TW	Temperature at which the water phase leaves the system
UGAS	Gas phase specific volumes
UOIL	Oil phase specific volumes
UWAT	Water phase specific volumes
V2	Vapor mole fraction at the system two to three phase transition

All sections

The following list contains those keywords that can only be used with the Thermal option, as well as general Eclipse 300 keywords (marked*) that have extra arguments or may be used in different ways in thermal runs. The other keywords are:

DEBUG3 *

Controls debug output. The following items are specific to the Thermal option

- 70 Debug from thermal
- 77 Debug of heat loss to analytic rocks
- 78 Debug of thermal cut-back

OPTIONS3 *

Activates special program options. The following items are either specific to the Thermal option or have different meanings:

- 1 Set to -1 to obtain implicit hydrostatic heads in the well model.
- 13 If < 0 the post-98A method of normalizing the solution change in the non-linear iterations is used.
- 18 If = 1 switch off Appleyard Chop.
- 21 If = -1 switch off attempts to stop Newton iterations oscillating between alternate states.
- 22 If > 0 causes the program to use an alternative set of tables from Haywood instead of the tables from Perry and Green.
- 27 If > 0 switch on temperature-based Appleyard chop.
- 40 Enables screening of the oil, water and gas saturations from a [GETSOL](#). SWAT values below 10^{-6} are set to 10^{-6} . This can resolve problems that arise in initializing the flash calculation in waterless cells.
- 46 A positive value sets the number of Newton iterations within a timestep in which a well can be revived after being closed because it is unable to flow given the current reservoir conditions. The default behavior for the 2001A release when not using the Thermal option is to revive such a well only during the first [NUPCOL](#) Newton iterations. When the Thermal option is used, the default is to allow 21 revivals within a timestep before the well is closed. If this switch is set to a negative value, no well revivals are permitted. If however the switch is set to -999 the simulator reverts to the pre-2001A treatment, whereby the number of revivals permitted within a timestep was fixed at 21.
- 52 In 2001A and later releases, the horizontal thermal transmissibility is based on the full cell cross-sectional area and does not include a net-to-gross factor. If this option is set to 1, the pre-2001A calculation, which includes net-to-gross, is used.
- 92 Alternative timestepping strategy will be implemented. In thermal runs (2006.1 onward) this option is automatically selected and can be turned off by setting a positive value.
- 135 If set to 1, an alternative implementation for predicting the energy density in the [HEATER](#) algorithm is activated
- 158 If set to 1, the Appleyard Chop is applied to components when they appear or disappear. This behavior is the default for simulations using the chemical reactions feature but can be disabled by setting this **OPTIONS3** switch to -1.
- 160 If set to 1, the pre-2008.2 thermal non-linear damping is restored.
- 161 If set to 1 an alternative procedure for simulations with rock-filled cells is activated. Mathematically this is a null change but variations due to numerical rounding may occur.
- 190 If set to a non-zero value, the pre-2009.1 setting of well enthalpies in thermal decks is activated. From 2009.1 onwards, the well enthalpies are set whenever an action is triggered.
- 214 If set to a non-zero value, this enables the pre-2009.2 John Appleyard Linear Solver ([JALS](#)).
- 217 If set to a non-zero value, the pre-2010.1 behavior for thermal heavy-oil equilibration will be restored.
- 218 If set to a non-zero value, the pre-2010.1 thermal temperature limits will be restored.

- 219 If set to a non-zero value, this enables the pre-2010.1 thermal rock-filled convergence method.
- 220 If set to 1, this enables the thermal oil viscosity-temperature correlation in the `OILVISC` keyword, which is based on $\log_{10}$.
- 235 The transmissibility and diffusivity for dual and multi-porosity connections use the simulation grid-cell bulk-volume and not the sum of the cell pore volume and cell rock volume. For most cases these will be equal, but for thermal and Coal Bed Methane runs, the assignment of rock volume might result in differences. The pre-2010.1 behavior can be restored by setting this item to 1.
- 236 The behavior when failing to converge in a thermal simulation has been improved. This includes not chopping on a minimum timestep until the maximum number of Newton iterations have been solved. The pre-2010.1 behavior can be restored by setting this item to 1.
- 260 If set to 1, the surface gas density will be determined by a surface flash, rather than using the data from the `DENSITY` keyword.

Examples

The following example files come with the Eclipse installation:

.DATA File	Description	Auxiliary files
THERM1	Based on problem 1A of <i>Aziz et al (1987)</i> single well cyclic steam injection radial grid 1 component dead oil	
THERM2	Based on problem 2A of <i>Aziz et al (1987)</i> steam drive in 1/8th of a nine spot pattern 1 component dead oil	THERM2 .GRF
THERM3	Based on problem 3A of <i>Aziz et al (1987)</i> steam drive in 1/8th of a nine spot pattern 3 component live oil	THERM3 .GRF
THERM4	Variant of THERM1 single well cyclic steam injection radial grid 2 component live oil equilibration with a gas cap mixed gas and steam injection	

.DATA File	Description	Auxiliary files
THERM5	Based on case 1 of Stone et al (2002) dual well SAGD 2 component live oil MSW	THERM5.GRF and WELL5.TXT
THERM5A	As above, but with perfectly horizontal wells.	
THERM5B	As above, but with inclined wells	
THERM6	Based on case 2 of Stone et al (2002) . single well SAGD 2 component live oil MSW	
THERM7	Based on case 3 of Stone et al (2002) single well cyclic steam injection 3 component live oil MSW	
THERM8	Based on case 3 of Stone et al (2002) single horizontal well, cyclic steam injection 2 component live oil heat loss from surface to reservoir MSW	
THERM9	Based on Coats (1980) laboratory scale combustion experiment 5 component live oil chemical reactions non-equilibrium initialization (NEI)	THERM9.GRF and OBS9.TXT
THERM10	Foamy oil	THERM10.GRF
THERM10A	As above, but without the foamy oil terms	
THERM10B	As above, but with reactions replacing the flash	
THERM10C	Foamy solids	
THERM11	Based on Coats (1980) combustion with coke 5 component live oil solid phase chemical reactions	THERM11.GRF

.DATA File	Description	Auxiliary files
THERM12	4 volatile components	
THERM12R	Heaters	
THERM12S	LGRs	
	Corner point geometry	
	Fast and flexible restarts	
THERM13A	Comparison of AIM, FULLIMP and IMPES runs.	THERM13.GRF
THERM13F	Data sets are based on THERM3 (problem 3A of <i>Aziz et al (1987)</i>), with 2000 cells and 9 wells.	
THERM13		
THERM14D	Variant of THERM1 (problem 3A of <i>Aziz et al (1987)</i>). Comparison of drift flux and homogeneous flow models in a multisegment well.	THERM14.GRF
THERM14H		
THERM15	Dual porosity and Joule-Thomson effect.	THERM15.GRF
THERM15J		
CASE3E100	Based on <i>Weinstein et al (1986)</i> . Conversion of a black-oil data set to a thermal data set	CASE3T.GRF
CASE3B		
CASE3T		
HEAT100	Conversion of a Black-oil Temperature option data set to a thermal data set	HEAT.GRF
HEAT300		
HEAT500		

Table 4.17: Thermal example files